

Before beginning:

Books:

We follow mainly William Feller Vol. 1 with excursions to

Alon and Spencer: The probabilistic Method

Michael Mitzenmacher - Eli Upfal: Probability and Computing

Paul Hoel - Sidney Port - Charles Stone: Vol 1.

You are free to consult any book you like in the library.

To do: Many people, now a days, loaned their brains to internet and social media, no one seems to think that they need to think – except those who want to brainwash you with false narratives.

Very important to think about the concepts and arguments till you understand. You should learn to think independently and question if you disagree. You will find the subject difficult if you pay attention only to the symbols instead of understanding what we are after; why we are doing in a particular way. Practice explaining steps without using symbols, of course to write we need symbols.

You should work out problem sets and write down solutions. You should practice writing proofs, especially proper justifications of steps. This is simple: read yourself what you have written! Remember, I can only evaluate what you write; Not what you have in mind.

Grading: Quiz 1 (Last class of Jan or first class Feb; 10 marks) Midsem (30) Quiz 2 (10) and semestral (50).

TA: no TA for the course.

What and Why of probability?

The concept of chance/uncertainty arises in several areas. Life is full of uncertainty and thus chance analysis can be applied to understand many practical problems. For instance, will it rain tomorrow? We are not certain, but need to know in order to advise farmers and fishermen. You can try to predict. Agreed, it is possible that sometimes we go wrong in predictions, no problem, good enough to be correct most of the time and miss very very few times. You get data about the necessary parameters — like wind speed, cloud concentration, pressure, temperature etc; — make a model to understand this uncertainty and predict.

Is it wise to release a particular medicine in the market? Here there is no data, you have to create. You need to conduct experiments; collect data regarding its effectiveness, side effects etc and then make a model and then take a decision.

Understanding chance phenomena was of supreme importance in physics too. It enters in several ways. You study several ‘laws’. For example, you know how a particle, with a given initial velocity and a given initial position, travels. But Does it really do so? If you see the rules you derived, they are under ideal conditions. In practice the wind will surely affect the motion and there will be fluctuations. In other words there is some uncertainty that creeps in.

Imagine a large container with water and also a few pollen particles. Assume that water is in equilibrium. Actually it is never in equilibrium. What we mean is that there are no (deterministically quantifiable) external forces that make pollen particle move — like water currents, air bubbles, evaporation etc. In practice you see the pollen particle still continues to perform motion. why does it move and how does it move. This motion is what you call Brownian Motion, because it was first observed by Robert Brown. Here is a lesson for us. Keep your eyes open and observe things carefully. Robert Brown, not only observed this motion, but also persisted that reasons must be found for this movement! He was even wondering if there is life in the pollen particles.

Many many years later Einstein explained the reason. It is all due to chance phenomenon: the surrounding water molecules keep hitting the pollen particle and displace it ‘this way, that way’ continuously and this causes the movement. Particles we are talking about being small, one molecular bombardment may not cause observable movement of the pollen particle; but the continuous bombardment of the large number of water molecules causes an observable displacement. Actually you do not consider *one* pollen particle. Throw some pollen particles. Ask: At time $t > 0$, what proportion of particles are at a distance x from their initial position. But let us not complicate life now. You should know that this chance (probabilistic) analysis was the basis for the first calculation of Avagadro number. Several years later Norbert Wiener proved that such motions can be mathematically modelled.

When you learn statistical mechanics, or quantum mechanics, you will see probability entering naturally. Now a days we use probability, in an essential way, in computer science too (randomized algorithms, simulation, $\dots$) ; and modelling of stock market prices and so on. Do you see the similarity: pollen particle moves this way/that way; stock price moves up/down.

Prediction of crop yield is another essential problem. After all, if there is going to be shortage of a produce, Government should initiate purchase orders now for future procurement. Can not wait till you see actual yield to take action, because then it may be too late – you may not get what you want or you may have to pay high prices.

Here is an interesting history. In 1940 the then Government wanted to estimate jute yield. Professor P C Mahalanobis said he can estimate based on a random sample of crop acreage. Inspite of widespread disbelief, he went ahead. Using 600-700 employees, Rs 8 Lakhs budget and random sampling, came up with an estimate of 7540 bales (1 bale $\sim$ 400 lbs). Government did complete enumerations with 33,000 employees and Rs 82 Lakhs and estimated 6304 bales. Actual yield : 7562 bales

[Digression: PCM was a physicist/ data analyst – statistician/planner. After the 1922 disastrous North Bengal floods, he made a statistical study of rainfall and floods extending a period of 50 years, suggested improvement of rapid drainage and not holding up of flood water – at a time when engineers were suggesting expensive retarding basins to hold up flood water. He carried out study of Brahmani-Mahanadi river basins too and again his conclusions from data analysis differed from what engineers were suggesting. At the opening of Hirakud dam, the then chief minister of Orissa Mr Kanungo wrote to PCM: “... recall the pioneering work you undertook in the twenties and thirties in collecting and compiling the data on the watershed of the Mahanadi which provided the base for designing the dam and reservoir. You had to do the work almost single handed without encouragement from any quarter. Rather, there was discouragement and sometimes ridicule for your work and its ultimate object. ...” In 1937 he made a report on Hooghly-Howrah Flushing and Irrigation scheme, to the Bengal government, which was helpful later for the Damodar Valley Hydro-Electric project. In the twenties, correcting meteorological data for observational errors, he established correlation between weather changes on the surface of earth and changes located 4 kms above sea level. He designed the first few of our five year plans.

During 1956 visit to the Indian Statistical Institute, Chinese Premier Zhou Enlai stayed well beyond schedule time discussing with PCM and concluded by “we want to learn from you”. PCM set up a powerful statistical infrastructure for the country. The famous American statistician Edward Deming commented: ‘No statistician ever had more influence on the entire profession and on governmental statistical offices the world over’.

He also made theoretical contributions like Mahalanobis D^2 statistic, Fractile graphical analysis (forerunner of what is now called non-parametric statis-

tics), interpenetrating subsamples (forerunner of what is now called Bootstrap technique).

I do not like to make such digressions as this. Wanted to mention few facts, before you start believing ill-mannered badmouthing of PCM by influential persons. PCM was a powerful thinker and made excellent use of randomness, topic of our course.]

uncertainty:

The origins of probability are indeed in gambling. Let us start with a concrete problem.

Neelabh and Siddharth are playing a match which consists of 9 games. The winner, one who wins a majority of games, will receive 800 Rs. Till now 5 games have been played: S won 3 and N won 2. Unfortunately the match has to be stopped for reasons beyond our control. How should the prize money be shared?

1. S says that he is the winner at this stage and so he should receive all. Thus S: 800 and N: 0.

2. N says that the match is stopped for no fault of his. So it should be regarded as draw and they should share equally. Thus N: 400 and S: 400.

3. Then S says: I won three out of five games, the money should be shared in the ratio 3:2. Thus S: 480 and N: 320.

4. Fermat and Pascal then think as follows. Why did this problem arise at all? Because of uncertainty regarding the remaining unplayed 4 games. So the solution should be gotten by trying to understand this uncertainty. Let us see. There are 16 possible scenarios in the remaining four games. Of these 5 scenarios make N winner ($NNNN, NNNS, NNSN, NSNN, SNNN$), while 11 scenarios make S winner. Therefore, assuming that the players are equally good to win any one game, it is reasonable to award the money in this proportion 11:5. Thus S: 550 and N: 250.

Thus entered the idea of understanding and analyzing uncertainty. Now-a-days you will reformulate the last argument as: the chances of S winning is $11/16$ and the chances of N winning is $5/16$ (given the results so far) and the money is shared in accordance to their chances of winning.

Suppose that an experiment has total number of scenarios n . Of these, you are interested in an event B having k scenarios. Then chances of the event B , (that is our chances of observing an outcome in the set B when we really do the experiment), equals $\frac{k}{n}$ — where we assumed that the scenarios are equally likely. Question: What is experiment and what is scenario? shall be made precise soon. Thought process always precedes definitions.

Modelling uncertainty:

So how do you model chance experiments. The first thing is the following: you must know what happens when you do the experiment, that is, the possible scenarios. They are called **outcomes**. For example when we toss a coin twice, one after other, the scenarios are HH, HT, TH, TT with obvious understanding of the symbols used. Each one of these is called an outcome of the experiment.

The set of all outcomes is called **sample space** of the experiment. Usually sample space is denoted by Ω . Any subset of the sample space is called an **event**.

Experiment: Toss coin twice

Sample space Ω : $\{HH, HT, TH, TT\}$.

Experiment: Roll a six faced die once.

Sample space Ω : $\{1, 2, 3, 4, 5, 6\}$

Experiment: Toss a coin till you get heads and then stop.

Sample space Ω : $\{H; TH; TTH; TTTH; \dots\}$.

So on and on.

In the first two examples there are finitely many outcomes. In the third example there are infinitely many outcomes, but countable. It is possible to think of experiments where there are more than countably many outcomes.

Experiment: Imagine a radio-active material disintegrating. The time till a beep occurs in the Geiger counter, that is, time till a particle is released.

We want to model the chance phenomenon in such experiments. Any modelling starts with understanding simple experiments. If they do not fit reality, then use the understanding thus far gained to make a realistic model; and so on. Basic philosophy: understand simple things first.

probability:

We have been using the word experiment and we shall describe several experiments shortly. But you might still wonder what exactly is 'experiment'. What exactly is an outcome. The idea of sample space is made precise as follows. As far as our mathematics is concerned, we simply start with a set (non-empty), and call this sample space and elements of Ω are called outcomes. Of course, the way you get the set Ω depends on the experiment you are considering. But what matters for us in modelling is the set and not who does the experiment, when etc.

In high school you have learnt ratios (of favourable outcomes to total number of outcomes or whatever your teacher taught you) as probabilities. So, are we developing something different? Not really, we are developing

more general models which include what you learnt, as special cases.

equally likely outcomes:

We first start understanding experiments which have finitely many outcomes. We further assume that the outcomes are equally likely; thus if there are 100 outcomes then each has a chance $1/100$ of occurring (when we perform the experiment). we have only intuitive idea of what this means and must make a mathematical model. For example in tossing a coin once, there are two outcomes H, T and we assume that they are equally likely, so each must have chance $1/2$ (something prompts us to believe chances are non-negative and add to one).

Definition: A finite nonempty set Ω is called sample space. Elements $\omega \in \Omega$ are called outcomes. The probability of each outcome is defined by: $P(\{\omega\}) = 1/|\Omega|$. Any subset $A \subset \Omega$ is called an event. We define probability of events by $P(A) = \text{sum of probabilities of outcomes in } A$. In other words $P(A) = |A|/|\Omega|$.

We denote $|S|$ for the number of points in a set, sometimes use $\#S$. Thus calculating probability reduces to counting number of outcomes in event of interest. This is one reason for considering finite sample spaces and equally likely outcomes. You can use your expertise in counting which you learnt in high school. Also, the formula above is exactly what you have learnt in high school. Did you see the words experiment and chance are not used in the definition.

You must understand two things. Most of the time we consider tossing coin or throwing balls into boxes. These are only symbolic and apply to many situations. For example any analysis involving coin tossing can be applied to experiments where there are just two outcomes; H/T or Success/Failure or On/Off or $+1/-1$ or good/defective etc. Secondly, ‘equally likely’ is only a first step. For example if you are manager of a company manufacturing bolts; each bolt could be good or defective. If these are equally likely, you should change your machine.

We shall now discuss several examples to see how the seemingly meaningless or trivial calculations you made in high school come to life.

Example 1: Matching

I have two usual decks of 52 playing cards each. I arrange one set of cards in a line, one after another. I now shuffle the second deck and place the cards in a line sequentially below each card in the first line. It is interesting to ask: what are the chances of a match; match at place i means

the same card (same denomination) appears in the two rows at place i . So the question is: what are the chances that there is match in at least one place.

What is the experiment here? Place the second deck below first one as stated. What is the sample space? All possible arrangements of the 52 cards, of the second deck, in a line. This constitutes Ω . How many outcomes are there? $52!$. is it possible to write them down and see how many of them are in our event? It takes a very very long time, so it is not the best way of doing it.

Let A be the set of all outcomes where there is at least one match. It is not easy to directly count $|A|$, number of elements in A . So we split this event into simpler events and use ‘inclusion-exclusion’ formula. Let, for $1 \leq i \leq 52$, A_i be the set of all outcomes in which the cards at the i -th place are same. Clearly $A = \cup A_i$.

■ Theorem 1: Inclusion-Exclusion principle:

$$|\bigcup_1^n A_i| = S_1 - S_2 + S_3 - S_4 + \cdots$$

$$S_1 = \sum_1^n |A_i|; \quad S_2 = \sum_{i < j} |A_i \cap A_j|; \quad S_3 = \sum_{i < j < k} |A_i \cap A_j \cap A_k|; \quad \cdots \quad \blacksquare$$

Is this formula correct? We shall see later.

Is this formula useful? After all, if we are unable to calculate the numbers S_i then this formula is also useless. We are lucky. I leave for you to check, in our example, $|A_i| = 51!$; and $|A_i \cap A_j| = 50!$ ($i \neq j$) and $|A_i \cap A_j \cap A_k| = 49!$ ($i < j < k$). and so on. Thus

$$S_1 = 52 \times 51! = 52!; \quad S_2 = \binom{52}{2} \times 50! = \frac{52!}{2!}; \quad S_3 = \binom{52}{3} \times 49! = \frac{52!}{3!}$$

Thus

$$P(A) = 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \cdots \pm \frac{1}{52!}$$

This is the chances that there is at least one match. I used $\pm$ for the last term, we know the sign is indeed minus (check). If you want chances of no match, $P(A^c)$ you will see

$$P(A^c) = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots \pm \frac{1}{52!}$$

Here are some comments on the power of technique/argument used above which is true of many mathematical arguments. Instead of 52 cards, if we

have two similar decks of n cards, then the chances that there is no match is given by

$$1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots \pm \frac{1}{n!}$$

In other words the argument is still applicable and gives an answer. We can also see that the answer is close to a number we already knew: $1/e$ if n is pretty large – enters Analysis!

symbols are unimportant. Consider the following problem: I have 52 letters to different persons and 52 envelopes with their addresses. I place the letters at random in the envelopes. What are the chances that at least one letter goes to its envelope. You will see the answer is exactly same as above. Convince yourself that this is so.

Did you realize that the word ‘shuffle’ quietly disappeared after statement of the problem. Where did it go? It went into the model! It is difficult to explain what is shuffling – who will do, how many times, by what method, etc. But what is its purpose? Its purpose is that all arrangements are equally likely. That is precisely what we used in our calculation. In other words what is understood by our mind in using the word ‘shuffling’ is precisely what is achieved by the mathematical modelling of equally likely outcomes.

This is called **matching experiment**. There are several other things you can do with this experiment.

Example 2: Sampling With Replacement

Let $S = \{1, 2, \dots, N\}$. Experiment consists of picking one after another, twenty times, points from this set A . Remember always selected from S , we are not excluding the already selected elements when selecting a later point. Thus we are making a list of 20 items from S , with repetitions allowed.

Sample space Ω consists of all sequences $(x_1, x_2, \dots, x_{20})$ where for each i , $x_i \in S$. Check $|\Omega| = N^{20}$. Each outcome has probability $1/N^{20}$.

This experiment is called **sampling with replacement**, of size 20. With replacement because when we take second item the first item is replaced in the set S ; the second selection is made as if you noted down the first item and put it back before second selection. Each outcome of this experiment is called a **sample (with replacement) of size 20**.

What are the chances that first observation (first point of the sample) is 1? Answer: there are N^{19} outcomes for which first item is 1. So the required probability is $N^{19}/N^{20} = 1/N$

What are the chances that second item is 1? Again there are N^{19} outcomes where second item is 1. So required probability is again $1/N$.

What are the chances that first and second item are 1? There are N^{18} such outcomes and so the probability is $N^{18}/N^{20} = 1/N^2$.

What are the chances that 1 is included in the sample, that is some x_i in the sample is 1? There are $(N - 1)^{20}$ outcomes where $x_i \neq 1$ for all i and hence the required probability equals

$$\frac{N^{20} - (N - 1)^{20}}{N^{20}} = 1 - \left(1 - \frac{1}{N}\right)^{20}$$

Of course you can talk about sample of size n instead of 20. The set from which you are picking, namely, $\{1, 2, \dots, N\}$ is called population. It need not be numbers. it could consist of all students of cmi.

Example 3: Sampling Without Replacement

Again let $A = \{1, 2, \dots, N\}$. Experiment consists of picking one after another, twenty times, points from this set A , each time NOT replacing the points selected earlier.

Sample space Ω consists of all sequences $(x_1, x_2, \dots, x_{20})$ where for each i , $x_i \in A$ and they are distinct. That is $x_i \neq x_j$ for $i \neq j$.

Check $|\Omega| = N(N - 1) \dots (N - 19)$.

We assume that $N \geq 20$. Otherwise the sample space is empty set.

Each outcome has probability

$$\frac{1}{N(N - 1) \dots (N - 19)}.$$

This experiment is called **sampling without replacement**, of size 20. Without replacement because when we take second item the first item is removed from the set A ; similarly at each stage selected item is removed and then next selection is made. **Each outcome of this experiment is called a sample (without replacement) of size 20.**

What are the chances that first observation (first point of the sample) is 1? Answer: there are $(N - 1)(N - 2) \dots (N - 19)$ outcomes for which first item is 1. So the required probability is $1/N$

What are the chances that second item is 1? Again there are $(N - 1)(N - 2) \dots (N - 19)$ outcomes where second item is 1. So required probability is again $1/N$.

What are the chances that first and second item are 1? Zero.

What are the chances that 1 is included in the sample, that is some x_i in the sample is 1? There are $(N - 1)(N - 2) \dots (N - 20)$ outcomes where

$x_i \neq 1$ for all i and hence the required probability equals $1 - \frac{N-20}{N} = \frac{20}{N}$

Example 4: committee making

Again let $A = \{1, 2, \dots, N\}$. Experiment consists of picking a subset of A of size 20.

Remember, we are not picking elements one by one. We grab a subset consisting of 20 elements.

Sample space Ω consists of all subsets of A which have exactly 20 elements. That is all $\omega \subset A$ with $|\omega| = 20$ Check $|\Omega| = \binom{N}{20}$

We assume that $N \geq 20$. Otherwise the sample space is empty set. Each outcome has probability $\frac{1}{\binom{N}{20}}$. This experiment is called **selecting a subset of size 20 or sampling a subset of size 20**. Each outcome is called a random subset of size 20.

What are the chances that first observation (first point of the sample) is 1? this question does not make sense. There is no first or second element. We have a subset consisting of 20 points, that is all. What are the chances that 1 is included in the selected set? that is $1 \in \omega$? There are $\binom{N-1}{19}$ such outcomes and hence the probability equals

$$\frac{\binom{N-1}{19}}{\binom{N}{20}} = \frac{(N-1)!}{19!(N-20)!} \frac{20!(N-20)!}{N!} = \frac{20}{N}$$

same as in the case of sampling without replacement.

For example when you want to make a committee of 3 students, it is the three that matters and not who is selected first etc.

Example 5: Maxwell-Boltzmann

I have 30 boxes numbered: $1, 2, \dots, 30$. I have 20 balls numbered: $1, 2, \dots, 20$. Experiment is to throw the balls into the boxes. $|\Omega| = 30^{20}$. all these outcomes are equally likely.

What are the chances that the first box is empty? $29^{20}/30^{20}$

What are the chances that boxes 1,5,7 contain respectively 10,7,3 balls? $\binom{20}{10}\binom{10}{7}/30^{20}$ Remember, any 10 balls could go in box 1.

This experiment is called **Maxwell-Boltzman experiment**. Boxes are energy levels and balls are elementary particles.

Example 6: Bose-Einstein

I have 30 boxes numbered: $1, 2, \dots, 30$. I have 20 balls all looking alike. There is no way to distinguish one ball from the other. Experiment is to throw the balls into the boxes. How many different arrangements can our eye perceive? $|\Omega| = \binom{49}{29}$.

This can be seen in several ways. Put 49 star marks in a row; select 29 of these and convert them into vertical lines. You see a picture of 20 balls (the remaining star marks) put in 30 boxes (created by the vertical lines). And every method of putting balls into the boxes is achieved this way. remember, that now it makes no sense to say: where did ball one go? all balls look alike. so two arrangements are different only when the ‘occupancy numbers’ are different, that is, the vector $(n_1, n_2, \dots, n_{30})$ where n_i is the number of balls in box i ; determines the arrangement. Two different vectors give two different arrangements.

Bose-Einstein Rule: These distinguishable arrangements are equally likely.

This experiment is called **Bose-Einstein experiment**. Again boxes are energy levels and balls are elementary particles.

What are the chances that box one is empty? $\binom{48}{28}/\binom{49}{29}$ You can simplify and see.

What are the chances that boxes 1,5,7 contain respectively 10,7,3 balls? $1/\binom{49}{29}$ This is because there is only one outcome satisfying the given condition.

Particles that obey this rule are called Bosons. Photons are known to obey this rule. To understand how outrageous is this rule, consider the following. I have a box with 1 green ball and 100 red balls numbered 1,2,..., 100. I pick a ball at random. What are the chances it is green: 1/101.

Suppose now I tell you that all red balls look alike, there are no numbers on them. How many outcomes your eye can perceive? only two: Red, green; Can I say that these are equally likely and conclude that chance of green ball is 1/2? Ridiculous. The B-E rule **does not** apply here. Thus you should be careful. Without knowing if B-E rule is ‘assumed’, you can not start calculating just because you are told ‘balls (or whatever)’ look alike.

Historically, making observations or collecting statistics we found that they obey this and hence the rule is invented. For this reason sometimes these probabilities are also called Bose-Einstein statistics.

[This was S N Bose; Satyendra Nath Bose. There was R C Bose, Raj Chandra Bose in combinatorics and coding. He along with S S Shrikhande and E T Parker were named ‘Euler’s Spoilers’ by NewYork Times; they ‘thoroughly’ disproved a conjecture of Euler on Orthogonal Latin Squares. There was J C Bose, Jagadish Chandra Bose famous for his work in Biology and Radio waves; the first transatlantic wireless signal transmission was received by Bose’s Diode Detector – Marconi used it without revealing.]

Example. 7: Fermi-Dirac

I have 30 boxes numbered: 1, 2, ..., 30. I have 20 balls all looking alike.

There is no way to distinguish one ball from the other. Experiment is to throw the balls into the boxes subject to the condition: No more than one ball in a box. $|\Omega| = \binom{30}{20}$ These are equally likely.

What are the chances that box one is empty? $\binom{29}{20}/\binom{30}{20}$. What are the chances that boxes 1,5,7 contain respectively 10,7,3 balls? Zero.

This is called **Fermi-Dirac experiment**. Protons obey this rule.

Example 8: Stick breaking

I have 23 sticks numbered $1, 2, \dots, 23$. Or 23 sticks of different colors. Break each into two parts one small part and one long part. Now I make a bag of 23 sticks again by pairing these 46 pieces. What are the chances of getting back original pairing (original sticks)? What is the probability that long pieces are paired with small parts?

Here

$$|\Omega| = \frac{46!}{2^{23} (23)!}$$

This is because, if c is the number of ways of pairing, consider the problem of pairing and arranging the pairs in a line. This can be done using two methods:

(Method 1)

make 23 pairs (c ways) and then arrange these 23 pairs in a row $(23)!$ ways. so the required number is $c (23)!$ ways.

(Method 2)

pick a pair out of the 46 pieces, put in place one; pick a pair out of the rest, put in place 2 etc. so can be done in

$$\binom{46}{2} \binom{44}{2} \cdots \binom{2}{2} = \frac{46!}{2^{23}}$$

As a result

$$c (23)! = \frac{46!}{2^{23}}; \quad c = \frac{46!}{2^{23} (23)!}$$

Thus each outcome has probability

$$\frac{1}{c} = \frac{2^{23} (23)!}{(46)!}$$

This is also the probability of getting original pairing.

If A is the set of all pairings where large parts are paired with short parts, then $|A| = (23)!$ — you need to put long pieces in a line, no matter how, and

then place short pieces one with each long piece. So the required probability equals

$$\frac{2^{23} (23)! (23)!}{(46)!} = \frac{2^{23}}{\binom{46}{23}}$$

As you could guess, these 23 sticks are the 23 chromosomes in human cell. They break at the centromere, giving you one long part and one small part – centromere is not exactly in the centre. Then they recombine. The first question asks the chances of getting original chromosomes back or original cell back. The second question asks for the chances of the cell dying; if two long pieces join there is not enough space in the cell to fit in. what happens if two short pieces join?

What happens if there are N sticks? For example, *Drasofila* has seven pairs of chromosomes.

Example 9: Reflection principle

In an election, there are two candidates A , B getting votes 16 and 24 respectively. Thus B won. What are the chances that throughout counting B is leading?

We discuss this problem for two reasons. Firstly, it teaches a new counting technique and the method has various applications. Secondly, the problem itself has a cute answer:

$$\frac{24 - 16}{24 + 16} = \frac{8}{40}.$$

What is vote counting? No, do not think of electronic machines; there is no counting you do there, you press a button and answer pops up. It is a sequence of letters $ABBBAAAB$ of length 40 reading the vote as you go along. But since we are interested in counting number of votes for each candidate, let us represent by numbers.

Let us count a vote for B as $+1$ and a vote for A as -1 . Thus vote counting means all possible sequences of ± 1 of length 40 which have 24 ones and 16 minus ones. This is the sample space. Thus $|\Omega| = \binom{40}{24}$

We assume that all these outcomes are equally likely. Thus the problem boils down to finding $|T|$ where T consists of sequences which have at every stage more $+1$ than -1 .

If we think of outcomes as simply sequences of ± 1 then to see if an outcome is in T or not, we need to add and check at every stage. So let us think of Ω in a different and convenient manner, already taking cumulative sums. Thus It is a sequence $(s_0 = 0, s_1, s_2, \dots, s_{40})$ where each $s_{i+1} - s_i = \pm 1$ and $s_{40} = 8$. Observe that if you have a sequence of ± 1 as above, then taking cumulative sums you get such a sequence (s_i) . Conversely, if you have a

sequence of (s_i) like this then their successive differences gives a sequence of ± 1 as above. Also this is a faithful conversion (bijection).

Now we want to picturize these points. A path is defined as a sequence $(0, s_0), (1, s_1), \dots, (k, s_k)$ where each s_i is an integer and $s_i - s_{i-1} = \pm 1$ for all $i \geq 1$. As a picture each successive points are joined by a straight line. The path is said to start at s_0 and end at s_k and is of length k . You can also talk about paths starting at $(1, 5)$ etc. Thus we can now identify our outcomes as paths starting at $(0, 0)$ and ending at $(40, 8)$.

You can draw the axes and join successive points by straight line and visualize a path. such a picturesque visualization suggests new ideas as we see now.

A vote counting corresponds to a path starting at $(0, 0)$ and ending at $(40, 8)$. Number of paths starting at $(0, 0)$ and ending at (n, r) are

$$\binom{n}{\frac{n+r}{2}} = \binom{n}{\frac{n-r}{2}}.$$

This is because, to have such a path the number of ones must be $(n+r)/2$ and number minuses must be $(n-r)/2$. If $k \geq 0$ is not an integer, we take $\binom{n}{k}$ to be zero. In the formula above if $(n+r)/2$ is not an integer (≥ 0) then value is by definition, zero.

You must understand a very important matter here. The formula is stated above as a fact with a proof. However the formula ' $\binom{n}{k} = 0$ when k is not integer' is a convention. It is something we adapted. You can not confuse conventions with facts. So the question arises, if $(n+r)/2$ is not an integer, can you show that the number of paths from $(0, 0)$ to (n, r) is indeed zero, so that the convention agrees with the stated formula. Yes.

Thus total number of vote countings are $\binom{40}{24}$. Let us note a simple but powerful fact.

Reflection Principle:

■ **Theorem 2:** Let $k \geq 1$ and $r \geq 1$ be integers.

number of paths from $(0, k)$ to (n, r) which touch or cross the x -axis (after starting) equals number of paths from $(0, -k)$ to (n, r) . ■

Note $(0, -k)$ is reflection of $(0, k)$ in the x -axis. The proof of this fact uses reflection in x -axis.

Let A be the first set of paths and B be the second set of paths. Take a path π in A : $(0, s_0), (1, s_1), \dots, (n, s_n)$; $s_0 = k$; $s_n = r$.

Since it touches the x -axis; let i be the first index with $s_i = 0$. Reflect the path till that point in x -axis. do not reflect the remaining segment of the

path. That is, consider the path π^* :

$$(0, -s_0), (1, -s_1), \dots (i, -s_i), (i+1, s_{i+1}), (i+2, s_{i+2}), \dots (n, s_n).$$

Remembering that $s_i = 0$, we see (i, s_i) is same as $(i, -s_i)$. We can easily verify that this is also a path, that is, successive s -differences are ± 1 . This path starts at $(0, -k)$ and ends at (n, r) . Hence $\pi^* \in B$.

Given path $\eta \in B$ it starts below x -axis and ends above; so must hit x -axis, take the first time it hits and reflect the part till then in x -axis, keeping later part as is. This will be a path $\pi \in A$ and $\eta = \pi^*$ and actually the map $\eta \mapsto \pi$ is inverse map of $\pi \mapsto \pi^*$.

This one-one map between A and B proves the result. ■

We are interested in

number of paths $(0, 0)$ to $(40, 8)$ that do not touch/cross x -axis.

[such a path should pass through $(1, 1)$]

= number of paths from $(1, 1)$ to $(40, 8)$ that do not touch x -axis

[subtract 1 from first coordinate to see]

= paths from $(0, 1)$ to $(39, 8)$ that do not etc

= [Total number of paths $(0, 1)$ to $(39, 8)$] minus [number of paths $(0, 1)$

to $(39, 8)$ that touch etc]

(Use reflection principle)

= [paths $(0, 1)$ to $(39, 8)$] minus [paths $(0, -1)$ to $(39, 8)$]

= [paths $(0, 0)$ to $(39, 7)$] minus [paths $(0, 0)$ to $(39, 9)$]

$$= \binom{39}{23} \text{ minus } \binom{39}{24}$$

$$= \binom{40}{24} \frac{24 - 16}{40}.$$

Since the total number of outcomes is $\binom{40}{24}$ we conclude that the required probability is $8/40$ as stated.

This is **path counting example** and there are several interesting consequences of the reflection principle.

Let us now prove the inclusion-exclusion principle.

Proof 1: By induction .

If A, B are disjoint then $|A \cup B| = |A| + |B|$ This is because a bijection on A to $|A|$ can be followed by a bijection of B , this gives a bijection of $A \cup B$ because A, B are disjoint. (Prove). If A, B are any two sets, first observe

$A \cup B = A \cup (B \setminus A)$ disjoint union; also $B = (B \setminus A) \cup (A \cap B)$ disjoint union so that

$$|A \cup B| = |A| + |B \setminus A| \quad |B \setminus A| + |B \cap A| = |B|$$

Adding these two we get $|A \cup B| = |A| + |B| - |A \cap B|$. Thus the statement is proved for $n = 2$. Suppose we proved for n . We prove it for $n + 1$. Take $A_1, \dots, A_{n+1}$. Induction hypothesis applied to $A_1, \dots, A_n$ gives

$$|\cup_1^n A_i| = T_1 - T_2 + T_3 - \dots \quad (\bullet)$$

where T_i are the sums $\sum_1^n |A_i|$, $\sum_{1 \leq i < j \leq n} |A_i \cap A_j|$ etc. Induction hypothesis applied to $B_1 = A_1 \cap A_{n+1}, B_2 = A_2 \cap A_{n+1}, \dots, B_n = A_n \cap A_{n+1}$ gives

$$|\cup_1^n (A_i \cap A_{n+1})| = U_1 - U_2 + U_3 - \dots \quad (\bullet\bullet)$$

where

$$U_1 = \sum_{1 \leq i \leq n} |A_i \cap A_{n+1}|$$

$$U_2 = \sum_{1 \leq i < j \leq n} |A_i \cap A_j \cap A_{n+1}|$$

$$U_3 = \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k \cap A_{n+1}|$$

etc. Applying $n = 2$ case for the sets $(\cup_1^n A_i)$ and A_{n+1} we get

$$\begin{aligned} |\cup_1^{n+1} A_i| &= |\cup_1^n A_i| + |A_{n+1}| - |\cup_1^n (A_i \cap A_{n+1})| \quad (\dagger) \\ &= (T_1 - T_2 + T_3 - \dots) + |A_{n+1}| - (U_1 - U_2 + U_3 - \dots) \\ &= (T_1 + |A_{n+1}|) - (T_2 + U_1) + (T_3 + U_2) - \dots \\ &= S_1 - S_2 + S_3 - \dots \end{aligned}$$

Proof 2: How many times one is added.

$|\cup_1^n A_i|$ is obtained by adding one as many times as there are points in $\cup_1^n A_i$. Want to show that in $S_1 - S_2 + S_3 - \dots$ one is added as many times as there are points in $\cup_1^n A_i$.

Firstly if $x \notin \cup_1^n A_i$ one is not added corresponding to this point in the calculation of any S_i and hence one is not added corresponding to this point in $S_1 - S_2 + S_3 - \dots$

So let $x \in \cup_1^n A_i$. If it is in exactly r of these sets then corresponding to this point, one is added r times in S_1 ; $\binom{r}{2}$ times in S_2 ; $\binom{r}{3}$ times in S_3 etc and

$\binom{r}{r}$ times in S_r and not added in the calculation of other S_i . Thus number of times one is added in $S_1 - S_2 + S_3 - \dots$ equals

$$\begin{aligned} r - \binom{r}{2} + \binom{r}{3} + \dots \pm \binom{r}{r} \\ = 1 - (1 - 1)^r = 1 \end{aligned}$$

as required. Thus on both sides of the inclusion-exclusion formula; one is added for each $x \in \cup A_i$ exactly once and not added for $x \notin \cup A_i$. ■

Let us summarize rules about probability.

Theorem 3: In any experiment with equally likely outcomes, sample space Ω

- (i) $P(\emptyset) = 0$ $P(\Omega) = 1$ $P(A^c) = 1 - P(A)$
- (ii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. More generally

$$P(\cup A_i) = \sum P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots$$

conditional probability:

One of the most important concepts is that of conditional probability. Most of the time we are not totally ignorant. There is a chance experiment, we do not know the outcome but we know some partial information about the outcome.

For example when you are predicting tomorrow's weather, you are not completely ignorant, you do not close your eye and say there are two possibilities etc. You will see the information about wind speed, temperature, pressure etc. Similarly, if a mother goes to a doctor and asks if her two year old baby is of correct weight, then the doctor does not blindly compare baby's weight with the average weight of two years olds. He will see several parameters like the birth weight etc and compares with the average weight of children with similar parameters.

Example: Consider rolling a fair die.

$\Omega = \{1, 2, 3, 4, 5, 6\}$ and $P(A) = |A|/6$.

Let A be the event $A = \{2, 3, 4\}$. Then $P(A) = 3/6$.

Some one told us that an even number appeared. How shall we define probabilities now? Let us denote $B = \{2, 4, 6\}$. It is meaningless to say that the probabilities of events are as earlier. strictly speaking, it is meaningful but practically useless. If we are now asked chances of the event A then we

feel that $P(A)$ should be $2/3$. Because, there are now only three possibilities: 2, 4, 6, of which the event A has two: 2, 4. Of course, if we say $P(A) = 2/3$ there will be total confusion because in the original experiment $P(A) = 3/6$. So we say $P(A|B) = 2/3$; read conditional probability of A given B or probability of A given B . Here ‘given B ’ means that an outcome in B occurred.

Similarly, if tomorrow someone tells that an odd number occurred, then you need to rethink and say that 1, 3, 5, are the possible outcomes with this information $P(A) = 1/3$ etc. It is a nuisance to keep on changing sample space and *then* calculate. One smart fellow found out the quantity $P(A|B)$ is nothing but $P(A \cap B)/P(B)$. You can also check this! This is what was used in high school.

This formula has the advantage that it expresses the conditional probability, that we felt, as ratio of usual probabilities. Thus you can calculate conditional probability without any sample space considerations, using only probabilities in the original experiment. We take this as the definition.

Definition: Suppose Ω be a probability space with equally likely outcomes (thus $p(\omega) = 1/|\Omega|$ for every outcome). Then we define for events A and B , conditional probability of event A given event B , by

$$P(A|B) = P(A \cap B)/P(B)$$

of course ‘given’ means we are told that an event in B occurred.

Warning: $A|B$ is NOT an event. Thus to say $P(A|B)$ is probability of the event $A|B$ is meaningless. It is conditional probability of the event A given the event B . Also whenever we use these notions it is understood that $P(B) \neq 0$.

Theorem: For any events the following holds

$$P(A \cap B) = P(A)P(B|A).$$

$$P(A \cap B \cap C) = P(A)P(B|A)P(C|A \cap B). \text{ More generally}$$

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \cdots P(A_n|\cap_1^{n-1} A_i)$$

Proof is easy. This gives a way of calculating probabilities using conditional probabilities. You might feel uncomfortable and say: what do you mean? we must be having probabilities already to talk about conditional probabilities. Yes, sometimes we know **some** probabilities and **some** conditional probabilities.

Polya urn scheme:

Consider an urn with 30 balls; 20 red and 10 green. here is the game which can be played forever.

Pick a ball at random, note its color and put back and add one ball of that color to the urn. Repeat...

We calculate some probabilities. What are the chances first draw is red?

$$P(R_1) = 20/30$$

What are the chances second draw is red? We do not know the result of the first draw and hence we do not know the composition of the urn before second draw and hence we can not calculate as a simple ratio. Even if we knew result of first draw, such a ratio gives only conditional probability given your assumption on the result of the first draw. We proceed as follows

$$\begin{aligned} P(R_2) &= P(R_2G_1) + P(R_2R_1) = P(G_1)P(R_2|G_1) + P(R_1)P(R_2|R_1) \\ &= \frac{10}{30} \frac{20}{31} + \frac{20}{30} \frac{21}{31} = \frac{20}{30} \end{aligned}$$

Our Theorems are at work! See how we split the event into disjoint sub-events by incorporating the lacking information about the first draw and use additivity of probability for disjoint events. Also you see how the product rule led to final calculation.

Thus chance of red ball remains same. In fact chance of red ball at any draw remains 20/30.

Note that $P(R_1) = 20/30$.

$$P(R_2|R_1) = 21/31 > P(R_1)$$

$$P(R_3|R_1 \cap R_2) = 22/32 > P(R_2|R_1).$$

If you see a red ball, then chances of seeing red ball is increased. If the second ball is also red, then chances of red at third are further increased. Though these conditional probabilities are changing, the chance of red ball at any draw equals the same, unchanging. This is where probability comes to life!

Our friend Naman came from very far. Suppose before starting, some one told him that there is flu epidemic in Chennai. Let us understand his feelings during a short duration after he lands in Chennai. If he right away met a person with flu, he tends to think that there must really be much flu here. The more and more flu people he meets soon after reaching here, he assigns higher and higher chances for flu in Chennai. On the other hand if the people he met had no flu, he tends to think that there is not that much flu here afterall. However, his experience has nothing to do with reality. There is a certain percentage who have flu and the chances of seeing a person with flu is just that fraction. [That is why I said in a short duration, you need not

worry about dynamics of spread]. Thus interpreting red as person with flu, the model reflects precisely this phenomenon.

Mathematics, like music, is amenable for diverse improvisations. Why add one ball? Why not 10 balls of that color? Yes, you can. Even then this phenomenon remains true. Chance of red ball at any stage remains same. Conditional probability of red increase very much now, if you saw a red ball – Almost like representing your nervousness.

Why ball of that color, why not ball of opposite color? yes, can be done, useful, but this phenomenon is no longer true. Why only two colors? Why not three colors? yes, can be done.

We continue a little more of these urn schemes and then take up the case of not necessarily equally likely outcomes, more realistic.

To show that the chances of red at any draw remain same. Let us start with r red balls and g green balls. Suppose S_n is the statement that n -th ball red has chances $r/(r+g)$; and hence n -th ball green has chances $g/(r+g)$. Suppose that S_{n-1} is shown. If E_n is the event n th draw red, then you can write

$$P(E_n) = P(E_n E_{n-1}) + P(E_n E_{n-1}^c) = P(E_{n-1})P(E_n | E_{n-1}) + \dots$$

Unfortunately this does not help. Even though you can substitute for $P(E_{n-1})$ from induction hypothesis, you still do not know the composition of the urn and hence are unable to find out $P(E_n | E_{n-1})$. So the best way to convert n -th draw to $(n-1)$ -th draw is to condition on the first draw

$$P(E_n) = P(E_n E_1) + P(E_n E_1^c) = P(E_1)P(E_n | E_1) + \dots$$

Now you can use $P(E_1) = r/(r+g)$ To calculate $P(E_n | E_1)$, given E_1 , there are $(r+1)$ red and g green balls and from now on we are looking at the $(n-1)$ -th draw. Unfortunately, since the data has changed, you can not use induction hypothesis to say $P(E_n | E_1) = (r+1)/(r+g+1)$. To allow this, our induction hypothesis should be strong. We proceed as follows.

(S_n) : whatever be $r, g \geq 1$; $P(R_n) = \frac{r}{r+g}$ in Polya scheme starting with r red and g green balls.

We are denoting by R_n the event that n -th draw is red, similarly for G_n . S_1 and S_2 are proved above: change 20 and 10 to r and g . Suppose S_n is proved. We shall prove for S_{n+1} . We need to calculate $P(R_{n+1})$.

$$\begin{aligned} P(R_{n+1}) &= P(R_{n+1} R_1) + P(R_{n+1} G_1) \\ &= P(R_1)P(R_{n+1} | R_1) + P(G_1)P(R_{n+1} | G_1) \end{aligned}$$

Note that given R_1 ; after the first draw you have a Polya scheme starting with $r+1$ red and g green balls and you want the chances of red **now** at n -th draw which by induction hypothesis is $(r+1)/(r+g+1)$. Similarly given G_1 also you can calculate getting

$$P(R_{n+1}) = \frac{r}{r+g} \frac{r+1}{r+g+1} + \frac{g}{r+g} \frac{r}{r+g+1} = \frac{r}{r+g} \quad \blacksquare$$

Let us make one more calculation.

$$P(R_2 G_1) = P(G_1)P(R_2 | G_1) = \frac{10}{30} \frac{20}{31}$$

$$P(R_1 G_2) = P(R_1)P(G_2 | R_1) = \frac{20}{30} \frac{10}{31}$$

Thus $P(R_1 G_2) = P(R_2 G_1)$. Since $P(G_2) = P(G_1)$ we see

$$P(R_1 | G_2) = P(R_2 | G_1)$$

$$P(\text{R at first} | \text{G at second}) = P(\text{R at second} | \text{G at first})$$

In other words there is time symmetry. Given information: G at time 2; ask: R at time 1, or, give the same information G , but at time 1, and ask the same question R , but at time 2. You get the same answer. Only the role of time has changed; given information is same and question asked is same. Here again probabilities come alive, because this is what we observe in practice.

Interpret red and green as genes and time in generations. Ask: what are the chances my father has a gene R given I have gene G . Or ask: what are the chances I have gene R given my father has gene G . Answer remains same under certain assumptions on the genetic structure of population. Note that information is same G , question is same R . In one case information is about me (second generation) and question is about my father (first generation). In the other case information is about my father (first generation) and question is about me (second generation) — a time reversal.

There is nothing special about R, G . You could have considered R, R or G, G . Nothing special about times 1, 2. We could have taken 10, 21 draws.

Friedman Urn model:

Indeed, the model of adding one ball of opposite color is also important and is called Friedman's urn model. Of course, the above phenomenon is no longer true, chances of red ball at draw n could change with n . Does this model also reflect something that we see? yes, this models 'safety campaign'.

Interpret red as accidents and green as safety measures. In general police are a relaxed lot and do nothing. When accidents increase, they suddenly become active and start implementing safety measures. Once you see more safety measures and accidents decline, then police become slack and accidents start increasing. Once they increase and you see more accidents, they once again wake up and implement safety measures.

Like simple harmonic motion, you can think of the after effect as restoring force. If there are too many red balls, you are likely to see a red ball and so add green ball. On the other hand if there are too many green balls, you are likely to see a green ball and you add a red ball.

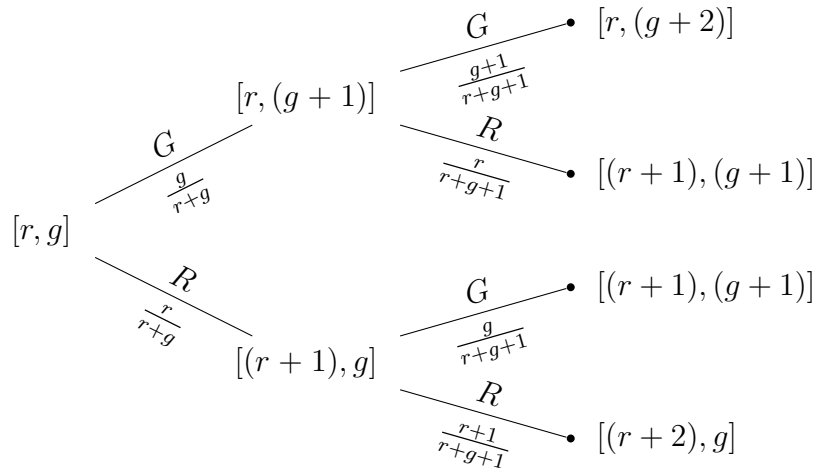
These urn models are invented with a view to modelling the phenomenon of aftereffects. In each of these urn models after you see a color, there is an

aftereffect due to change in the composition of the urn.

Tree diagrams:

Returning to Polya Urn scheme, we can pictorially represent the outcomes. We can draw a tree diagram depicting the composition of the urn at each stage and the outcomes and conditional probabilities. This is what we shall do below starting with general (r, g) urn.

If you read the letters along any particular path, you will get outcomes. Probability of the event is simply product of the numbers on the branches along the path. On each branch is written conditional probability of that event given the past till then. At the end of each branch is written the composition of the urn to help you calculate probabilities for the next branch. Initially composition is r red and g green. After green draw urn has become $[r, g + 1]$ etc.



On each branch, the numbers given are not the probabilities, but the conditional probabilities given what happened so far, as seen from the connection to the root.

Tree diagrams are a convenient way of picturing a sequential experiment as it unfolds. Outcomes of the experiment are just the branches of the tree (a path from root to leaf) and probability of each outcome is the product of numbers along the branch – that is, along the edges of the path. The earlier theorem on $P(A \cap B \cap C)$ etc justifies this calculation.

Discrete experiments - All over again:

So far we have been considering equally likely outcomes and finite sample space. Having got some experience, we shall now deal with experiments

where the outcomes may not be equally likely and also the number of outcomes may not be finite. Again as a first step we only consider sample spaces with countably many points.

Definition: A probability space is (Ω, P) where Ω is a finite or countably infinite set and P is a function on Ω which associates a non-negative number $P(\omega)$ with each $\omega \in \Omega$ such that $\sum_{\omega \in \Omega} P(\omega) = 1$. As earlier events are subsets of Ω and for an event $A \subset \Omega$, we define $P(A) = \sum_{\omega \in A} P(\omega)$.

The idea is that we are doing an experiment and points of Ω are just the outcomes. For outcome ω the number $P(\omega)$ is the probability of the outcome ω . Since we feel that chances are non-negative (chances of rain equals minus $1/2$ does not sound good) and also something or other must surely happen; that is, $P(\Omega) = 1$. Note that symbols like $\sum_{\omega \in A} P(\omega)$ are meaningful; enumerate $A = \{a_1, a_2, \dots\}$ and this symbol $\sum_{\omega \in A} P(\omega)$ is defined as sum of the series $\sum_n P(a_n)$. This is well defined because the numbers $P(\omega)$ are non-negative and so if someone enumerates A in a different way and adds the answer will not change.

Also note that we are using the same symbol P for the probability of outcomes and also for probability of events. This does not cause confusion because if the event A consists of just one outcome ω_0 , that is, $A = \{\omega_0\}$, then our definition $P(A)$ is same as the given $P(\omega_0)$.

Also note that there is no clash with the earlier equally likely case. If Ω is finite and $P(\omega) = 1/|\Omega|$ then equally likely case results.

Example: $\Omega = \{H, T\}$ and $P(H) = 1/3$, $P(T) = 2/3$.

Example: $\Omega = \{1, 2, 3, \dots\}$ and $P(n) = 1/2^n$ for $n \geq 1$.

Example: Take same $\Omega = \{1, 2, 3, \dots\}$. Fix any nonnegative numbers $p_n \geq 0$ you like which add to one and put $P(n) = p_n$.

Earlier theorem remains valid.

Theorem Let (Ω, P) be any probability space.

- (i) $P(\emptyset) = 0$ $P(\Omega) = 1$ $P(A^c) = 1 - P(A)$.
- (ii) $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$

More generally, for finitely many events, $P(\cup A_i) = S_1 - S_2 + S_3 - \dots$ where

$$S_1 = \sum P(A_i), \quad S_2 = \sum_{i < j} P(A_i \cap A_j), \quad S_3 = \sum_{i < j < k} P(A_i \cap A_j \cap A_k), \dots$$

(iii) For infinitely many disjoint events; $P(\cup A_n) = \sum P(A_n)$. ■

To prove (ii) you can either use induction or show that for each $\omega \in \Omega$, the number $P(\omega)$ is added exactly once on both sides if $\omega \in \cup A_i$ and otherwise not added both sides.

We talk about conditional probabilities only when $P(B) \neq 0$. The same argument we used in the equally likely case, allows us to define $P(A|B) = P(A \cap B)/P(B)$. Let us convince ourselves with an example.

Example: $\Omega = \{1, 2, 3, 4, 5, 6\}$; $p(k) = k/21$.

Thus the die is loaded/biased, chances of a face are proportional to the number on that face. As earlier let $A = \{2, 3, 4\}$ and $B = \{2, 4, 6\}$ Suppose that we are told B occurred. How should we redefine our probabilities? Clearly then $p^*(1) = p^*(3) = p^*(5) = 0$. Suppose $p^*(2) = a$ then we must have $p^*(4) = 2a$ because for our die chances of 4 are double the chances of 2. Similarly $p^*(6) = 3a$ but since we must have $\sum_{\omega \in \Omega} p^*(\omega) = 1$ we must have $a = 1/6$. Thus when we are told B occurred

$$p^*(1) = p^*(3) = p^*(5) = 0; p^*(2) = 1/6; p^*(4) = 2/6; p^*(6) = 3/6$$

Hence under the information given, we feel

$$P(A|B) = p^*(2) + p^*(3) + p^*(4) = \frac{1+2}{6} = \frac{1}{2} \quad (\text{feel})$$

The suggested formula gives

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{6/21}{12/21} = \frac{1}{2} \quad (\text{suggested formula})$$

Thus the feeling agrees with the suggested formula. As I said, you can justify the suggested formula in all experiments, not only in this example.

The definition of conditional probability can be used to calculate $P(A \cap B)$ if you knew $P(B)$ and $P(A|B)$; simply using $P(A \cap B) = P(B)P(A|B)$. As earlier, writing the definition and cancelling denominators we get the following theorem.

Theorem: Let (Ω, P) be any probability space and B an event.

- (1) $P(\emptyset|B) = 0$ $P(\Omega|B) = 1$ $P(A^c|B) = 1 - P(A|B)$
- (2) $P(A_1 \cup A_2|B) = P(A_1|B) + P(A_2|B) - P(A_1 \cap A_2|B)$

(3) For finitely many events $P(\cup A_i | B) = S'_1 - S'_2 + S'_3 - \dots$ where

$$S'_1 = \sum P(A_i | B), \quad S'_2 = \sum_{i < j} P(A_i \cap A_j | B), \quad S'_3 = \sum_{i < j < k} P(A_i \cap A_j \cap A_k | B), \dots$$

(4) For infinitely many disjoint events, $P(\cup A_n | B) = \sum P(A_n | B)$

(5) $P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2 | A_1)P(A_3 | A_1 \cap A_2)$

Bayes formula:

There are interesting and useful calculations we can do with conditional probabilities. Suppose in a population 1 percent have a disease. Thus the chances that I have the disease is $1/100$. D stands for the event that I have the disease. Thus $P(D) = 0.01$ and $P(D^c) = 0.99$. When I go to the doctor he suggests a test to be done. So I get the test done. The test result says I have the disease. Question: do I have? Unfortunately no test is foolproof, there is always a possibility of false positive (I do not have disease but the test says I have) and false negative (I have but the test says I do not have the disease). Usually depending on the lab or the accuracy of machines used, we know the chances of going wrong. Let us use $T-$ for the event that test says NO and $T+$ for the event test says yes. Thus we know

$P(T- | D) = a$ and $P(T+ | D^c) = b$.

As a result

$P(T+ | D) = 1 - a$ and $P(T- | D^c) = 1 - b$.

With this notation, my question is $P(D | T+) = ?$

$$\begin{aligned} P(D | T+) &= \frac{P(D \cap T+)}{P(T+)} = \frac{P(D \cap T+)}{P(D \cap T+) + P(D^c \cap T+)} \\ &= \frac{P(D)P(T+ | D)}{P(D)P(T+ | D) + P(D^c)P(T+ | D^c)} \\ &= \frac{0.1(1 - a)}{0.1(1 - a) + 0.99(b)} \end{aligned}$$

This is called Bayes formula.

Bayes formula: (Ω, P) is a probability space. $\Omega_i, 1 \leq i \leq k$ are disjoint events with $\cup \Omega_i = \Omega$. Let E be an event. Then for $1 \leq i \leq k$

$$P(\Omega_i | E) = \frac{P(\Omega_i)P(E | \Omega_i)}{P(\Omega_1)P(E | \Omega_1) + P(\Omega_2)P(E | \Omega_2) + \dots + P(\Omega_k)P(E | \Omega_k)}$$

This is also called Bayes formula for causes. E is an event which can occur due to several causes either Ω_1 or Ω_2 and so on. Now I have observed

the event E . I want to know: What are the chances that Ω_5 is the cause for occurrence of the event E ? The formula answers precisely this question. For example an event E (may be vibration of a rod) could be due to several causes: Gravitational waves/wind/ heat/ movement of Earth/A bird flew nearby/ a plane passed overhead/etc. I have observed E . What are the chances it is due to Gravitational wave? This is outrageous way of putting things, its purpose is only to bring out the spirit. Matters concerning Gravitational waves are extremely complex. Firstly situation is not at all discrete, secondly there are several parameters. You must first have a model to start with. And so on. I do not know how Bayes method is applied in Gravitational wave detection.

Independence:

Example: Consider the usual deck of cards: It has four ‘suits’: Clubs ♣; Diamonds ♦; Hearts ♥; and spades ♠ and in each suit 13 cards: $A, 2, 3, \dots, 10, J, Q, K$. I pick a card at random. Thus $|\Omega| = 52$. Let A be the event selected card is Ace (A). Then $P(A) = 4/52$. I tell you that the selected card is spades and ask for probability of A now. Thus if B is the event: selected card is spades, then we want $P(A|B)$. Clearly it equals $1/13$. Thus

$$P(A|B) = P(A)$$

The information that the event B occurred did not alter/influence the probability of A . When this happens, that is, $P(A|B) = P(A)$; equivalently $P(A \cap B) = P(A)P(B)$; it appears reasonable to say that the events are independent.

Example: $\Omega = \{HH, HT, TH, TT\}$

Probability of each outcome is $1/|\Omega| = 1/4$. If A is the event ‘first toss is H ’ and B is the event ‘second toss is H ’ then A, B are independent. Let us now define the event C : ‘exactly one H ’. Then we see $P(A \cap B) = 1/4 = P(A)P(B)$ and also $P(B \cap C) = P(TH) = 1/4 = P(B)P(C)$ and similarly $P(A \cap C) = P(HT) = 1/4 = P(A)P(C)$. In other words A, B are independent; B, C are independent and C, A are independent.

Question: Should we agree to say A, B, C are independent?

if you know about any two of these, then you know about the third one. For example if both of A, B occur or none of A, B occurs, then C can not occur. If exactly one of A, B occurs, then C must. Thus we should not really say that the three are independent. After all if you say certain things are independent, they have nothing do with each other; if you have any information

on any of them it should not reveal about the remaining. So what fails in this example?

$P(A \cap B | C) \neq P(A \cap B)$, equivalently $P(A \cap B \cap C) \neq P(A \cap B)P(C)$ or equivalently $P(A \cap B \cap C) \neq P(A)P(B)P(C)$.

If we put this extra condition, we can show the intuitive feeling outlined above is correct and A, B, C are independent. More generally we make the following definition.

Definition: Events $A_1, A_2, \dots, A_n$ in a probability space are independent if for any number of them probability of their intersection equals product of their probabilities. In symbols, the following is true: for every $k \leq n$ and every $1 \leq i_1 < i_2 < \dots < i_k \leq n$.

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = P(A_{i_1})P(A_{i_2}) \dots P(A_{i_k})$$

Does this definition reflect what we had in mind? We first show that the definition is equivalent to the following:

■ if for each i , the event B_i be either A_i or A_i^c , then
 $P(B_1 \cap B_2 \cap \dots \cap B_n) = P(B_1)P(B_2) \dots P(B_n)$ (♠) ■

The advantage of this is that the double suffixes, $i_1, i_2, \dots$ disappear. But we consider for each of the events either itself or its complement. First we show that definition implies (♠). This is done by induction on the number of complements that appear. For example, to prove

$$P(A_1 \cap A_2 \cap \dots \cap A_{n-1} \cap A_n^c) = P(A_1)P(A_2) \dots P(A_{n-1})P(A_n^c)$$

we consider, using the definition,

$$P(A_1 \cap A_2 \cap \dots \cap A_{n-1}) = P(A_1)P(A_2) \dots P(A_{n-1})$$

$$P(A_1 \cap A_2 \cap \dots \cap A_{n-1} \cap A_n) = P(A_1)P(A_2) \dots P(A_{n-1})P(A_n)$$

and subtract the second from the first.

Conversely, if (♠) is given, to show that the definition holds, write down

$$P(A_1 \cap A_2 \cap \dots \cap A_{n-1} \cap A_n) = P(A_1)P(A_2) \dots P(A_{n-1})P(A_n)$$

$$P(A_1 \cap A_2 \cap \dots \cap A_{n-1} \cap A_n^c) = P(A_1)P(A_2) \dots P(A_{n-1})P(A_n^c)$$

and add to get

$$P(A_1 \cap A_2 \cap \dots \cap A_{n-1}) = P(A_1)P(A_2) \dots P(A_{n-1})$$

For any given $i_1, \dots, i_k$ you can proceed in a similar fashion eliminating unwanted suffixes.

This can be used to show any information on some of these does not influence the others.

Given a probability space and events, we can use the above equations to check if they are independent events. Or we can use the requirement of independence to assign probabilities.

For example, if we say: toss a fair coin twice independently, then it means,

assign probabilities so that questions about first and second tosses are independent. For example chances of HH is product of chances that first toss is H multiplied by chances second toss is H . Hence $P(HH) = 1/4$. We are repeating, independently, the experiment of tossing the coin. This leads to more general concept of repeating experiments independently.

Definition: Suppose we have experiments (Ω_i, P_i) for $1 \leq i \leq n$. We define independent performance of the experiments, or, also called the product experiment (Ω^*, P^*) as follows.

$$\Omega^* = \prod_1^n \Omega_i = \{(\omega_1, \dots, \omega_n) : \omega_i \in \Omega_i; \forall i\}; \quad P^*(\omega_1, \dots, \omega_n) = \prod_1^n P_i(\omega_i).$$

This space represents performing the n experiments, one after the other, independently. Fix any events $A_i \subset \Omega_i$, and define the events: A_i^* = the set of all outcomes in Ω^* such that i -th coordinate is in A_i . Note that the event A_i^* in the product depends only on i -th experiment, that is, whether an outcome of Ω^* is in A_i^* or not depends only on i -th experiment. Then, you can verify easily, in the space (Ω^*, P^*) the events $A_1^*, \dots, A_n^*$ are independent.

You must keep in mind that in defining P^* we multiplied the probabilities because we want events depending on different i to be independent. You must pause and think about matters.

if each (Ω_i, P_i) is the same experiment (Ω, P) then Ω^* is the set of all n -tuples of points from Ω and P^* value of such an n -tuple is the product of P values of its coordinates. This experiment is called independent repetitions of the original experiment (Ω, P) .

For instance, let us consider the experiment of tossing a coin once. $\Omega = \{H, T\}$ and $P(H) = p$ and $P(T) = 1 - p$ where $0 < p < 1$. Thus chance of heads is p . The experiment of tossing the coin n times independently means the product experiment where the sample space is Ω^* and probabilities are given by $p(\epsilon_1, \dots, \epsilon_n) = p^i(1 - p)^{n-i}$ where i is the number of H in the outcome $(\epsilon_1, \dots, \epsilon_n)$.

Let us calculate probabilities of some interesting events. A is the set of all outcomes having exactly k heads. You see that each such outcome has probability $p^k(1 - p)^{n-k}$ and there are $\binom{n}{k}$ many such outcomes. Thus

$$P(A) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

Here we have tossed a coin n -times and counted number of heads. This is a recurring phenomenon, we perform an experiment and make a measurement. For instance, if you want to decide whether to accept a lot of bolts supplied to your company, you need to know whether they confirm to the specifications. You can not test all items. You take a sample of 100 items one by one, test and count the number of defective found. Base your decision on this number. it is unimportant whether the first is defective or the tenth etc. If you are trying to estimate chance of heads for a given coin, you can toss it, say n times and if you find k heads, then k/n is a good estimate of the chance of heads. It is unimportant where exactly the k heads appeared among the n tosses. These measurements are called random variables.

Definition: Let (Ω, P) be a probability space. A random variable on the space is a real valued function. The distribution of the random variable is a list or table giving all values of the random variable and the corresponding probabilities; that is against the value a write the probability $P(X = a)$, that is $P\{\omega : X(\omega) = a\}$.

Sometimes you do not want to use the term table etc. You define **probability mass function** of the random variable X to be the function f defined on the real line by $f(a) = P\{\omega : X(\omega) = a\}$ for each $a \in R$. Since our basic sample space is countable, clearly the set $\{\omega : X(\omega) = a\}$ is empty set for all but countably many values a . Thus $f(a) = 0$ for all but countably many values a . That is why we normally write those values and probabilities. If random variable is simply a function on Ω why do you need a new term, could have called it a function. Well, by using the word random variable we draw your attention to the fact that there is a probability on Ω .

Example: In the above example of tossing a coin n times (when nothing is said, the tosses are independent), if X is the number of heads then distribution of X is

values	0	...	k	...	n
probabilities	$(1-p)^n$	...	$\binom{n}{k}p^k(1-p)^{n-k}$	...	p^n

This distribution is called Binomial distribution; denoted $B(n, p)$ and we write $X \sim B(n, p)$ – read X has binomial distribution with parameters n and p . Instead of writing $(1-p)$ we write q from now on.

Why is this called Binomial? Because the probabilities are terms in the binomial expansion of $(p+q)^n$.

Example: Same coin. Toss till you get one Head and stop. X is the number of Tails obtained.

Sample space: $\{H, TH, TTH, TTTH, \dots\}$

Probabilities: $P(T^k H) = q^k p$.

Random variable: $X(T^k H) = k$.

Distribution: value k with probability $q^k p$ for $k = 0, 1, 2, \dots$.

This is called Geometric random variable and this distribution is called Geometric distribution. We write $X \sim G(p)$ — X has Geometric distribution with parameter p .

(You might wonder if $TTTTT\dots$, that is all T is also a possible outcome. We can show that its chances are zero and so we shall not count it as an outcome. Why is its chance zero? Getting all the first 100 Tails has probability q^{100} ; surely getting all Tails is a smaller event and so must have probability at most q^{100} . Nothing special about hundred. Similar consideration applies in the next example.)

Why is it geometric? Because probabilities are terms of a geometric progression.

(Warning: The random variable Y = total number of tosses is called Geometric random variable by some authors. Clearly $Y = X + 1$. There should be no confusion, For us X is $G(p)$)

Example: Same coin. Fix integer $r \geq 1$. Toss till you get r Heads and then stop. X is the number of Tails obtained.

Sample space: All finite sequences of H, T with last letter H and before that exactly $(r - 1)$ letters are H . Probabilities: $P(\omega) = q^k p^r$ if ω is of length $(r + k)$.

Random variable: $X(\omega) = k$ if ω is of length $(r + k)$.

Distribution: value k with probability $\binom{r+k-1}{r-1} q^k p^r$ for $k = 0, 1, 2, \dots$.

This distribution is called negative Binomial distribution and any random variable with this distribution is called a negative Binomial random variable: $X \sim NB(r, p)$ — X has negative binomial distribution with parameters r and p . Why is it called negative binomial? Because the probabilities are terms in the expansion of $(1 - q)^{-r}$ times p^r . Recall, since $0 < q < 1$,

$$(1 - q)^{-r} = 1 + \binom{-r}{1}(-q) + \binom{-r}{2}(-q)^2 + \dots + \binom{-r}{k}(-q)^k + \dots$$

Here

$$\begin{aligned} \binom{-r}{k} &= \frac{(-r)(-r-1)(-r-2)\dots(-r-k+1)}{k!} \\ &= (-1)^k \frac{r(r+1)(r+2)\dots(r+k-1)}{k!} = (-1)^k \binom{r+k-1}{k} \end{aligned}$$

(Warning: Sometimes the random variable $Y =$ total number of tosses is called negative Binomial random variable. Clearly $Y = X + r$. There should be no confusion, For us X is $NB(r, p)$)

Suppose our coin has a very very small chance of heads, say 0.0001 and we toss the coin 10000 times. What are the chances of 5 heads? of course you can give binomial probability which is difficult to compute. When p is very small and n is very large and $np = \lambda > 0$ then chances of value k approximately equals $e^{-\lambda} \lambda^k / k!$. Here is the precise result:

Theorem: Let $X_n \sim B(n, p_n)$. Suppose $np_n \rightarrow \lambda > 0$ as $n \rightarrow \infty$. Then for each k

$$P(X_n = k) \rightarrow e^{-\lambda} \frac{\lambda^k}{k!}$$

Note that since $n \rightarrow \infty$, and np_n converges to a finite quantity, we have $p_n \rightarrow 0$. Proof of the theorem is very simple: Let us denote $\lambda_n = np_n$ so that $\lambda_n \rightarrow \lambda$.

$$P(X_n = 0) = (1 - p_n)^n = \left(1 - \frac{\lambda_n}{n}\right)^n \rightarrow e^{-\lambda}$$

$$P(X_n = 1) = np_n(1 - p_n)^{n-1} = \lambda_n \left(1 - \frac{\lambda_n}{n}\right)^n (1 - p_n)^{-1} \rightarrow \lambda e^{-\lambda}$$

You can prove by induction on k or directly that for every k the result holds.

Example: Let $\lambda > 0$. The distribution:

Values: k . for $k = 0, 1, 2, \dots$

probabilities: $e^{-\lambda} \frac{\lambda^k}{k!}$ for $k = 0, 1, 2, \dots$

is called Poisson distribution and if X has this distribution then we say X is a Poisson random variable; $X \sim P(\lambda)$.

Unlike the previous examples, we have not given an experiment that produces this random variable. We have obtained this distribution as a limit. But if you want me to show a sample space and a random variable with this distribution, I can surely do.

Sample space $\Omega = \{0, 1, 2, 3, \dots\}$ Probabilities: $P(k) = e^{-\lambda} \lambda^k / k!$.

These P values are non-negative and add to one. So (Ω, P) is a sample space. The random variable is $X(k) = k$ for $k \in \Omega$.

In practice this is supposed to fit well in several phenomena. The number of accidents during a month at a busy intersection (because, a very large number of cars pass through and each car can potentially cause an accident but with a very very small chance); the number of particles emitted by a

radioactive material during an hour (because there are very large number of atoms and each can potentially fly but has a very small chance — all other atoms are not allowing it to fly) and so on. Thus this is very useful model for several phenomena.

Example: I have N_1 oranges and N_2 Apples. I pick a set of n items and count the number of oranges in the set. Here sample space is the collection of all n element subsets of the entire lot of fruits. There are $\binom{N_1+N_2}{n}$ sample points. The random variable X is the following. If ω is a sample point, that is, a subset of size n of the fruits, then $X(\omega)$ is the number of oranges in the subset ω . Possible values of X are $0, 1, 2 \dots, n$ and

$$P(X = k) = \binom{N_1}{k} \binom{N_2}{n-k} / \binom{N_1 + N_2}{n}$$

There is a subtle point to notice. Suppose for instance $N_1 = 4$, $N_2 = 3$ and $n = 4$. Then surely any sample contains at least one orange because there are only three apples. Thus the value zero is not possible. But no need to worry. Even though we showed the value zero, the probability we had written is also zero — numerator equals $\binom{4}{0} \binom{3}{4} = 0$. Try to write down exact possible values of the random variable X .

This is called Hypergeometric distribution and the random variable a Hypergeometric random variable with parameters $(N_1, N_2; n)$. The reason is that the probabilities can be matched with terms of a Hypergeometric series; we shall not get into the details.

Expectation and Variance:

We play a series of games. I toss a coin. Heads up, I give you one Rupee; Tails up, you give me one Rupee. Thus if X is your profit, then $X = +1$ if Heads up and $X = -1$ if Tails up.

If chance of Heads is $2/3$ and Tails is $1/3$; will you play? Yes you will. The argument is that if we play 30 games you are likely to win about 20 games and loose 10 so that you gain 10 Rs; or on the average $1/3$ Rs per game. Of course instead of 30 games and so on we can calculate $(+1)(2/3) + (-1)(1/3) = 1/3$ as expected profit per game.

If chance of Heads is $1/3$ and Tails is $2/3$; will you play? No, you will not. Similar argument applies: You expect to win $(+1)(1/3) + (-1)(2/3) = -1/3$ per game.

What if the coin is fair. You will think, looks fair, neither you win, nor loose on the average. you may decide to play, hoping some profit.

Let us consider fair coin game again, but now $X = +1000$ if Heads up and $X = -1000$ if Tails up. Clearly this is also fair, expected gain is zero. Will you play? probability not, because it has 'high risks': you may loose the first three games (though you may win later games). In other words the random variable X has larger spread. The average value and spread is made precise below.

Definition: If X is a random variable taking values $\{x_i : i \geq 1\}$ with respective probabilities $\{p_i, i \geq 1\}$; then we define the Expected value/Average value/Mean value by the formula $E(X) = \sum x_i p_i$ provide $\sum |x_i| p_i$ is finite. If this last sum is not finite, we shall not define $E(X)$.

We define the variance of X by $Var(X) = \sum x_i^2 p_i - (\sum x_i p_i)^2 = \sum x_i^2 p_i - (E(X))^2$ if $\sum x_i^2 p_i$ is finite.

Standard deviation or risk or uncertainty of X is positive square root of $Var(X)$. Usually mean is denoted by μ_X or μ and variance is denoted σ_X^2 or σ^2 with $\sigma > 0$.

Observe that if $\sum x_i^2 p_i$ is finite, then $\sum |x_i| p_i$ is also finite, so that variance is meaningful. This is because $|x| \leq x^2 + 1$ for any real number x . Further Variance is a non-negative quantity, so that standard deviation makes sense.

This is because, by Cauchy- schwarz inequality

$$(\sum |x_i|p_i)^2 = (\sum |x_i|\sqrt{p_i} \sqrt{p_i})^2 \leq (\sum x_i^2 p_i)(\sum p_i).$$

We calculate some means and variances.

In these examples I am again repeating sample space and the random variable definition, just to make sure that you understand what the random variable is — though we could do the calculations recalling only the distributions and not the sample space etc. All our sample spaces are finite or countable. Accordingly, all our random variables take countably many values. Random variables which take finite or countably infinite number of values are called discrete random variables. When we deal later with more general random variables, you do not always have the luxury of seeing the sample space.

Example 1 (Bernoulli variable): Toss a coin once, chance of Heads in a toss is p . Let X be the number of Heads.

Sample space: $\{H, T\}$. Probabilities: $P(H) = p; P(T) = 1 - p = q$.

$X(H) = 1$ and $X(T) = 0$

Values of X are 1, 0 with probabilities p, q respectively.

Expected value: $E(X) = p$.

Variance: $V(X) = p - p^2 = pq$.

This is called Bernoulli random variable. Sometimes the random variable X taking values ± 1 is called Bernoulli random variable. Sometimes *any* random variable that takes two values is called Bernoulli variable.

Example (Binomial variable): Same coin above. Toss n times. Let X be the number of Heads.

Sample space: All sequences of H, T of length n .

Probability: $P(\omega) = p^k q^{n-k}$ if k is the number of Heads in the sequence ω .

random variable: $X(\omega) = \text{number of } H \text{ in the sequence } \omega$.

Distribution: values k with probability $\binom{n}{k} p^k q^{n-k}$ for $0 \leq k \leq n$.

Expected value: $E(X) = np$. Variance: $V(X) = npq$. Here is the calculation:

$$\begin{aligned} E(X) &= \sum_0^n k \binom{n}{k} p^k q^{n-k} = \sum_1^n k \binom{n}{k} p^k q^{n-k} = np \sum_1^n \binom{n-1}{k-1} p^{k-1} q^{n-k} \\ &= np(p+q)^{n-1} = np. \end{aligned}$$

and

$$\sum_0^n k^2 \binom{n}{k} p^k q^{n-k} = \sum_0^n k(k-1) \binom{n}{k} p^k q^{n-k} + \sum_0^n k \binom{n}{k} p^k q^{n-k}$$

$$\begin{aligned}
&= \sum_2^n k(k-1) \binom{n}{k} p^k q^{n-k} + \sum_1^n k \binom{n}{k} p^k q^{n-k} \\
&= n(n-1)p^2(p+q)^{n-2} + np = n^2p^2 - np^2 + np
\end{aligned}$$

so that

$$\text{var}(X) = n^2p^2 - np^2 + np - n^2p^2 = np(1-p).$$

Example (Geometric variable): Same coin $0 < p < 1$. Toss till you get one Head and X is the number of Tails obtained.

Sample space: $\{H, TH, TTH, TTTH, \dots\}$ Probabilities: $P(T^k H) = q^k p$.

Random variable: $X(T^k H) = k$.

Distribution: value k with probability $q^k p$ for $k = 0, 1, 2, \dots$.

Expectation: $E(X) = q/p$. Variance: $V(X) = q/p^2$.

$$E(X) = \sum_0^\infty k q^k p = p \sum_1^\infty k q^k$$

But if $q + 2q^2 + 3q^3 + \dots = S$ then

$$S - qS = \frac{q}{1-q} = \frac{q}{p}; \quad S = \frac{q}{p^2}$$

so that $E(X) = q/p$.

Try repeating similar argument to calculate variance. However We use a different method which uses power series and differentiation. Let $f(x) = \sum_0^\infty q^k x^k$ This is a power series with radius of convergence $1/q$, larger than one. This can be differentiated term by term any number of terms. Observe $f(x) = 1/(1-qx)$ so that

$$f'(x) = q/(1-qx)^2 \quad f'(1) = q/p^2$$

But by term by term differentiation $f'(1) = \sum k q^k$. Thus $\sum k q^k p = pq/p^2$ or $E(X) = q/p$ as seen above. Again

$$f''(x) = 2q^2/(1-qx)^3 \quad f''(1) = 2q^2/p^3$$

Thus comparing with term by term differentiation, we see $\sum k(k-1)q^k = 2q^2/p^3$. or $\sum k(k-1)P(k) = 2q^2/p^2$. Hence

$$\text{Var}(X) = \frac{2q^2}{p^2} + \frac{q}{p} - \frac{q^2}{p^2} = \frac{q}{p^2}.$$

Example (Poisson Variable): This is obtained as a limit. I shall not be able to give a simple sample space. We settle for the trivial setup.

Sample space: $\{0, 1, 2, 3 \dots\}$ Probability: $P(k) = e^{-\lambda} \lambda^k / k!$.
Random variable $X(\omega) = \omega$.

$$E(X) = \sum_0^{\infty} k e^{-\lambda} \frac{\lambda^k}{k!} = \lambda \sum_1^{\infty} e^{-\lambda} \frac{\lambda^{k-1}}{(k-1)!} = \lambda$$

$$\sum_0^{\infty} k(k-1) e^{-\lambda} \frac{\lambda^k}{k!} = \lambda^2 \sum_2^{\infty} e^{-\lambda} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2$$

$$Var(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Note that Poisson is obtained as a limiting case of Binomial as $np_n \rightarrow \lambda$. Since $B(n, p_n)$ has mean np_n , we see Poisson has mean λ . Again $B(n, p_n)$ has variance $np_n(1 - p_n)$ which again converges to λ since $p_n \rightarrow 0$.

We shall now see properties of expectation.

Theorem: (Ω, P) be a (discrete) experiment. X, Y random variables on Ω

(i) $E(X) = \sum_{\omega \in \Omega} X(\omega)P(\omega)$. More precisely If $E(X)$ is defined then the right side is also absolutely convergent and equality holds; further if the right side is absolutely convergent, then $E(X)$ is defined and equality holds.

(ii) $E(X + Y) = E(X) + E(Y)$. More precisely, if the expected values of X and Y are defined then so is that of $X + Y$ and the equality holds.

(iii) $E(cX) = cE(X)$ where c is a real number. More precisely, if $E(X)$ is defined then $E(cX)$ is also defined and the equality holds.

(iv) If $f : R \rightarrow R$ and $Y(\omega) = f(X(\omega))$ and if $E(Y)$ exists, then $E(Y) = \sum f(x_i)p_i$. Here the distribution is given by: value x_i with probability p_i for $i \geq 1$.

Note that $X + Y$ is the random variable defined by $(X + Y)(\omega) \equiv X(\omega) + Y(\omega)$ and cX is the random variable defined by $(cX)(\omega) \equiv cX(\omega)$.

(i) says that the action of taking expected value can be carried out on Ω , if you have not yet calculated the distribution of X . This is useful in proving (ii) because if you know distributions of X and Y , you can not decide on the distribution of $X + Y$; so knowing distributions alone of X, Y , you will not be able to calculate $E(X + Y)$, just using definition. But (i) allows you to say

$$\begin{aligned} E(X + Y) &= \sum [X(\omega) + Y(\omega)]P(\omega) \\ &= \sum X(\omega)P(\omega) + \sum Y(\omega)P(\omega) = E(X) + E(Y) \end{aligned}$$

To see (iii) you can proceed similarly

$$\begin{aligned} E(cX) &= \sum (cX)(\omega)P(\omega) = \sum cX(\omega)P(\omega) \\ &= c \sum X(\omega)P(\omega) = cE(X) \end{aligned}$$

Of course you can also use distribution of cX because if X takes values x_i with probability p_i then cX takes values cx_i with probability p_i — when $c \neq 0$. However if $c = 0$ then you can do it directly (?).

Let us now prove (i). suppose X takes values $\{x_k : k \geq 1\}$ with probabilities $\{c_k : k \geq 1\}$ respectively. Let us partition the sample space Ω as follows

$$A_k = \{\omega \in \Omega : X(\omega) = x_k\}; \quad k = 1, 2, 3, \dots$$

Note that

$$c_k = P(X = x_k) = P\{\omega : X(\omega) = x_k\} = \sum_{\omega \in A_k} P(\omega).$$

Also

$$\sum_{\omega \in A_k} |X(\omega)|P(\omega) = |x_k| \sum_{\omega \in A_k} P(\omega) = |x_k|c_k.$$

Remember that a series of positive numbers can always be added; at the worst the sum may be infinity. Moreover we can add in any order we like — the grand sum is afterall supremum of finite sums. Thus

$$\sum |X(\omega)|P(\omega) = \sum_k |x_k|c_k \quad (\bullet)$$

Hence, one side is finite iff the other side is finite. Thus a necessary and sufficient condition for expectation to be defined is that the left side above be finite. Thus if any one side of $(\bullet)$ is finite, then both series $\sum X(\omega)P(\omega)$ and $\sum_k x_k c_k$ are absolutely convergent and adding in two stages of the first sum gives the stated equality to prove (i).

Proof of (iv): Note that this is not a tautology. For example the formula is not definition because there is no mention of distribution of Y . It is not from part 1, because there is no ω . It is a formula to calculate $E(Y)$ by using the distribution of X : no need to look at the sample space, no need to calculate the distribution of Y either.

As we go along we shall not make too much fuss about existence etc, we assume that the expectations we are talking about exist and proceed with

calculations.

Proof of the formula is simple. By part 1,

$$E(Y) = \sum_{\omega} Y(\omega)P(\omega) = \sum_i \sum_{\omega: X(\omega)=x_i} Y(\omega)P(\omega) = \sum_i f(x_i)p_i \quad \blacksquare$$

Here is an example to appreciate (i).:

I toss a fair coin twice and X is the number of heads obtained and Y is the number of Tails obtained then $X \sim B(2, 1/2)$ and $Y \sim B(2, 1/2)$. Also $P(X + Y = 2) = 1$, thus $X + Y$ is constant 2. On the other hand, consider tossing a fair coin 4 times and X is the number of heads in the first two tosses and Y is the number of Heads in the last two tosses. Even now $X \sim B(2, 1/2)$ and $Y \sim B(2, 1/2)$ as earlier. However now $X + Y \sim B(4, 1/2)$.

Thus you cannot calculate the distribution of $X + Y$ from those of X and Y . So if you depended on the definition of expectation via distribution ($\sum x_k p_k$) then there is no way to conclude the above.

Definition: Let X be a random variable taking value x_i with probability p_i for $i \geq 1$. For any integer $n \geq 1$ we define the n -th moment of X as $E(X^n) = \sum x_i^n p_i$ provided $\sum |x_i|^n p_i$ is convergent. If this is not convergent, we shall not define the n -th moment. This is denoted $\mu_n(X)$ or simply μ_n .

Observe that the statement $E(X^n) = \sum x_i^n p_i$ in the definition makes use of the previous theorem. We have neither calculated the distribution of X^n nor used the space of outcomes. We used previous theorem with $f(x) = x^n$. Note that μ_1 is simply the mean and usually the suffix is omitted, denoted μ . Observe variance is nothing but $\mu_2 - \mu_1^2 = \mu_2 - \mu^2 = E(X^2) - [E(X)]^2$.

Suppose we have a random variable X with mean μ . Then $V(X) = E[(X - \mu)^2]$. To see this, We simply take $f(a) = (a - \mu)^2$ in the above theorem:

$$\begin{aligned} E[(X - \mu)^2] &= \sum (x_i - \mu)^2 p_i = \sum x_i^2 p_i + \mu^2 \sum p_i - 2 \sum \mu x_i p_i \\ &= \sum x_i^2 p_i + \mu^2 - 2\mu^2 = \sum x_i^2 p_i - \mu^2 = \mu_2 - \mu^2 = V(X) \end{aligned}$$

Expected number of matches:

The fact sum of expectations equals expectation of sum is a very very useful result. Let us consider the matching problem with 52 cards and 52 envelopes. There are $(52)!$ outcomes. Let X be the number of matches. that is, if you take an outcome ω then $X(\omega)$ is the number of matches according to the placement ω . This variable can take integer values $\{k : 0 \leq k \leq 52\}$. It is

not easy, though we did, to calculate $P(X = k)$. It is not easy to calculate $E(X)$ using the definition. We shall cleverly express X as sum of variables for each of which expectation can be calculated in a painless manner.

For $1 \leq i \leq 52$, let X_i be the following random variable on our space: $X_i(\omega) = 1$ or zero according as there is match at place i or not in the arrangement ω . Thus

$$X(\omega) = \sum_{i=1}^{52} X_i(\omega)$$

Note that X_i indicates whether there is match at place i or not and is hence called indicator random variable. The beauty is that it takes only two values zero and one. Thus for any i ,

$$E(X_i) = P(X_i = 1) = \frac{51!}{52!} = \frac{1}{52}$$

and

$$E(X) = 1.$$

A hopeless situation is rescued by the linearity of expectation. Also note that even if you have 1000 cards instead of 52, the expected number of matches still equals one.

Chebyshev's inequality: Let $a > 0$.

- (i) For any non-negative rv X we have $P(X \geq a) \leq \frac{E(X)}{a}$
- (ii) For a random variable X

$$P(|X| \geq a) \leq E(|X|)/a; \quad P(|X| \geq a) \leq E(|X|^2)/a^2.$$

- (iii) X be a rv with mean μ and variance σ^2 .

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

It is assumed that the required Expectations are finite.

The inequality (i) is also referred to as Markov Inequality.

Proof of Chebyshev is simple.

- (i) Let $a_1, a_2, \dots$ be all the values of X with respective probabilities (p_i) .

$$P(X \geq a) = \sum_{i: a_i \geq a} p_i \leq \sum_{i: a_i \geq a} \frac{a_i}{a} p_i \leq \frac{1}{a} \sum a_i p_i = \frac{1}{a} E(X)$$

- (ii) follows by applying (i) to $|X|$ and $|X|^2$. Note that the events $(|X| \geq a)$ and $(|X|^2 \geq a^2)$ are same.
- (iii) follows by applying (i) to $(X - \mu)^2$. Note that $E[(X - \mu)^2] = \sigma^2$. ■

height of bus:

Here is a simple special case. suppose $a = 3\sigma$ then the above inequality says, by taking complements,

$$P\{\omega : \mu - 3\sigma < X(\omega) < \mu + 3\sigma\} \geq 1 - \frac{1}{9} = \frac{8}{9} \sim 90\%.$$

For example if average height of the adult (?) population of Chennai is five feet with standard deviation equal to 1/6 (two inches), then 90% of adults have height at most five and half feet. In particular, if you are asked to design height of a bus so that ninety percent of adults can travel comfortably, then five and half feet is a good suggestion.

Weak Law of Large Numbers for coin tossing:

We shall now ask whether the mathematics reflects the intuitive ideas with which we started. For example, when we say p ($0 < p < 1$) is the chance of heads for the coin, our intuitive feeling is that in a large number of tosses a proportion p will be heads. Carefully note, I am not saying expected number of proportion of Heads is p ; which is true and we calculated above.

We defined mathematical models, concept of independence and so on. Can we show that actually a proportion p will be heads or proportion of heads is getting close to p ? We immediately realize that the question is not well formulated. The proportion of heads is a random variable; there is a non-zero chance that all tosses may result in heads. In fact the proportion of heads takes all values $0/n, 1/n, 2/n, \dots, n/n$ each with positive probability.

Thus a better question is: can you show that when the number of tosses is large then the proportion of heads is close to p with very very high probability? Yes.

Theorem WLLN for coin tossing:

Let Y_n be the proportion of heads in n tosses of a coin whose chance of heads in a toss equals p ($0 < p < 1$). Given any $\epsilon > 0$, $\lim_n P(|Y_n - p| \geq \epsilon) = 0$. More precisely,

$$P(|Y_n - p| \geq \epsilon) \leq \frac{p(1-p)}{n\epsilon^2} \leq \frac{1}{4\epsilon^2} \frac{1}{n} \rightarrow 0 \quad \blacksquare$$

Observe that ϵ being fixed, the first conclusion follows from the second. Proof is simple. If X_n is the number of Heads in n tosses, then we know $X_n \sim B(n, p)$ variable, $\mu = np$ and variance $\sigma^2 = npq$. Since $Y_n = X_n/n$ we see, by Chebyshv

$$P(|Y_n - p| \geq \epsilon) = P(|X_n - np| \geq n\epsilon) \leq \frac{npq}{n^2\epsilon^2} \leq \frac{1}{4\epsilon^2} \frac{1}{n} \rightarrow 0,$$

where we used $p(1-p) \leq 1/4$ if $0 < p < 1$.

What does this mean? You give me any error limit for p , say $\epsilon > 0$ and also give me an $\eta > 0$, then I can show you an N such that if you toss n times ($n > N$) then

$$P(p - \epsilon < Y_n < p + \epsilon) \geq 1 - \eta$$

You give me any error limit, say, 0.01. Demand that observed proportion should be in $(p - 0.01, p + 0.01)$ with very high probability, say you want with probability at least 0.999. Yes, I can prescribe to you an N so that if you toss at least so many times then the observed proportion is within the limits you prescribed with probability at least 0.999. Think about it.

This is called Weak Law of Large Numbers for coin tossing. This is a satisfactory answer confirming that maths is on right track. Since we are considering coin tossing now, this theorem is for coin tossing. Since this says something about what happens in large number of tosses, this is law of large numbers. This is ‘weak’ because there is a better theorem ‘strong’ law. Strong Law answers the following question: Ok, you are saying that in very very large number of tosses the proportion is close to p ; can you show that if I really toss **infinite number of times** then the proportion **actually equals** p . It may look like a meaningless question since we can never toss infinitely many times; it is not silly. The answer is yes, with probability one.

There is another important byproduct of this. In reality, no one tells us the chance of heads. What this says is the following: If you toss it a large number of times and take the observed proportion of heads, then that is a good estimate of the unknown p . The estimate can be made close to the actual value p with very high probability. Carefully understand this.

You might wonder, why would any one play with coin? As I said earlier, the same happens whenever there are two alternatives. For example let p be the proportion of people supporting a political party and $(1 - p)$ the proportion that does not support. The same philosophy above helps you to estimate p .

Weierstrass:

We can use Chebyshev to obtain a theorem of Weierstrass. It says that every continuous function on a closed bounded interval is very close to a polynomial. If you tell how close you want, I can give a polynomial which is so close. Here is the precise statement.

Weierstrass approximation Theorem:

Let f be a real continuous function on $[0, 1]$ and $\epsilon > 0$. We can find a polynomial P (in one variable x) such that $|f(x) - P(x)| < \epsilon$ for every $x \in [0, 1]$.

Given the function f , let us define some polynomials, one for each $n \geq 1$.

$$P_n(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}.$$

These are called Bernstein polynomials associated with f and this proof is due to Bernstein. Note that even though you see f in the above definition; we only used the values of f at certain points.

Observe whatever x we take

$$f(x) - P(x) = \sum_k \left[f(x) - f\left(\frac{k}{n}\right) \right] \binom{n}{k} x^k (1-x)^{n-k} \quad (*)$$

We used here that binomial probabilities add to one. Note that if k is such that k/n is close to x then the first term $[f(x) - f(\frac{k}{n})]$ is small because f is continuous. If x is far from k/n , that is, k is far from nx , then Chebyshev tells the second term $\binom{n}{k} x^k (1-x)^{n-k}$ is small — these are binomial $B(n, x)$ probabilities far from its mean. Here is the execution.

We are given $\epsilon > 0$. Pick, using uniform continuity of f , a number $\delta > 0$ so that

$$|u - v| < \delta \Rightarrow |f(u) - f(v)| < \epsilon/2.$$

Also, using the fact that a continuous function on $[0, 1]$ is bounded fix a number $M > 0$ so that $|f(x)| < M$ for all $x \in [0, 1]$.

Then fix a N such that

$$\frac{1}{\delta^2 N} < \epsilon; \quad i.e. \quad N > \frac{1}{\epsilon \delta^2}$$

In what follows $n > N$, where N is as chosen above. How did we know such choices are to be made? Because of the explanation after (*). To proceed with the proof, From (*)

$$|f(x) - P(x)| \leq \sum_k \left| f(x) - f\left(\frac{k}{n}\right) \right| \binom{n}{k} x^k (1-x)^{n-k}$$

(used $|\sum a_i| \leq \sum |a_i|$)

$$\begin{aligned} &\leq \sum_{k: |\frac{k}{n} - x| < \delta} [***] + \sum_{k: |\frac{k}{n} - x| \geq \delta} [***] \\ &\leq \frac{\epsilon}{2} \sum_{k: |\frac{k}{n} - x| < \delta} [\text{binomial prob.}] + 2M \sum_{k: |k - nx| \geq n\delta} [\text{binomial prob}] \end{aligned}$$

(Used choice of δ and M) First sum, being addition of some binomial probabilities, is at most one. Second sum is nothing but $P(|Z - nx| \geq n\delta)$ for a $B(n, x)$ variable and is hence at most

$$\frac{nx(1-x)}{n^2\delta^2} \leq \frac{1}{4\delta^2n}$$

(used that $x(1-x) \leq 1/4$ for each $x \in [0, 1]$)

$$|f(x) - P(x)| \leq \frac{\epsilon}{2} + 2M \frac{1}{4\delta^2n} \leq \epsilon$$

(choice of N and $n > N$).

This completes the proof. Note that our choices of N , δ , and M have nothing to do with x and so the above holds for all $x \in [0, 1]$.

You see that the theorem has no probability in it. It is a theorem of Analysis, but we used probability to prove it.

Graphs with large χ :

Most of the applications that we saw involve some calculations, not too many. If you are clever and have patience to use the results you know carefully; you can achieve more complicated things. I want to show you one such calculation of Erdos. You should get ready and pull your energies to climb a peak.

We are looking for a graph satisfying certain properties and there is nothing probabilistic in the question. But we use probability to show such graphs exist. If you use your inequalities cleverly and show $P(A) > 0$ then obviously the set A is not empty.

Recall that a graph is a pair $g = (V, E)$ where V is a finite set of points, called vertices. E is a collection of unordered pairs of distinct vertices. Pictorially you can imagine V to be a set of points; join a pair of points if that pair is in E — join by a line (or curve), called edge. We do indeed say vw are joined if the pair is in E .

If you imagine a graph pictorially, then you can imagine loops, a vertex joined to itself by a loop. But our definition excludes because we said pairs of distinct vertices. We said unordered pairs, thus the edge (vw) is same as (wv) . Thus there is no direction for the edges. (Remember Tournament, it is ordered). Also when you think pictorially, you can imagine the same pair (vw) joined by several lines, multiple edges. Our definition excludes that because E is a subset of pairs. These are called undirected graphs without loops and multiple edges.

For a graph g , its chromatic number is the smallest number of colours needed to colour the vertices so that vertices of same colour are not joined. This is denoted by $\chi(g)$. Thus if $\chi(g) = 6$ we can colour the vertices using six colours so that if vw is an edge then v and w get different colours. Equivalently, you can partition the set of vertices V into six disjoint sets $\{V_i; 1 \leq i \leq 6\}$ so that no two vertices in the same V_i are joined. In fact if you can colour by six colours, then vertices receiving colour i gives you V_i . Conversely if such a partition is given colour vertices in V_i with colour i .

Chromatic number $\chi(g)$ is very difficult to handle in calculating probabilities. We define the notion of independent set which is easier to handle. For a graph g , a set S of its vertices is called independent set if no two vertices in S are joined. By $\alpha(g)$ we denote the cardinality of largest possible independent set. More precisely, if $g = (V, E)$ is the graph, then

$$\alpha(g) = \max\{|S| : S \subset V; S \text{ independent}\}$$

Since we are considering graphs with finite number of vertices, the above quantity is well defined. Before we proceed we make a simple observation to convince you that information about α tells you some thing about χ .

We claim that for any graph $g = (V, E)$

$$|V| \leq \alpha(g)\chi(g); \quad \text{i.e.,} \quad \chi(g) \geq \frac{|V|}{\alpha(g)} \quad (\star)$$

Indeed, if $\chi(g) = c$ then colour the vertices with c colours and put V_i to be the set of vertices receiving colour i for $1 \leq i \leq c$. Thus we have a partition of V into c sets, in particular $|V| = \sum |V_i|$. Each V_i is an independent set because no two vertices of the same colour are joined. Thus $|V_i| \leq \alpha(g)$ for each i . adding over i we get $|V| \leq c \alpha(g)$ as stated.

Let $k \geq 3$. A sequence of vertices $(v_1, v_2, \dots, v_k)$ is called a cycle if all the pairs $(v_1, v_2); (v_2, v_3); \dots, (v_{k-1}, v_k); (v_k, v_1)$ are edges. It is called a cycle

of length k . Note that v_k, v_1 are joined. Also note $k \geq 3$. Thus edges are not cycles.

Graphs with few edges can be coloured with few colours. If the chromatic number is very large then there must be lots of connections in the graph and it is natural to wonder if the graph must then possess small cycles. However this is not the case. there is no relations between these two.

Theorem (Erdős):

Let $k \geq 1$ and $l \geq 3$ be integers. then it is possible to make a graph whose chromatic number is at least k and has no cycles of length smaller than l .

This is one of the first results that led to the rich theory of Random Graphs or Erdős-Renyi graphs. In a sense this is the beginning of random graph theory, a subject on which you can spend a lifetime.

You can see how clueless we are. How many vertices should we start with? which ones to join? We shall bring in probability as follows. Consider the vertex set $V = \{1, 2, \dots, n\}$ and a number $0 < p < 1$. Let $\mathcal{G}_n$ be the set of all possible graphs on V . This will be our sample space. How many points are there in this set?

$$|\mathcal{G}_n| = 2^{\binom{n}{2}}$$

This is clear because there are $\binom{n}{2}$ pairs; you select any subset of these pairs; join them and do not join other pairs. Or equivalently, for each pair you have to make up your mind: to join (=1) or not to join (=0).

We now put a probability on this sample space $\mathcal{G}_n$. Take a pair, toss your coin, Heads-join; tails-do not. continue for each pair tossing independently. equivalently if $g \in \mathcal{G}_n$ then

$$P(g) = p^k(1-p)^l; \quad g = (V, E) \quad k = |E| \quad l = \binom{n}{2} - k$$

Indeed if your graph has k edges then the coin should have come up Heads exactly k times when you are tossing for those edges, and Tails when you are tossing for all the other edges. Remember at each pair you toss to decide to make it an edge, that is, join it or not.

We meet Wednesday 28th January, 10:30.

We are now proving that there are graphs with chromatic number at least k and it has no cycles of length l or smaller.

Why are we doing this? I already told you that this was to dispel a belief that if the chromatic number is large, there must be too many connections and there are likely to be small cycles. This example shows that there is in fact no relation between the two: chromatic number and girth (length of smallest cycle). This is another example (you saw Weierstrass Theorem) where there is no probability in the question, but we use probability to answer the question. This is also an example where we use nothing more than the simple things you already know. It tells you that the ability to think, to cleverly use what you know, to keep hitting the problem (trial and error and improvising) are essential for success. The calculations also tell you how, at times, you can use liberal bounds to make expressions neat and life simple. Incidentally this is, in a sense, the beginning of the theory of Random graphs. We showed that for any graph $g = (V, E)$

$$|V| \leq \alpha(g)\chi(g); \quad \text{i.e.,} \quad \chi(g) \geq \frac{|V|}{\alpha(g)} \quad (\star)$$

$\chi(g)$ is chromatic number (vertex colouring) and $\alpha(g)$ is the size of the largest independent set of vertices in the graph. The plan of the proof is as follows.

For $n \geq 10$, consider the collection of all graphs on n the vertices $\{1, 2, \dots, n\}$. Let $\mathcal{G}_n$ be the bag containing these graphs. Line up all these bags.

$$\mathcal{G}_{10}; \quad \mathcal{G}_{11}, \quad \dots, \dots, \mathcal{G}_n, \dots$$

Nothing special about 10, you can start with 4 or 20!

(♣). We show that after some stage every bag contains plenty of graphs without small cycles.

(♦). We show that after some stage every bag contains plenty of graphs which have only small independent sets. Remember that $(\star)$ implies these graphs have large χ value.

(♥). We make sure that both the above hold for some graphs.

(♠). Take one such graph g as above. Kill all unwanted cycles by removing one vertex from each such cycle, you will be removing only few vertices because there are only few unwanted cycles. The resulting graph g^* is still large. Since independent sets will still be small, hopefully chromatic number will be large for this graph g^* .

What is this ‘plenty’ of graphs? On each $\mathcal{G}_n$ we put a probability P_n by using a number $0 < p_n < 1$. Remember, whatever n you take, and whatever $0 < p < 1$ you take we have a probability P on $\mathcal{G}_n$: For $g \in \mathcal{G}_n$

$$P(g) = p^k(1-p)^l; \quad g = (V, E) \quad k = |E| \quad l = \binom{n}{2} - k$$

This gives nonzero probability to each g – this is just $B(2^{\binom{n}{2}}, p)$. This is called Erdős-Renyi random graph model (with parameter p).

But remember unless I tell you what p you should take, you have no model. To execute the plan mentioned, fix a number θ with $0 < \theta < 1/l$. In other words fix any number strictly between zero and $1/l$ and name it θ . this number will not change till the proof ends.

Given $n \geq 10$ we take $p_n = n^{\theta-1}$. Thus we have a probability space $(\mathcal{G}_n, P_n)$ with this p_n . If we are arguing with a fixed n , we take the liberty of omitting suffix p_n and just use p , which means p_n .

♣:

for any graph g , let $X(g)$ = number of cycles of length at most l in the graph g . These are unwanted cycles. Observe that X can be regarded as a random variable on each $\mathcal{G}_n$. Strictly speaking such a sentence makes no sense because by definition a random variable is a function defined on a probability space. We should restrict definition of X to $\mathcal{G}_n$, name it X_n and regard this as a random variable on the probability space $(\mathcal{G}_n, P_n)$. But no need to tax our brains as long as we understand.

The first step is made precise by showing the following: There is an n_0 such that for $n > n_0$ we have

$$\text{on } \mathcal{G}_n; \quad P_n(X \geq n/2) < 1/2. \quad (\clubsuit)$$

Let us understand X in simpler terms. If $N_i(g)$ is the number of cycles of length i in g (thus N_i is a function on $\mathcal{G}_n$ and so a random variable), then we have

$$X(g) = \sum_{i=3}^l N_i(g).$$

In turn, the random variable N_i can be expressed in simpler terms as follows – using indicator variables, an idea we used in calculating expected number of matches. Take a sequence $s = (v_1, \dots, v_i)$ of i vertices. Define $I_s(g)$ to be one or zero according as $(v_1, \dots, v_i)$ is a cycle in the graph g or not. Then

we have

$$N_i(g) \leq \sum_{s:|s|=i} I_s(g)$$

Here the sum is over all sequences s of length $|s| = i$ consisting of distinct vertices. The reason for inequality is that on right side one i -cycle is counted several times. For example $(v_1, v_2, \dots, v_i)$ is the same cycle as $(v_2, \dots, v_n, v_1)$ (warning: $v_1 v_2 v_3$ is not same as $v_1 v_3 v_2$).

$$E(I_s) = P(I_s = 1) = p_n^i$$

because each of the pairs (v_j, v_{j+1}) , including (v_i, v_1) is joined with probability p_n and independently. Also the number of summands is less than n^i . Hence

$$E(N_i) \leq \sum_{s:|s|=i} E(I_s) \leq n^i p_n^i = n^i n^{\theta i - i} = n^{\theta i} \leq n^{\theta l}.$$

Finally,

$$E(X) \leq \sum E(N_i) \leq \sum_3^l n^{\theta i} \leq l n^{\theta l}.$$

By Chebyshev, on $\mathcal{G}_n$,

$$P(X \geq n/2) \leq \frac{E(X)}{n/2} \leq 2l n^{\theta l - 1} \rightarrow 0$$

because $\theta l - 1 < 0$. Thus there is an n_0 such that

$$n \geq n_0 \quad \Rightarrow \quad P(X \geq n/2) < 1/2. \quad (\clubsuit)$$

($\diamond$):

Now let us understand α . Keep in mind we fix n and discuss $\mathcal{G}_n$ now. Thus α is random variable on $\mathcal{G}_n$. Let us take an integer $x > 1$. We shall suggest, later, what x to take since we want to achieve something. Note that $\alpha(g) \geq x$ iff there is an independent set of size x in g .

$$\{g : \alpha(g) \geq x\} = \bigcup_{S:|S|=x} A_S; \quad A_S = \{g : S \text{ independent in } g\}$$

$$P(A_S) = (1 - p_n)^{x(x-1)/2}$$

(each of the $x(x-1)/2$ pairs in S are not joined)

$$\leq e^{-p_n x(x-1)/2}$$

(used the inequality $1 - p \leq e^{-p}$) How many terms are there in the union?
 At most $\binom{n}{x} \leq n^x$. Using $P(\cup A_S) \leq \sum P(A_S)$, we get

$$P(\alpha \geq x) \leq n^x e^{-p_n x(x-1)/2} = [n e^{-p_n(x-1)/2}]^x.$$

I want to make a choice of x so that the quantity in brackets above is smaller than half. That is want

$$n e^{-p_n(x-1)/2} < 1/2; \quad i.e. \quad x > 1 + \frac{2}{p_n} \log(2n).$$

Take

$$x_n = \lfloor 3 + \frac{2}{p_n} \log(2n) \rfloor \quad \lfloor c \rfloor = \text{greatest integer} \leq c. \quad (\bullet)$$

Clearly $\lfloor 3 + \frac{2}{p_n} \log(2n) \rfloor > 1 + \frac{2}{p_n} \log(2n)$ as required.

Since

$$2 + \frac{2}{p_n} \log(2n) \geq \frac{2}{p_n} \log(2n) = 2n^{1-\theta} \log(2n) \rightarrow \infty$$

we can safely pick n_1 so that on $\mathcal{G}_n$;

$$n \geq n_1 \Rightarrow P_n(\alpha \geq x_n) \leq (1/2)^{x_n} < 1/2 \quad (\diamond)$$

($\heartsuit$) :

Let us now on consider $n \geq n_0 \vee n_1$. Then clearly both the above hold. Since the events above each has probability smaller than $1/2$, we conclude:

$n \geq n_0 \vee n_1$ implies there are graphs $g \in \mathcal{G}_n$ such that $\alpha(g) < x_n$ and it has at most $n/2$ cycles of length $\leq l$.

($\spadesuit$) :

Fix any $n > n_0 \vee n_1$. Take one such graph $g \in \mathcal{G}_n$ as above. Note that g has n vertices; it has at most $n/2$ many unwanted cycles (cycles of length $\leq l$). We shall now destroy these unwanted cycles. Take each such cycle and remove one vertex that appears in that cycle. We have thus removed at most $n/2$ many vertices. Let g^* be the graph g restricted to these vertices. it has the undeleted vertices as the vertex set and any pair here is joined iff it was joined in g .

This graph g^* has no unwanted cycles – Then it would have been a cycle earlier too and how come no vertex in this cycle was removed?

Moreover, an independent set here is independent in g as well because we have neither deleted nor added any edges among the remaining vertices. Thus

$\alpha(g^*) \leq \alpha(g)$; number of vertices in g^* is at least $n/2$; g^* has no unwanted cycles;

$$\alpha(g^*) \leq \alpha(g) \leq x_n \leq \frac{3}{p_n} \log(2n) = 3n^{1-\theta} \log(2n).$$

Let us now see $\chi(g^*)$. By (★)

$$\chi(g^*) \geq \frac{n/2}{x_n} \geq \frac{n}{6n^{1-\theta} \log(2n)} = \frac{n^\theta}{6 \log(2n)}.$$

Remember this is true for any $n > \max(n_0, n_1)$. Since

$$\frac{n^\theta}{6 \log(2n)} \rightarrow \infty,$$

we can choose such a large n and $g \in G_n$ so that that $\chi(g^*) > k$. This completes proof of the Theorem.

Note that as n increases p_n gets smaller, giving only few connections, yet chromatic number is maintained large. How did someone think of $n^{\theta-1}$? he did not, it came out! you do with one p and realize you can not do. Change p with n , you do not know how to change and get this p_n . You do not expect life to be too complicated to suggest a nasty function of n . Trial and error with power function, you need patience.

joint distribution:

If we have two random variables on a probability space and look at their distributions separately, We can not understand any relationships that may exist between the random variables.

Scenario 1:

Let us toss a fair coin independently twice. Then outcomes are

$$HH, HT, TH, TT$$

each probability $1/4$.

Let X be number of Heads and Y be number of tails. Then:

Distribution of X :

Values: 0, 1, 2. probabilities: $1/4$, $1/2$, $1/4$.

Distribution of Y :

Values: 0, 1, 2. probabilities: $1/4$, $1/2$, $1/4$.

The fact that $X + Y = 2$ is not clear when we look at the distributions.

To understand two random variables fully, we need to consider their joint distribution. Just as distribution of one random variable is a table giving values along with respective probabilities; joint distribution of (X, Y) is table giving values of the pair and corresponding probabilities.

Method 1:

Just like one r.v., suppose we plot the values of the pair (X, Y) along with their corresponding probabilities.

values	probabilities
$(0, 2)$	$1/4$
$(1, 1)$	$1/2$
$(2, 0)$	$1/4$

Then it is clear that numbers in each pair add to two.

Method 2:

You can present the same table as a bivariate table as follows. the values of X are in the left vertical margin. The values of Y are in the top horizontal margin. The (i, j) -th entry in the table is the probability that $(X = i, Y = j)$. The bottom horizontal margin are the column totals and the right vertical margin are the row sums.

If you read the top and bottom margins then you see the distribution of Y . If you read the left and right vertical margins then you see the distribution of X .

$X \backslash Y$	0	1	2	total
0	0	0	$1/4$	$1/4$
1	0	$1/2$	0	$1/2$
2	$1/4$	0	0	$1/4$
total	$1/4$	$1/2$	$1/4$	1

Even now it is clear that only those pairs of values that add to two have positive probability and others have probability zero.

Method 1 has the advantage that only those pairs which have non-zero probability are listed, unnecessary pairs are not listed. However if some one asks you what are the possible values of X alone; you need to see each pair and pick up the first coordinate, time consuming. Of course, in the above

example it is simple but in general it is not so.

Method 2 appears a neater presentation, though there are several zeros in the matrix of probabilities. However you can immediately understand, without any calculation, the possible values of X and their probabilities: just read the left and right margins. Similarly for Y . Thus by looking at this table, you can not only understand the pair (X, Y) but also X alone and Y alone too without any further work. You will be able to detect some interesting phenomena too by just staring at the table. We see now. Consider tossing a fair coin four times. let X denote the number of heads in the first two tosses and Y the number of heads in the last two tosses. We shall not describe method 1; it has nine tuples with corresponding probabilities. here is method 2.

$X \backslash Y$	0	1	2	total
0	1/16	1/8	1/16	1/4
1	1/8	1/4	1/8	1/2
2	1/16	1/8	1/16	1/4
total	1/4	1/2	1/4	1

The table immediately tells you another interesting feature: product of the marginals gives the corresponding entry in the matrix. More precisely

$$i\text{-th row } j\text{-th column entry} = (i\text{-th row total}) \cdot (j\text{-th column total}).$$

Such a feature would not be easily detectable if we used method 1. Also note that this X, Y have the same distributions as the previous X, Y .

Definition: joint distribution of two rvs X, Y defined on a space is a bivariate table giving values of X along the first vertical margin and values of Y along the top horizontal margin and the (a, b) -th entry of the table is the probability $p_{ab} = P(\omega : X(\omega) = a, Y(\omega) = b)$

If you look at the row sums $\sum_b p_{ab}$ you get the probability $P(X = a)$. Thus if you enlarge your table by adding row and column totals, then reading the vertical margins gives you the distribution of X , called marginal distribution of X . Similarly the horizontal margins of the table gives you the distribution of Y , called the marginal distribution of Y .

You must keep in mind that the marginal distributions are indeed distributions. The adjective ‘marginal’ draws your attention to the fact that there are other variables and they had a joint distribution and by looking at the ‘appropriate’ marginal we got the distribution of X .

Independence of rv:

This is a new definition but there is no new idea.

Two rvs X, Y defined on a space are independent if they have nothing to do with each other; meaning, answer to a question about one rv is not influenced when information about the other is revealed. Equivalently, if A is an event described by X and B is an event described by Y , then A, B are independent. Here is the precise definition.

Two rvs X, Y defined on a space are independent if for every values a of X and b of Y ; $P(X = a, Y = b) = P(X = a)P(Y = b)$

We do not have to say values of X and value of Y , because in the contrary case both sides are zero, equality holds. Of course, did we put down what we have in mind? After all, if you take any set of values of X , say S and any set of values of Y , say T then we can define two events: $A = \{\omega : X(\omega) \in S\}$ and $B = \{\omega : Y(\omega) \in T\}$. Does our definition show that these two events are independent? Yes

$$\begin{aligned} P(X \in S; Y \in T) &= \sum_{a \in S, b \in T} P(X = a, Y = b) = \sum_{a \in S, b \in T} P(X = a)P(Y = b) \\ &= \sum_{a \in S} \sum_{b \in T} P(X = a)P(Y = b) = \sum_{a \in S} P(X = a)P(Y \in T) \\ &= P(X \in S)P(Y \in T) \end{aligned}$$

finitely many rvs:

Suppose we have finitely many rvs $X_1, \dots, X_k$ on a probability space. We can talk about the joint distribution of $(X_1, \dots, X_k)$. It is a table giving all the k -tuples $(a_1, \dots, a_k)$ of possible values of these rvs and against such a tuple, its probability $P(X_1 = a_1, \dots, X_k = a_k)$.

The rvs $(X_1, \dots, X_k)$ defined on a space are independent if for every k -tuple as above

$$P(X_1 = a_1, \dots, X_k = a_k) = P(X_1 = a_1) \cdots P(X_k = a_k)$$

Of course, even if a k -tuple of numbers is not a possible value of our rvs, the above equation remains true because both sides then are zero.

Just as in the case of two rvs, we can show that if $S_1, \dots, S_k$ are subsets of the values of the rvs then independence implies

$$P(X_1 \in S_1, \dots, X_k \in S_k) = P(X_1 \in S_1) \cdots P(X_k \in S_k)$$

. Indeed,

■ The r.v. $X_1, X_2, \dots, X_k$ are independent iff for any $S_1, S_2, \dots, S_k \subset R$ where S_i is a subset of the possible values of X_i

$$P(X_i \in S_i \ \forall \ i) = \prod_1^k P(X_i \in S_i) \quad \blacksquare$$

■ If $X_1, \dots, X_k$ are independent then any sub collection is also independent. For example X_2, X_4, X_7 are independent. ■

This follows by taking some of the sets S_i above to be Ω , the sample space.

Digression: table

We have been using the word table to describe distributions. It is not necessary to use this word.

As in the case of one rv, if we have k random variables $X_1, \dots, X_k$ then their distribution as the function defined on R^k by:

$$f(a_1, \dots, a_k) = P\{\omega : X_i(\omega) = a_i \ \forall \ i\} \quad (a_1, \dots, a_k) \in R^k$$

This is called **joint probability mass function** (in order not to be confused with another function we come across later) of the k variables $X_1, \dots, X_k$. Since Ω is countable, we see that for only countably many k -tuples, the event in the display above is non-zero. Our earlier description used only these tuples in the table. In the second method we used a ‘box’ of values for the tuple. Think about it.

Examples: you can give many examples of multivariate distributions. toss a coin 4 times and X = number of heads in first three tosses and Y = number of heads in the last three tosses. Or X = number of heads in the four tosses and Y = number of head runs. Or you can roll a die twice and X = number on first throw and Y = maximum of the two throws. Here is a useful example.

Multinomial distribution: Roll a die, with k faces, n times. face i appearing in a roll is p_i . here $p_i > 0$ for each i and $\sum p_i = 1$. For $1 \leq i \leq k$, let X_i be the number of times face i appears. Then the possible values of $(X_1, X_2, \dots, X_k)$ are all k -tuples $(n_1, n_2, \dots, n_k)$ of non-negative integers with $\sum n_i = n$. It is easy to see that

$$p(n_1, n_2, \dots, n_k) = \frac{n!}{n_1! n_2! \cdots n_k!} p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$$

these values are terms in the expansion of $(p_1 + p_2 + \cdots + p_k)^n$ and this distribution is called **Multinomial distribution**. When $k = 2$ this is binomial distribution, in that case if we keep track of heads is enough. Here also if we keep track of $(k - 1)$ faces it is enough.

expectation of $\varphi(X_1, \cdots, X_k)$:

Suppose we have r.v. $X_1, \cdots, X_k$ defined on a probability space (Ω, P) . Suppose that $\varphi : R^k \rightarrow R$ is a function. We can define a random variable Z on Ω as follows.

$$Z(\omega) = \varphi(X_1(\omega), \cdots, X_k(\omega)) = \varphi(X_1, \cdots, X_k)(\omega).$$

We can think of calculating $E(Z)$ in several ways.

In what follows we assume that all sums, we are dealing with, are absolutely convergent.

■ **(A)** We can calculate the distribution of Z ; say takes values (z_n) with respective probabilities (α_n) . Then compute

$$E(Z) = \sum_n z_n \alpha_n$$

(B) You already have the joint distribution of $(X_1, \cdots, X_k)$ before you — say this tuple takes values $\{(a_1, \cdots, a_k)\}$ with respective probabilities $\{p(a_1, \cdots, a_k)\}$. No need to calculate distribution of Z . Just compute

$$E(Z) = \sum_{(a_1, \cdots, a_k)} \varphi(a_1, \cdots, a_k) p(a_1, \cdots, a_k)$$

(C). Do not calculate any distribution whatsoever. Look at sample space and compute

$$E(Z) = \sum_{\omega} Z(\omega) p(\omega) = \sum_{\omega} \varphi(X_1(\omega), X_2(\omega), \cdots, X_k(\omega)) p(\omega)$$

■
Are these equivalent? Do they give same answer? Note that **(A)** is definition and the others are not.

Yes. they all give the same answer. We had already seen, in one variable set-up that **(A)** and **(C)** are equivalent. Similar proof shows that **(B)** and **(C)** are also equivalent. Here is how. Start with **(C)**. Do the sum in two ways. First for each fixed tuple $(a_1, \cdots, a_k)$; sum or subtotal only over those ω in the set

$$\{X_1 = a_1, \cdots, X_k = a_k\}$$

Then add all these subtotals, you get (B). ■

if you are prepared to use a little analysis, you can actually show the following: if one series above converges absolutely, then so do the others. Thus absolute convergence of any one of the above three series can be taken towards the existence of $E(Z)$ and then any one of the three series above can be used as definition of $E(Z)$.

The equivalence of the above three will be referred to as **substitution formula** or **change of variable formula**.

■ If X and Y are independent random variables on a probability space, $E(X)$ and $E(Y)$ defined; Then $E(XY)$ is also defined and

$$E(XY) = E(X)E(Y)$$

Let us not spend time on existence part. Towards the proof of the equality, use the function $\varphi(a, b) = a.b$ in substitution rule to see

$$E(XY) = \sum_{a,b} \varphi(a, b) P(X = a, Y = b)$$

Use independence

$$= \sum_{a,b} a b P(X = a) P(Y = b)$$

sum w.r.t. b ;

$$= \sum_a a P(X = a) E(Y) = E(X) E(Y).$$

Covariance, correlation:

if two random variables X, Y are independent then we have

$$E(XY) = E(X)E(Y)$$

It appears reasonable to take $E(XY) - E(X)E(Y)$ as a measure of non-independence or relatedness.

Definition: covariance between X, Y is defined as

$$Cov(X, Y) = \sigma_{X,Y} = E(XY) - E(X)E(Y)$$

Observe that $\sigma_X^2 = \sigma_{XX}$. Correlation between X and Y is defined as

$$\text{correlation}(X, Y) = \rho_{X,Y} = \frac{\sigma_{X,Y}}{\sigma_X \sigma_Y}$$

when $\sigma_X \neq 0$ and $\sigma_Y \neq 0$. If either σ_X or σ_Y is zero, we define $\rho_{XY} = 0$. This is justified when you see soon that in such a case one random variable is indeed constant random variable.

The relation between two quantities should not depend on the units used to measure the quantities. Suppose I measure heights X in inches and weights Y in Kilograms and find correlation. You find heights in feet and weights in pounds. Then all my quantities X are 12 times yours and you will quickly see, if I did not have the denominator we get different answers.

If X, Y are independent then $cov(X, Y) = 0$. However $cov(X, Y) = 0$ does not mean that they are independent. We shall see later.

theorem:

1. $\sigma_X^2 = 0$ iff there is a number a such that $P(X = a) = 1$
2. If $X_1, \dots, X_n$ are independent then $Var(\sum X_i) = \sum Var(X_i)$
3. More generally, for any random variables $X_1, \dots, X_n$ defined on a space $Var(\sum X_i) = \sum_i Var(X_i) + 2 \sum_{i < j} Cov(X_i, X_j)$
4. $\sigma_{aX+bY} = ab\sigma_{XY}$ and $\sigma_{X+a, Y+b} = \sigma_{XY}$
5. $\rho_{aX+c, bY+d} = \rho_{XY}$
6. If ρ is the correlation between X and Y , then $-1 \leq \rho \leq 1$.

Remember when we talk about variances etc, we assume they exist.

Proof:

- (1) Let $E(X) = \mu$. Since $(X - \mu)^2$ takes nonnegative values, the hypothesis $E[(X - \mu)^2] = \sigma^2 = 0$ implies $P\{(X - \mu)^2 = 0\} = 1$. Thus $P(X = \mu) = 1$. Conversely, if $P(X = a) = 1$, then $\mu = E(X) = a$ and $(X - \mu)^2 = 0$ showing $\sigma^2 = 0$.
- (2) If $E(X_i) = \mu_i$, then $E(\sum X_i) = \sum \mu_i$. Also by independence, $E(X_i X_j) = \mu_i \mu_j$ for $i \neq j$, so that

$$E[(X_i - \mu_i)(X_j - \mu_j)] = E(X_i X_j) - E(X_i \mu_j) - E(X_j \mu_i) + \mu_i \mu_j = 0$$

$$\begin{aligned} Var(\sum X_i) &= E\{(\sum X_i - \sum \mu_i)^2\} = E\{[\sum (X_i - \mu_i)]^2\} \\ &= E\{\sum (X_i - \mu_i)^2 + 2 \sum_{i < j} (X_i - \mu_i)(X_j - \mu_j)\} \\ &= \sum Var(X_i) \end{aligned}$$

- (3) The same proof above shows this also, use $E\{(X_i - \mu_i)(X_j - \mu_j)\} = Cov(X_i, X_j)$
- (4) and (5) are routine calculations.

(6) Suppose $P(X = x_i, Y = y_j) = p_{ij}$. Assume $E(X) = E(Y) = 0$ (See 5). By change of variable formula and Cauchy-Schwarz

$$\begin{aligned} |E(XY)| &= \left| \sum_{i,j} x_i y_j p_{ij} \right| \leq \sum_{i,j} |x_i| |y_j| p_{ij} \leq \sum_{i,j} (|x_i| \sqrt{p_{ij}}) (|y_j| \sqrt{p_{ij}}) \\ &\leq \sqrt{\sum_i \sum_j x_i^2 p_{ij}} \sqrt{\sum_j \sum_i y_j^2 p_{ij}} = \sigma_X \sigma_Y. \end{aligned}$$

Thus $|Cov(X, Y)| \leq \sigma_X \sigma_Y$ showing $|\rho| \leq 1$. ■

Here is a very useful consequence:

Theorem WLLN

Suppose $X_1, \dots, X_n$ are independent random variables with common mean μ and common variance $\sigma^2 > 0$, then

$$P \left\{ \left| \frac{X_1 + \dots + X_n}{n} - \mu \right| > \epsilon \right\} \leq \frac{\sigma^2}{n\epsilon^2}$$

In particular suppose we have an infinite sequence $\{X_i, i \geq 1\}$ such that for each n , the variables $(X_i, 1 \leq i \leq n)$ satisfy the above hypothesis. Then we have, for each $\epsilon > 0$,

$$P \left\{ \left| \frac{X_1 + \dots + X_n}{n} - \mu \right| > \epsilon \right\} \rightarrow 0 \quad \text{as } n \rightarrow \infty \quad \blacksquare$$

This is known as the Weak Law of Large numbers. Of course if the random variables all have same distribution (which has finite variance) then they have the same mean and variance.

The content of the theorem, just as in coin tossing case, is this: the average gets as close to the mean as you want with as high probability as you want provided n is large.

In particular if you take sample (with replacement) $X_1, \dots, X_n$ from a population, then the average $\sum X_i/n$ has the interpretation of observed average. Thus if the sample size is large then the observed average is close to population mean. In particular, if you did not know the mean, then the observed average is a good estimate of the population mean.

The earlier WLLN for coin tossing is a special case of this, by taking each X_i as one with probability p and 0 with probability $q = 1 - p$.

Before returning to WLLN, let us make an observation. If we have independent random variables and if we make new variables depending on disjoint collections then these new variables are also independent. What does this mean? if $X_1, \dots, X_n$ are independent and Y depends on (X_1, X_9) and Z depends on (X_3, X_5) and W depends on (X_2, X_4, X_8) then Y, Z, W are independent. Here is a sample of such a result.

If $X_1, X_2, X_3, \dots, X_9$ are independent random variables on a sample space (Ω, P) and if $f : R^3 \rightarrow R$ and $g : R^2 \rightarrow R$ and $h : R^4 \rightarrow R$ then the random variables $Y = f(X_1, X_2, X_9)$ and $Z = g(X_3, X_7)$ $W = h(X_4, X_5, X_6, X_8)$ then Y, Z, W are independent.

I hope you understand the rvs Y, Z, W . For example if you take $\omega \in \Omega$, then value of Y at that point is $Y(\omega) = f(X_1(\omega), X_2(\omega), X_9(\omega))$. Proof is simple. If you take any three numbers a, b, c then

$$\begin{aligned} P(Y = a, Z = b, W = c) &= P[\omega : Y(\omega) = a, Z(\omega) = b, W(\omega) = c] \\ &= \sum_{i_1, \dots, i_9}^* P[X_1 = i_1, X_2 = i_2, \dots, X_9 = i_9] \end{aligned}$$

where the sum is over all 9-tuples such that $f(i_1, i_2, i_9) = a$ and $g(i_3, i_7) = b$ and $h(i_4, i_5, i_6, i_8) = c$. But by independence of X variables

$$= \sum_{i_1, \dots, i_9}^* P(X_1 = i_1)P(X_2 = i_2) \dots P(X_9 = i_9)$$

Now sum over i_1, i_2, i_9 then over i_3, i_7 and then the others to get

$$P(Y = a, Z = b, W = c) = P(Y = a)P(Z = b)P(W = c)$$

Returning to WLLN, here is the more precise explanation of population and so on, I was saying last time. Suppose there are one lakh people: $1, 2, 3, \dots, N$ ($N = \text{one Lakh}$). We denote this set by S and call it population. Suppose $f(i) = \text{age of person } i$, then $\mu = \sum f(i)/N$ is the average age of the population. If $f(i) = \text{monthly income of person } i$, then $\mu = \sum f(i)/N$ is the average monthly income of the population. If $f(i) = 1$ when person i supports a particular political party, zero otherwise then $\mu = \sum f(i)/N$ is the proportion of population that support that party.

Population does not necessarily mean people, anything, for example one lakh bolts supplied for construction of metro. $f(i)$ is one or zero according bolt i is

defective or not. Then of course μ is proportion of defective bolts. Population could be villages and for each village i , $f(i)$ is the jute acreage in that village.

Let us consider SRSWR, simple random sampling with replacement, of size n . Thus sample space Ω consists sequences $\omega = (i_1, i_2, \dots, i_n)$ of length n from S . and the probability is $P(\omega) = 1/N^n$ for each $\omega \in \Omega$. Thus i_1 represents the first person selected, i_2 , the second person selected and so on. We define $X_k(\omega) = i_k$, the k -th person selected, or k -th sample observed. we have already seen that $X_k, 1 \leq k \leq n$ are independent rvs. Let $Y_k = f(X_k)$, the observation of data (income, age, preference etc) on the k -th sample object observed. Since Y_k depends only on X_k and the variables (X_k) are independent, we also know that (Y_k) are independent too. the quantity $\bar{Y} = \sum_1^n Y_k/n$ denotes the observed average of the quantity we are interested. What is $E(Y_k)$? It is μ because, as discussed earlier X_k takes each of the values 1 to N with probability $1/N$ and so $E(f(X_k)) = \sum_1^N f(i) \frac{1}{N} = \mu$.

What does WLLN say in this context. What it says is that $\bar{Y}$ gets closer and closer to μ as n becomes large. In other words the observed average gets closer to the unobserved population average. In particular, the observed average is a ‘good’ estimate for the unobserved μ .

[Harold Hotelling famous American Statistician: “No technique of random sample has, so far as I can find, been developed in the United States or elsewhere, which can compare in accuracy with that described by Professor Mahalanobis”. Statistician Edward Deming: “No country, developed, underdeveloped or over-developed has such a wealth of information about its people as India in respect of expenditures” Economics Nobel Laureate Angus Deaton: “Where Mahalanobis and India led, the rest of the world has followed, so that today, most countries have a recent household income or expenditure survey. Most countries, can only envy India in its statistical capacity.” Now we have destroyed everything PCM set up by false narrative that they are not giving timely reports. The fact is that we suppressed timely reports: unemployment survey, consumer expenditure survey. We even found a reason to dismiss the family health survey scientist when he came out with timely report. I know how the country was when we got independence, you do not. I know the efforts by stalwarts to bring us to this state, you do not. In the zeal to usurp credit for everything, current rulers are even shamelessly discrediting great Scientists and thinkers like PCM. No one will probably ever tell you how effectively PCM used randomness and data analysis for the benefit of our nation.]

sums of independent rvs:

In many contexts you need to know distribution of sum of variables if you know the distribution of each variable in the sum.

If $X \sim B(n, p)$; $Y \sim B(m, p)$ independent then $X + Y \sim B(n + m, p)$.
Indeed, for $k \leq m + n$

$$\begin{aligned} P(X+Y = k) &= P(X = i, Y = j \text{ for some } 0 \leq i \leq n; 0 \leq j \leq m; i+j = k) \\ &= \sum_{i+j=k} \binom{n}{i} p^i q^{n-i} \binom{m}{j} p^j q^{m-j} = p^k q^{m+n-k} \binom{m+n}{k} \end{aligned}$$

You can generalize: $X \sim B(n, p)$; $Y \sim B(m, p)$; $Z \sim B(r, p)$ then $X + Y + Z \sim B(m + n + r, p)$. Indeed we knew from earlier discussion, $X + Y$ is independent of Z and we know what these two are. You can use induction. You should not be in a hurry to say 'trivial because apply what we did to $X + Y$ and Z '. you need to first observe that $X + Y$ and Z are independent, otherwise you can not apply what you did.

On a particular day, If the number X of accidents in Chennai is $P(\lambda)$ and the number Y of accidents in Bombay is $P(\mu)$; one would like to know the distribution of the total number of accidents: $X + Y$. Stop and think; the problem is NOT simply saying if 5 accidents in Chennai and 6 in Bombay then the total is 11. We do not know, we are trying to model, we have model for Chennai and Bombay, that is all. It is reasonable to assume that X and Y are independent.

If $X \sim P(\lambda)$, and $Y \sim P(\mu)$ are independent then $X + Y \sim P(\lambda + \mu)$
Indeed

$$\begin{aligned} P(X + Y = n) &= P(X = k, Y = n - k \text{ for some } 0 \leq k \leq n) \\ &= \sum_{k=0}^n e^{-\lambda} \frac{\lambda^k}{k!} e^{-\mu} \frac{\mu^{n-k}}{(n-k)!} = e^{-(\lambda+\mu)} \frac{(\lambda + \mu)^n}{n!} \end{aligned}$$

We have already seen

$X \sim G(p)$; $Y \sim G(p)$ independent, then $X + Y \sim NB(2, p)$.

Conditional distributions and Conditional expectations:

You now have all the basic material: probability and conditional probability; random variables; expectation and variance; joint distribution of random

variables.

What comes next is combination of conditional probability with other concepts leading to conditional distribution, conditional expectation. This is a truly probabilistic concept and is indeed fundamental to several topics in probability (as well as its applications). But remember this is not a new concept but just the concept of conditional probability applied to the context of random variables. And then applying the known concept of expectation with conditional probabilities rather than probabilities.

Consider the experiment: Take sample of size two with replacement from $\{0, 1, 2\}$, equivalently, pick two numbers independently at random from these three. Let X be the maximum of the sample and Y is the second chosen point. Sample space is

$$\Omega = \{00; 01; 02; 10; 11; 12; 20; 21; 22\}$$

Each outcome has probability $1/9$.

The joint distribution of (X, Y) is given below, convince yourself that this is so.

$X \backslash Y$	0	1	2	total
0	$1/9$	0	0	$1/9$
1	$1/9$	$2/9$	0	$3/9$
2	$1/9$	$1/9$	$3/9$	$5/9$
total	$3/9$	$3/9$	$3/9$	1

Sometimes we have partial information. For example you have info that $X = 2$. You would wonder what could Y be? Of course, when you are dealing with chance variables such a question does not mean you should tell a particular value of Y . You would like to know the distribution of Y . Of course Y takes values 0, 1, 2 and against each value we should now list the conditional probabilities: $P(Y = 0|X = 2)$; $P(Y = 1|X = 2)$; $P(Y = 2|X = 2)$. Thus conditional distributions of Y are as follows:

Given $X = 0$:

values of Y :	0	1	2
Conditional probabilities:	1	0	0

Given $X = 1$:

values of Y :	0	1	2
Conditional probabilities:	1/3	2/3	0

Given $X = 2$:

values of Y :	0	1	2
Conditional probabilities:	1/5	1/5	3/5

These are called conditional distributions of Y for the values of X given as shown. Note that these conditional probabilities do add to one in each case. You can calculate conditional expectations which means expectations under these conditional distributions. Thus $E(Y | X = 0)$, conditional expectation of Y given $X = 0$ is 0.

Similarly $E(Y | X = 1) = 2/3$ and $E(Y | X = 2) = 7/5$.

Remember distribution of Y is only one; whereas conditional distributions are many; one for each given value of X . Similarly expectation is just a number whereas conditional expectations of Y are many numbers; one for each given value of X .

similarly you can talk about conditional distributions of X given Y and conditional expectation $E(X | Y)$.

We want to summarize all these conditional expectations into one quantity; not a number but a function Z . It will be defined as a function of X . When X takes a value x this function takes the number $E(Y | X = x)$ as its value. Thus for all sample points ω for which $X(\omega) = a$ we have the same value for the function Z . This Z is called conditional expectation of Y given X , denoted $E(Y | X)$. You think of it as a function of X ; say, $\varphi \circ X$.

For instance in the above example, for the sample point 00, X value is zero, so value of Z is $Z(00) = E(Y | X = 0) = 0$. Thus in the above example: $E(Y | X)$ is the function which takes value zero for sample points $\{00\}$ and value $2/3$ for sample points $\{10, 01, 11\}$ and takes value $7/5$ for sample points $\{20, 21, 22, 12, 21\}$. This is the function $E(Y | X)$.

So keep in mind $E(Y | X)$, read as Expectation of Y given X , is not a number; it is a function of X . There is again no confusion because we only said 'expectation of Y given X ' — sounds like an incomplete sentence, we only said given X but did not say what is it that is given about X . Obviously it must depend on what is given about X and so is a function of X , it is defined on the sample space.

Since it is a function of X , it is a random variable. So it makes sense to take its expectation once again!

Let us quickly do all these considerations in generality.

Suppose (Ω, P) is a probability space and X, Y are random variables. Suppose their joint distribution, given in a bivariate table is as follows.

X takes values $\{x_i : i = 1, 2, \dots\}$ may be finite or infinite set.
 Y takes values $\{y_j : j = 1, 2, \dots\}$ may be finite or infinite set.

$$p_{ij} = P(X = x_i, Y = y_j)$$

The so called marginal totals are the following.

$$p_{i\bullet} = \sum_j p_{ij}; \quad p_{\bullet j} = \sum_i p_{ij}; \quad i, j \geq 1$$

Thus the distribution of X and Y are given by

$$P(X = x_i) = p_{i\bullet} : \quad i = 1, 2, \dots$$

$$P(Y = y_j) = p_{\bullet j} : \quad j = 1, 2, \dots$$

Now we can define the conditional distribution of Y given $X = x_i$:

$$P(Y = y_j | X = x_i) = \frac{p_{ij}}{p_{i\bullet}} : \quad j = 1, 2, \dots$$

this is defined for each $i = 1, 2, \dots$ and whatever be i all the numbers above add to one when summed over j . You must remember i and j are on different footing here. In the above display, i is fixed and j varies.

$$E(Y | X = x_i) = \sum_j y_j \frac{p_{ij}}{p_{i\bullet}}$$

Finally $E(Y | X)$ is the following function: When X takes value x_i , it takes the value $E(Y | X = x_i)$. If you want what exactly is the function, it is the following on the sample space.

$$E(Y | X)(\omega) = E(Y | X = a); \quad \text{where } a = X(\omega)$$

Thus the function $E(Y | X)$ takes same value at two sample points ω and η if $X(\omega) = X(\eta)$. As a consequence it is a function of X , namely

$$E(Y | X) = \varphi \circ X; \quad \text{where } \varphi(x_i) = E(Y | X = x_i)$$

Of course we have not defined the function φ at real numbers other than the x_i s. But if you are worried about that put $\varphi(x) = 0$ if x is not any of the numbers $\{x_i\}$. These do not matter. After all when you compose two functions $f(g(x))$, values of f on points in the range of g matter. Think about it.

As a consequence of the change of variable rule

$$\begin{aligned} E[E(Y|X)] &= \sum_i \varphi(x_i) P(X = x_i) \\ &= \sum_i \left(\sum_j y_j \frac{p_{ij}}{p_{i\bullet}} \right) p_{i\bullet} = \sum_i \sum_j y_j p_{ij} = \sum_j \sum_i y_j p_{ij} = \sum_j y_j p_{\bullet j} \\ &= \sum_j y_j P(Y = y_j) = E(Y) \end{aligned}$$

Thus we have proved

$$\blacksquare \quad E[E(Y|X)] = E(Y) \quad \blacksquare$$

This is a very useful formula. It allows you to calculate expected values in complicated situations. here are some examples.

Example: You own a big store. From experience you know that a random number of customers enter the shop on a given day. Each customer spends certain amount which is again random. You want to know expected daily sales. First observe that this is not sum of two or three random variables, sum of random number of random variables. It is unrealistic to model that a certain fixed number enter on a day.

Let N be the number of customers and assume $E(N) = \nu$. Let us suppose X_i is the sales for person i . we assume all X_i have same distribution with mean μ . Then total sales is $Y = \sum_1^N X_i$. You can not say $E(Y) = N\mu$, does not make sense because N is random. But Given $N = k$ you know $Y = \sum_1^k X_i$ and hence has expectation $k\mu$. In other words $E(Y|N) = N\mu$. hence

$$E(Y) = E(E(Y|N)) = E(N\mu) = \nu\mu$$

Example: For another example, Suppose a random number N of customers enter a bank, the manager – from experience – knows the distribution of the amount of money a given customer withdraws. We assume that this distribution is same for all customers. He wants to know the expected total withdrawl (he should be ready with reasonable amount of cash!) Thus if Y is the total disbursal, then $Y = \sum_1^N X_i$ where N is the random number of

customers and X_i is the amount i -th person withdraws. The same argument as above gives $E(Y) = E(N)E(X)$.

For another example let N be the number of earthquakes occurring in a day (they occur almost all the time, just that they are not strong for us to feel). Suppose that the chances that an earthquake is of intensity above 3 is p , $0 < p < 1$. What is the expected number of earthquakes above level 3 in a day? Answer: $pE(N)$. Proved as above, define X_i is one or zero if the i -th quake is above this level or not. Observe $E(X_i) = p$. Then required expectation is $pE(N)$.

In fact if we assume more we can even derive the actual distribution, instead of just expected value. Let us assume $N \sim P(\lambda)$. (X_i) are independent 0-1 variables with probabilities $(1 - p)$ and p . We want distribution of $Y = \sum_1^N X_i$. Note that for Y to take value k we must have N at least k . Thus

$$P(Y = k) = \sum_{n \geq k} P(Y = k, N = n) = \sum_{n \geq k} P(Y = k | N = n) P(N = n)$$

Note that given n earth quakes, the chances that exactly k of them are above the threshold is given by binomial probability. So

$$\begin{aligned} P(Y = k) &= \sum_n \binom{n}{k} p^k q^{n-k} e^{-\lambda} \frac{\lambda^n}{n!} = \frac{1}{k!} e^{-\lambda} \lambda^k p^k \sum_{n \geq k} \frac{1}{(n-k)!} (q\lambda)^{n-k} \\ &= \frac{1}{k!} e^{-\lambda} \lambda^k p^k e^{\lambda q} = \frac{1}{k!} e^{-\lambda p} (\lambda p)^k \end{aligned}$$

Thus $Y \sim P(\lambda p)$.

The same situation manifests in the following example. An insect lays eggs, a random number, say $P(\lambda)$ many. Each egg matures with probability p and dies with probability $(1 - p)$. What is the distribution of mature eggs? $P(\lambda p)$. Think about it.

Example:

Imagine there is a maze with five doors: 1,2,3,4,5.

If rat exits through 1, it is out in one minute. If exits through 2, then it is out in 2 minutes. If exits through 3, it travels for 3 minutes and then returns to the starting place. similarly, If it exits through 4 (or 5), then returns to the starting place after 4 (or 5) minutes.

Assume that the rat always chooses one of the four doors at random. What is the expected time to exit from the maze?

You can write down the sample points, for each sample point calculate its probability $p(\omega)$ and the time $Y(\omega)$ to exit for that sample point ω and calculate $\sum Y(\omega)p(\omega)$.

Every sample point is a finite (possibly empty) sequence of symbols 3,4,5 followed by a last symbol which is either 1 or 2.

Here is a simpler way. Let Y be the time to exit and X be the first choice of door. Suppose $E(Y) = a$. Clearly

$$E(Y|X=1) = 1; \quad E(Y|X=2) = 2. \quad E(Y|X=3) = 3 + a$$

because the conditional distribution of Y given $X=3$ is same as the distribution of $3+Y$. similarly $E(Y|X=4) = 4 + a$. $E(Y|X=5) = 5 + a$. Thus using the earlier formula,

$$a = E(Y) = E[E(Y|X)] = \frac{1 + 2 + (3 + a) + (4 + a) + (5 + a)}{5} = (15 + 3a)/5$$

or

$$a = 15/2$$

Of course you might wonder if this is an artificial example. No. If some one says that rats learn from experience, how do you test if this is right? Here is one way. you conduct above experiment on several rats several times on each. If they always take time close to above value then obviously they are not learning. If the time is visibly shorter after some repetitions, then you have reason to believe that they are learning.

The conditional distribution of Y given $X = x_i$ is a new definition, not a new concept. it is just concept of distribution and conditional probability of events. Conditional expectation $E(Y|X = x_i)$ is a new definition, not a new concept; it is just expectation w.r.t. the conditional; distribution. However $E(X|Y)$ is a new concept, we are thinking of this as a function on the sample space. Unless you are careful, this will cause confusion.

Here is an interesting question. Consider again $X \sim P(\lambda)$ and $Y \sim P(\mu)$, the number of accidents in Chennai and Bombay respectively. They are independent. Unfortunately our recorder did not maintain separate records, he noted only the total $X + Y$. He told us that the total number is 10. Can we find out how many were in Chennai? Yes.

$$P(X = k|X + Y = 10) = \frac{P(X = k, X + Y = 10)}{P(X + Y = 10)} = \frac{P(X = k, Y = 10 - k)}{P(X + Y = 10)}$$

$$\begin{aligned}
&= e^{-\lambda} \frac{\lambda^k}{k!} e^{-\mu} \frac{\mu^{10-k}}{(10-k)!} \frac{10!}{e^{-(\lambda+\mu)}(\lambda+\mu)^{10}} \\
&= \binom{10}{k} \left(\frac{\lambda}{\lambda+\mu} \right)^k \left(\frac{\mu}{\lambda+\mu} \right)^{10-k}
\end{aligned}$$

Thus given $X + Y = n$ the conditional distribution of X is $B(n, \frac{\lambda}{\lambda+\mu})$.

$$E(X | X + Y = n) = n \frac{\lambda}{\lambda + \mu}$$

pgf:

Let X be a non-negative integer valued random variable, say takes the value k with probability p_k for $k = 0, 1, 2, \dots$

Definition: The probability generating function (pgf) is defined by

$$\varphi_X(s) = p_0 + p_1 s + p_2 s^2 + \dots = \sum p_n s^n \quad 0 \leq s \leq 1.$$

This is an infinite degree polynomial or power series. Note it converges for at least $-1 \leq s \leq 1$. Of course if the random variable takes only finitely many values, then this is a polynomial and defined for all s .

This is a way of remembering the sequence of probabilities. you can see that

$$p_n = \varphi^{(n)}(0)/n!$$

We are using differentiability term-by-term of the power series, I hope you remember your analysis. Thus the function φ generates the probabilities.

$$X \sim B(n, p); \quad \varphi_X(s) = (q + ps)^n$$

$$X \sim P(\lambda); \quad \varphi(s) = e^{-\lambda(1-s)}$$

$$X \sim G(p); \quad \varphi(s) = p/(1 - qs)$$

Since

$$\begin{aligned}
&(p_0 + p_1 s + p_2 s^2 + \dots)(a_0 + a_1 s + a_2 s^2 + \dots) = \\
&p_0 a_0 + (p_0 a_1 + p_1 a_0) s + (p_0 a_2 + p_1 a_1 + p_2 a_0) s^2 + \dots
\end{aligned}$$

we see that if X, Y are independent (non-negative integer valued) then

$$\varphi_{X+Y} = \varphi_X \varphi_Y \quad (\spadesuit)$$

This is one way of computing the distribution of sum of independent random variables. You can identify the distribution of $X + Y$ by looking at $\varphi_X \varphi_Y$. You can observe $(\spadesuit)$ as follows too: First note $\varphi_X(t) = E(t^X)$ for every t

(always $0 \leq t \leq 1$.) If X, Y are independent then so are t^X and t^Y and hence $E(t^X t^Y) = E(t^X)E(t^Y)$.

You can also get ‘factorial moments’ by differentiating at one.

$$\varphi'(1) = \sum p_k k = E(X)$$

$$\varphi''(1) = \sum p_k k(k-1) = E[X(X-1)]$$

$$\varphi'''(1) = \sum p_k k(k-1)(k-2) = E[X(X-1)(X-2)]$$

and so on. We are differentiating at $t = 1$. Is it differentiable? Firstly we should only talk about left derivative because we do not know if φ is defined for $t > 1$. Actually, since we are considering non-negative integer rvs, it is not difficult to show,

$$\sum np_n = \lim_{t \uparrow 1} \varphi'(t)$$

whether the left side is finite or not. Note that $\varphi'_X(t)$ is indeed defined for $0 < t < 1$, by properties of power series.

Branching process:

There is an organism. it gives raise to certain number of offspring (could be zero), called generation 1. These in turn give raise to offspring, call generation 2. Those in generation 2 give raise to offspring, to be called generation 3 and so on. What are the chances that the generations continue for ever? How are they growing?

To discuss a model of Bhabha, consider a high energy particle (cosmic ray?) entering earth’s atmosphere. it bumps into several particles in the earth’s atmosphere and by collision creates several high energy particles, call them generation 1. These in turn bump into several things in the atmosphere and by collisions create new high energy particles, call them generation 2. And so on it goes. How do you understand this process?

In nuclear chain reactions, the process starts with a neutron released. This hits several nuclei and releases neutrons, they form generation 1. These neutrons, in turn, hit several nuclei and release more neutrons, generation 2. Will the process build up (and make the bomb explode when the neutron build up exceeds a critical mass) or will it die off before reaching a critical mass?

There is cell which is infected. This in turn infects some cells, generation 1. These newly infected cells infect some more, called generation 2. And so

on it continues. How do you understand the process?

There is a graph. Consider a vertex v . this is generation zero. Consider all vertices joined to v , called G_1 . All vertices joined to vertices in G_1 , but not occurred in G_0, G_1 . this is G_2 and so on. Of course our graph being finite, this process soon stops. But when you consider large graphs and asymptotics (limits as number of vertices goes to infinity) this is a good way of analysing.

There is one commonality in all the above processes. A particle is giving raise to some new particles and these in turn give raise and so on. Such a process is called Branching Process. To get to grips with the phenomenon, we have to understand the mechanism of off-spring production. These matters are best handled by the analytical tool, probability generating function.

generation size pgf:

Let us fix a progeny generating function

$$\varphi(t) = p_0 + p_1t + p_2t^2 + p_3t^3 + \dots$$

This is the generating function of the number of individuals produced by one particle. We assume the following:

(•) generation n consists of offspring of particles in generation $(n - 1)$ and not others.

(••) Each particle produces offspring, independent of other particles and the offspring distribution is governed by φ .

The number of fellows in generation n is denoted by X_n , for $n \geq 0$ and its pgf by φ_n . Here then is the process:

We start with one particle, generation 0. Thus $X_0 = 1$. Clearly $\varphi_0(t) \equiv t$.

Its offsprings consist of generation 1. Thus X_1 takes the value j with probability p_j . So X_1 has pgf φ and thus $\varphi_1 = \varphi$.

The total number of fellows produced by those in generation 1 consists of generation 2 and their number is X_2 . Thus, if $X_1 = 5$ then X_2 consists the number of offspring of these 5 fellows. More precisely, if $Y_1, \dots, Y_5$ are the number of offspring of these five fellows, then X_2 has the same distribution as $\sum_{i=1}^5 Y_i$. This last sum has pgf $[\varphi(t)]^5$.

On the other hand if there are 10 fellows in generation 1, then X_2 would have the same distribution as $\sum_{i=1}^{10} Y_i$ where each Y_i has the pgf φ and are independent. In other words it would have pgf $[\varphi(t)]^{10}$.

In other words $P(X_2 = k | X_1 = j)$ is just the coefficient of t^k in $[\varphi(t)]^j$
How do we find the pgf of X_2 ?

$$\begin{aligned}
P(X_2 = k) &= \sum_j P(X_2 = k, X_1 = j) \\
&= \sum_j P(X_2 = k | X_1 = j) P(X_1 = j) \\
&= \sum_j p_j \times \text{coefficient of } t^k \text{ in } [\varphi(t)]^j \\
&= \text{Coefficient of } t^k \text{ in } \sum_j p_j [\varphi(t)]^j \\
&= \text{Coefficient of } t^k \text{ in } \varphi(\varphi(t))
\end{aligned}$$

Thus pgf of X_2 is given by

$$\varphi_2(t) = \sum_k [\text{Coefficient of } t^k \text{ in } \varphi(\varphi(t))] \times t^k = \varphi(\varphi(t))$$

It is easy to guess

$$\varphi_3(t) = \varphi(\varphi(\varphi(t)))$$

Let us define ‘iterated compositions’ as follows:

$$\varphi_0(t) = t; \quad \text{for } n \geq 1, \quad \varphi_n(t) = \varphi(\varphi_{n-1}(t)) = \varphi_{n-1}(\varphi(t))$$

Note that φ takes values in $[0, 1]$ when t is in $[0, 1]$. Hence composition makes sense and you can show by induction each φ_n makes sense and takes values in $[0, 1]$.

ignoring earlier convention that φ_n denotes pgf of X_n ; let us temporarily agree to define φ_n as above. Let us proceed to justify that

φ_n so defined is indeed the pgf of X_n .

What we argued above is just that this claim is true for $n = 0, 1, 2$. Suppose that this is true for $n = 0, 1, 2, \dots, m-1$. we show it is true for $n = m$. this will then complete the proof of claim. Let us observe that if the first generation has j persons, then the m -th generation consists of the $(m-1)$ -th descendants of each of these j persons. Thus the total number in the m -th generation, given $X_1 = j$ has pgf $[\varphi_{m-1}(t)]^j$.

$$P(X_m = k) = \sum_j P(X_m = k, X_1 = j)$$

$$\begin{aligned}
&= \sum_j P(X_m = k | X_1 = j) P(X_1 = j) \\
&= \sum_j p_j \text{ coefficient of } t^k \text{ in } [\varphi_{m-1}(t)]^j \\
&= \text{coefficient of } t^k \text{ in } \sum_j p_j [\varphi_{m-1}(t)]^j \\
&= \text{coefficient of } t^k \text{ in } \varphi([\varphi_{m-1}(t)]) \\
&= \text{coefficient of } t^k \text{ in } \varphi_m(t)
\end{aligned}$$

as claimed.

Average generation size:

Let us denote the mean progeny size of a fellow is m . That is, the expected number of offspring of a fellow is m . We assume m is finite. Thus $\varphi'(1) = m$ (more precisely limit $\varphi'(t)$ as t increases to 1 equals m).

Thus there are ‘on the average’ m fellows in generation 1. Each of them has ‘on the average’ m children and so second generation has ‘on the average’ m^2 . Each of these has ‘on the average’ m children and so third generation has ‘on the average’ m^3 children. This is made precise below.

$$\begin{aligned}
E(X_n) &= \left. \frac{d}{dt} \varphi_n(t) \right|_{t=1} \\
\varphi'_n(t) &= \varphi'(\varphi_{n-1}(t)) \varphi'_{n-1}(t).
\end{aligned}$$

Hence by induction, we see $\varphi'_n(1) = m^n$. Thus

$$E(X_n) = m^n$$

You can calculate the variance of X_n too. But we shall not.

The next question is whether the generations continue or extinct at some stage. That is, whether some generation size is zero. Thus we are interested in

$$\alpha = P(X_n = 0 \text{ for some } n)$$

Though there is much that can be discussed about branching processes, we shall close our discussion with proving a nice theorem about α .

We are discussing whether the generations continue forever or extinct at some stage. That is, whether some generation size is zero. Thus we are interested in

$$\alpha = P(X_n = 0 \text{ for some } n)$$

that is,

either $X_1 = 0$ or $X_2 = 0$ or $X_3 = 0 \dots\dots\dots$.

We are asking for probability of an event that depends on all the infinitely many random variables. Of course, if there is none in generation 3, then there is none in future too. In other words the events $(X_n = 0)$ are increasing. Thus

$$\alpha = \lim_n P(X_n = 0)$$

if $p_0 = 0$ then the generations continue forever. If $p_0 = 1$ then there is none from generation one itself.

Theorem: assume $0 < p_0 < 1$.

1. α is the smallest solution in $[0, 1]$ of the equation $\varphi(t) = t$.
2. The equation $\varphi(t) = t$ has at most two solutions in $[0, 1]$ and $t = 1$ is a solution.
3. $\alpha = 1$ if $m \leq 1$. And $\alpha < 1$ if $m > 1$.

Of course even when $p_0 = 0$ or $p_0 = 1$ the statements are true, but trivial. Thus if mean offspring is at most one, then the population is extinct sooner or later. If the mean offspring is larger than one then there is a chance of survival forever.

We shall now prove the theorem about extinction probability. We assume below that $0 < p_0 < 1$.

(a) Consider the events $A_n = (X_n = 0)$. They are increasing as noted already above, if there is none in generation n then there is none later. And their union is precisely the event that some X_n is zero. In other words

$$P(A_n) \uparrow \alpha$$

Clearly, $P(A_1) = \varphi(0)$. $P(A_2) = \varphi(\varphi(0))$. In other words, if $a_n = P(A_n)$ then

$$a_{n+1} = \varphi_{n+1}(0) = \varphi(a_n)$$

simply because φ_n is the pgf of X_n and $\varphi_{n+1} = \varphi(\varphi_n(t))$. Since $a_n \uparrow \alpha$ and φ is continuous on $[0, 1]$, taking limits in the above equality leads to

$$\varphi(\alpha) = \alpha$$

Let now $c \in [0, 1]$ satisfy, $\varphi(c) = c$. since φ is increasing and $c \geq 0$ we see

$$\varphi(0) \leq \varphi(c) = c; \quad a_1 \leq c$$

Proceeding by induction, if we proved $a_n \leq c$ then the same reasoning as above gives

$$a_{n+1} = \varphi(a_n) \leq \varphi(c) = c$$

Thus for all n , we have $a_n \leq c$, showing $\alpha \leq c$.

This proves (1) of the theorem.

(b). Since $\varphi(1) = 1$, we only need to show that there can be at most one solution of $\varphi(t) - t = 0$ in $[0, 1]$.

If there are two solutions say $a < b$ then

$$\varphi(a) - a = \varphi(b) - b = \varphi(1) - 1 = 0.$$

Mean value theorem applied to $\varphi(t) - t$ on $[a, b]$ and $[b, 1]$ leads to points c, d such that $a < c < b < d < 1$ and

$$\varphi'(c) - 1 = \varphi'(d) - 1 = 0$$

One more MVT leads to t_0 with $0 < t_0 < 1$ such that

$$\varphi''(t_0) = 0$$

In case $p_0 + p_1 < 1$, this is a contradiction because

$$\varphi''(t) = (2!)p_2 + (3!)t + \cdots, \quad 0 < t < 1.$$

Since at least one term on right side is non-zero and all are non-negative, we can not have $\varphi''(t_0) = 0$.

On the other hand if $p_0 + p_1 = 1$, then $y = \varphi(t) = p_0 + p_1 t$ is a straight line and meets the straight line $y = t$ in at most one $t \in R$ and $t = 1$ is visible. So meets at no other point.

Thus in any case there is at most one solution in $[0, 1]$ proving (2) of the theorem. You can argue this by using convexity of φ also.

(c) If $\varphi(t) = t$ has a solution $a \in [0, 1)$ then MVT applied on the interval $[a, 1]$ to $\varphi(t) - t$ will give a point $c \in (0, 1)$ with $\varphi'(c) = 1$. Note that now $p_0 + p_1 < 1$, because in the contrary case we know there is no solution of $\varphi(t) = t$ in $[0, 1)$. and as noted above $\varphi'' > 0$ in $(0, 1)$ so that φ' is strictly increasing there. This shows

$$m = \lim_{t \rightarrow 1^-} \varphi'(t) > 1.$$

Recall that m is the mean offspring of an individual

if $\varphi(t) = t$ has a solution in $[0, 1)$ then $m > 1$. (•)

Assume now 1 is the only solution of $\varphi(t) = t$. Clearly $p_0 > 0$, otherwise zero is a solution of the above equation, anyway we assumed $0 < p_0 < 1$ to avoid trivialities. Thus the graph $y = \varphi(t)$ is above graph $y = t$ at the point $t = 0$ and since they do not meet anywhere till 1, we conclude

$$\varphi(t) > t \quad (i.e.) \quad 1 - \varphi(t) < 1 - t \quad \forall t < 1 \quad \text{or} \quad \frac{1 - \varphi(t)}{1 - t} \leq 1$$

Taking limit as $t \uparrow 1$ you get

$$m = \varphi'(1) \leq 1$$

[even though each difference quotient is smaller than one, unless I show that the limit exists, I can not conclude that $m \leq 1$. Though we can do this using our assumption that mean is finite, it is not necessary to justify. We can argue as follows. No matter how close to 1 you take $t < 1$ the above inequality with MVT tells you that $\varphi'(u) \leq 1$ for some $u \in (t, 1)$. Since φ' is increasing this is enough to conclude $\lim_{t \rightarrow 1^-} \varphi'(t) \leq 1$. But this is m . It is not necessary to talk about $\varphi'(1)$.] Thus

if $\varphi(t) = t$ has no solution in $[0, 1)$ then $m \leq 1$. (••)

It is easy to conclude from (•)(••) part (3) of the theorem.

As Soham was saying we did not show any sample space and random variables X_n denoting size of the n -th generation. We can, for any given n , describe sample space and probabilities but it is unnecessary. Our questions of interest are about probabilities involving these X_n , which can be answered without looking at the sample space. Further, constructing sample space of outcomes will not help in our calculations. In fact this is one of the points, later we will not be able to see sample spaces, yet we shall be able to calculate probabilities using the available information. This is easily achieved if you have enough information about distributions.

To be a little more specific, suppose I toss a coin (chance of heads p) 100 times. Then there are 2^{100} outcomes. We did go through all the outcomes without writing down, knew probability of each outcome and calculated the distribution of X , number of heads. Once this is available to us to answer questions about X , there is no need to look at the sample space.

You may still want to know: are we unable to define sample space? No, we can construct, for each n , the sample space describing the outcomes upto

and including the n -th generation. This can be done by specifying a tree explaining number of children and their children etc. You can also describe outcomes by carefully-coded sequences of integers too. At this stage we can not construct the sample space for the entire infinite sequence (X_n) . This is because we are only handling countable sample spaces and constructing for the entire infinite sequence can not be done.

Recall, when we discussed WLLN, I said we are, with our present knowledge, unable to ‘show’ an infinite sequence of coin tossing variables. But I also promised to show you an infinite sequence of coin tossing variables later. This is our first instance where we could answer complicated questions without looking at the sample space. Also this is really an instance where a question truly involving an infinite sequence of rvs is answered.

Hardy Weinberg:

We shall consider an urn problem. Have two boxes: Mbox MB and Fbox FB. There are balls in each box and on each ball one of the following pair is written: AA, Aa, aa . Suppose these are in the proportion

$$AA : Aa : aa = u : v : w; \quad u, v, w > 0; u + v + w = 1$$

in each box. I Take one ball from each box and pick one letter from each ball. What are the chances that I get AA : what are the chances I get Aa ? what are the chances I get aa ? We treat Aa and aA as same.

Here is a way of doing it. There are nine possible pairs of balls you could have picked up. For example

$$P(\langle AA, AA \rangle) = u^2; \quad P(\langle aa, aa \rangle) = w^2; \quad P(\langle AA, Aa \rangle) = P(\langle Aa, AA \rangle) = uv$$

$$P(\langle AA, aa \rangle) = P(\langle aa, AA \rangle) = uw \text{ and so on. Also}$$

$$P(AA|\langle AA, AA \rangle) = 1; \quad P(AA|\langle AA, Aa \rangle) = 1/2$$

$$P(AA|\langle Aa, Aa \rangle) = 1/4 \quad P(AA|\langle AA, aa \rangle) = 0$$

Thus conditioning on the nine cases and doing the needed algebra, you get the following

$$AA : Aa : aa = p^2 : 2pq : q^2; \quad p = u + \frac{1}{2}v; \quad q = 1 - u = \frac{1}{2}v + w.$$

You can do in another way too. Pick a ball from MB and a letter from the ball then you get $A : a = p : q$ where p, q are as above. Same happens if you pick from FB; and for the pair you get the above proportions.

So if you pretend that first generation urns have above proportion and next generation urns are formed in the manner described above then the proportions will be as derived $AA : Aa : aa = p^2 : 2pq : q^2$. Let us call this second

generation urns. With same rule how does the third generation urns look like?

So suppose now you start with urns MB2 and FB2 with these new proportions and repeat the above process. Then what happens? Nothing new, the same proportions are maintained, simply because $p^2 + pq = p$.

Thus from now on; these proportions $p^2 : 2pq : q^2$ are maintained forever.

This conclusion above is called **Hardy-Weinberg Law** in genetics. What are the boxes, what are the letters and what is all this? Humans have 23 pairs of chromosomes. They are nicely identified and numbered. Let us consider the first 22 chromosome pairs. Each pair consists of similar things in structure but possibly different in chemical content. Let us concentrate on one of these pairs and a particular location on this pair. Let us assume that the chemical at this place on this chromosome can occur in two types A and a .

Thus the possible combination is one of the three types AA, Aa, aa . We identify Aa and aA . The above problem refers to the proportions of these types and the phenomenon of how they change over generations. According to Mendelian laws child gets one letter from M and one from F. This is precisely what was happening in the problem.

The main conclusion is that the same proportions are maintained from the third generation, thus stability is attained soon in the population and you also know how the stable frequencies look like. One can argue using the above law that mutations must be going on.

You can also consider the situation when the proportions in the two urns are different: MB is $u : v : w$ and FB is $u_1 : v_1 : w_1$. Similar phenomenon occurs. We can also consider the 23rd pair of chromosomes. They are different both in structure and content. they are named X and Y . Those with XY are called M and those with XX are called F. Thus offspring always gets X from mother. The offspring is M if it gets Y from father; it is F if it gets X from father. We can consider a chemical present at a specific location on the X chromosome and derive a similar law. This chemical can occur in two types. Thus M has only one dose and F has two doses. The H-W law gives in this case limiting proportions. Here is a specific example of such chemical: one that controls 'colour blindness', another is Hemophilia. One can deduce if a proportion '1 in 1000' of males are colour blind, then among females only '1 in 1000,000' will be colour blind. We are assuming limiting proportions, but do not worry, so many generations have elapsed.

Of course, several assumptions are to be made, and one should also discuss the situation when some of the assumptions are not met. But this is not a course in genetics. Then why did I do this? Probability pervades all subjects.

This is an important law and you should know.

Chandrasekhar Model:

I have a huge supply of balls, as many as I want. I have two numbers: $0 < p < 1$ and $\lambda > 0$. Here is a game I play. I start with a box having a certain number of balls. Every morning, I take the balls in the box and for each ball decide to keep it or throw out. Having done that, I add a certain number of balls.

Question: what happens in the long run?

of course, to make sense of the question and answer it, you should know the mechanism of removing and adding balls. Here is how I remove: Take a ball, toss coin; Heads up decide to remove, Tails up decide to keep. Do this for each of the balls in the box. Here is the mechanism of adding the balls: Add $P(\lambda)$ many balls. This means, select an integer in such a way that chance of selecting n is

$$e^{-\lambda} \frac{\lambda^n}{n!}; \quad n = 0, 1, 2, \dots$$

Having selected an integer as above add so many balls. remember there is no limit for the number of balls being added but it is some finite number. So there never are infinitely many balls in the box.

Now the mechanism is completely specified. The only thing that needs to be told is: how did the game start? With how many balls did I start on day zero?

For simplicity let us assume that we started with zero balls.

Much later you will see that it does not matter. Even if you decide to roll a die and start with as many balls as the face that shows up, you will have exactly the same answer as we get below.

so let $p_k(n)$ be the probability of having k balls on day n . Thus if X_n is the number of balls on day n then you agree that it is a random variable.

$$p_k(n) = P(X_n = k).$$

Thus $p_k(0)$ is one if $k = 0$ and zero if $k \neq 0$ (because we started with zero balls on day zero).

$$p_k(1) = e^{-\lambda} \lambda^k / k!$$

because there is nothing to remove on day one, all the balls are those added that morning.

$$p_k(2) = P(X_2 = k) = \sum_{i=0}^{\infty} P(X_2 = k, X_1 = i)$$

$$= \sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} P(X_2 = k | X_1 = i)$$

Let us denote

$$p_{ij} = P(X_2 = j | X_1 = i)$$

Thus we have i balls today. To have j balls tomorrow you should keep certain number of balls of today. this number can not exceed i because that is all you have. It also can not exceed j because you want to have j tomorrow. And then add some number of balls to make a total of j balls. Thus

$$p_{ij} = \sum_{l=0}^{i \wedge j} \binom{i}{l} q^l p^{i-l} e^{-\lambda} \lambda^{j-l} \frac{1}{(j-l)!}.$$

In passing let us note that p_{ij} is also $P(X_{n+1} = j | X_n = i)$ whatever be n . Returning to earlier calculation

$$\begin{aligned} p_k(2) &= \sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \sum_{l=0}^{i \wedge k} \binom{i}{l} q^l p^{i-l} e^{-\lambda} \lambda^{k-l} \frac{1}{(k-l)!} \\ &= e^{-\lambda} e^{-\lambda} \lambda^k \frac{1}{k!} \sum_{l=0}^k \binom{k}{l} q^l \sum_{i=l}^{\infty} p^{i-l} \lambda^{i-l} \frac{1}{(i-l)!} \\ &= e^{-\lambda} e^{-\lambda} \lambda^k \frac{1}{k!} (1+q)^k e^{\lambda p} \\ &= e^{-\lambda(1+q)} [\lambda(1+q)]^k / k!. \end{aligned}$$

Exactly the same argument shows

$$p_k(3) = e^{-\lambda(1+q+q^2)} [\lambda(1+q+q^2)]^k / k!.$$

$$p_k(n+1) = e^{-\lambda(1+q+\dots+q^n)} [\lambda(1+q+\dots+q^n)]^k / k!.$$

Lo and behold

$$p_k(n) \rightarrow e^{-\lambda/p} (\lambda/p)^k \frac{1}{k!}.$$

Thus in the long run you expect to have $P(\lambda/p)$ many balls in the box. In the long run, as usual means: rigorously ‘as $n \rightarrow \infty$ ’ and in practice ‘for all large n ’.

This model is called Chandrasekhar model. What is this model for and why are we removing and adding balls? Well, actually balls are not balls and box is not box and day is not day and we are doing nothing either. Then is this all a hoax?

Imagine a Huge glass vessel filled with liquid. You put some coloured pollen particles (simply referred to as particles) into it. You fix, a small piece of volume in the middle of the vessel. This small piece of volume is our box (not the big thing). Day is not a day, but $(1/200)$ th of a second. During every time interval, some of the particles (in the small volume, I fixed my attention on) leave that volume. Of course, some particles, from the Jar outside this volume, enter during this period.

What we discussed is precisely modelling this phenomenon. This model is due to the Astrophysicist S Chandrasekhar. Data was actually collected, he was testing Brownian motion calculations in the theory of molecular fluctuations.

Why Poisson for the number of particles entering the volume under focus? There are so many particles in the Huge jar, each having a small chance of entering our region. Now you know what should be a good model for the number of particles entering.

The above process is an example of, what is called, Markov chain. This is probabilistic analogue of the deterministic ‘dynamics’ or ‘motion’ of physics. We shall return to this later. Having seen interesting calculations with conditional probabilities, let us discuss another problem of practical importance.

Best Predictions:

I have a sample space (Ω, P) and two random variables X and Y with finite means, variances and known covariance.

We wish to estimate Y . Here are specific questions.

Question 1: What is the best constant fit to Y ?

Question 2: What is the best ‘linear in X ’ fit to Y ?

Question 3: What is best ‘function of X ’ fit to Y ?

Q1:

Meaning of the question is the following. What is the best number a so that $E\{(Y - a)^2\}$ is the least possible. If your guess is the number a and if then I observe Y ; you will incur a loss $(Y - a)^2$. Which number a minimizes your loss? This loss is called squared error loss. $(Y - a)$ is the error. Of course, minimizing error is meaningless because you do not know it till you make observation. So you minimize expected squared error.

If μ and σ^2 are the mean and variance of Y , then

$$E\{(Y - a)^2\} = E\{[(Y - \mu) + (\mu - a)]^2\}$$

$$\begin{aligned}
&= E\{(Y - \mu)^2\} + E\{(\mu - a)^2\} + 2E\{(Y - \mu)(\mu - a)\} \\
&= \sigma^2 + (\mu - a)^2
\end{aligned}$$

So best guess and expected penalty for this guess are

$$\mu_Y; \quad \sigma_Y^2 \quad (\bullet)$$

Q2:

Again, ‘best’ is interpreted as earlier. you have to prescribe a linear function of X , say, $a + bX$ (if you are fussy and do not want to use the word linear for $a + bX$, agreed, use the word affine). Of course you know the joint distribution of (X, Y) . Remember, you have to tell me numbers a, b so that when I substitute for X , the observation, then the number $a + bX$ will be your prediction for Y . In other words, a, b can not depend on the observation X , they must be numbers determined in advance by you.

Now we make observation (x, y) . Thus observed value of Y is y and your guess is $a + bx$. The penalty is $(y - a - bx)^2$. We must minimize penalty, makes no sense because it is random. So minimize expected penalty $E\{(Y - a - bX)^2\}$. Since we want to minimize over (a, b) we set gradient equal to zero. Interchanging differentiation and expectation, the equations are

$$\begin{aligned}
2E(Y - a - bX) &= 0; \quad \text{or} \quad \mu_Y = a + b\mu_X. \\
2E\{X[Y - a - bX]\} &= 0 \quad \text{or} \quad E(XY) - a\mu_X - bE(X^2) = 0
\end{aligned}$$

Substituting for a from the first equation we get

$$\begin{aligned}
E(XY) - \mu_X[\mu_Y - b\mu_X] - bE(X^2) &= 0 \\
\sigma_{XY} - b\sigma_X^2 &= 0; \\
b &= \frac{\sigma_{XY}}{\sigma_X^2}; \quad a = \mu_Y - \frac{\sigma_{XY}}{\sigma_X^2}\mu_X.
\end{aligned}$$

Strictly speaking you need to find second derivative to decide whether this is max or min or none! let us skip that part. Also we are assuming the case $\sigma_X^2 \neq 0$.

What is the expected penalty for such a guess?

$$\begin{aligned}
&E \left\{ \left(Y - \mu_Y - \frac{\sigma_{XY}}{\sigma_X^2}(X - \mu_X) \right)^2 \right\} \\
&= \sigma_Y^2 + \frac{\sigma_{XY}^2}{\sigma_X^2} - 2\frac{\sigma_{XY}}{\sigma_X^2}\sigma_{XY} = \sigma_Y^2 - \frac{\sigma_{XY}^2}{\sigma_X^2}
\end{aligned}$$

Remembering that $\sigma_{XY} = \rho\sigma_X\sigma_Y$, the above equals $\sigma_Y^2[1 - \rho^2]$. Thus the best guess and expected penalty for this guess are

$$\mu_Y + \frac{\sigma_{XY}}{\sigma_X^2}(X - \mu_X); \quad \sigma_Y^2[1 - \rho^2] \quad (\bullet\bullet)$$

comparing with $(\bullet)$ and using that $|\rho| \leq 1$ you see that this guess using X reduces the penalty and the percentage reduction is in terms of the correlation between the variables. As expected if X, Y are independent, then $\rho = 0$ and there is no reduction and indeed your prediction is just μ_Y .

Q3:

Going by the experience of Q1; the best guess, if we know the X value should be conditional expectation. we can show this just like the earlier ones. Thus let φ be the function $\varphi(a) = E(Y|X = a)$, defined for every value of X . We believe that $\varphi(X)$ (you make your own notation whether you want to use $\varphi(X)$ or composition notation $\varphi \circ X$) should be the best guess.

In passing observe that we know the joint distribution of (X, Y) and hence can calculate $E(Y|X = a)$ for each number a in advance and the actual value to be predicted is obtained after the observed value of X is announced.

The exact calculation is not too difficult but has some subtle issues. I shall briefly sketch. Want to show that if you take any other function of X , say $\Psi(X)$; then

$$E\{[Y - \Psi(X)]^2\} \geq E\{[Y - \varphi(X)]^2\}.$$

We can assume that $\Psi(X)$ has finite variance, otherwise left side is infinity and so the above inequality holds. We proceed as in Q1. add and subtract $\varphi(X)$ in right side

$$\begin{aligned} E\{[Y - \Psi(X)]^2\} &= E\{[Y - \varphi(X)]^2\} + \\ &E\{[\varphi(X) - \Psi(X)]^2\} + 2E\{[Y - \varphi(X)][\varphi(X) - \Psi(X)]\} \quad (\dagger) \end{aligned}$$

We need a very important fact. If you have any function of X , say $h(X)$ then

$$E[h(X)Y | X] = h(X)E(Y | X) \quad (\clubsuit)$$

In particular $E(h(X) | X) = h(X)$.

(This stands to reason. Easy to see $E[5Y | X] = 5E[Y | X]$. Thus when you are told $X = a$, we see that $h(X) = h(a)$ is just a number, so $E[h(X)Y | X = a] = E[h(a)Y | X = a] = h(a)E[Y | X = a]$; this being true for any number a , we conclude $E[h(X)Y | X] = h(X)E[Y | X]$. This intuitive feeling can be made precise by simply going to the joint distribution of (X, Y) , conditional

distribution given X and using the definition of conditional expectations.)

Returning to $(\dagger)$; if $Z = [Y - \varphi(X)][\varphi(X) - \Psi(X)]$ then last term of $(\dagger)$ (apart from factor 2) equals $E(Z)$ which we understand now. In view of $(\clubsuit)$, taking $h = \varphi - \Psi$ we see

$$\begin{aligned} E[Z|X] &= [\varphi(X) - \Psi(X)] E\{[Y - \varphi(X)]|X\} \\ &= [\varphi(X) - \Psi(X)] \{E[Y|X] - E[\varphi(X)|X]\} \end{aligned}$$

If you look at the second term on right side you realize, by an application of $(\clubsuit)$ and definition of φ ;

$$E[Y|X] - E[\varphi(X)|X] = \varphi(X) - \varphi(X) = 0$$

Thus $E(Z|X) = 0$. and hence

$$E(Z) = E\{E[Z|X]\} = E\{0\} = 0$$

Did you see how all the rules of conditional expectation and the simple formula we proved earlier: $E(Z) = E[E(Z|X)]$, helps. Thus third term on RHS of $(\dagger)$ is zero. Second term is nonnegative. This proves the required inequality. further equality holds in $(\dagger)$ when and only when the following holds,

$$E\{[\varphi(X) - \Psi(X)]^2\} = 0.$$

Thus the best guess and expected penalty for this guess are

$$E(Y|X); \quad E\{[Y - E(Y|X)]^2\} \quad (\bullet \bullet \bullet)$$

Calculus analogy:

Do you see that the first two questions are exactly like in Calculus? Of course, you need to interpret ‘best’ as in calculus, not expected squared error etc, there is no probability. Given a continuous function $x \mapsto y(x)$ on R and a point a what is the ‘best constant function guess’, say, $h(x)$ for y near a ? Best means $h(x)$ converges to $y(a)$ as $x \rightarrow a$. But of course h being constant function, this simply means constant function $y(a)$.

Answer: the constant function $h(x) \equiv y(a)$.

Given a differentiable function $x \mapsto y(x)$ on R and a point a what is the ‘best linear function $ax + b$ guess’ for y near a ? Best means not only has same value at a but same slope.

Answer: the tangent function at a ; $h(x) = y(a) + y'(a)(x - a)$.

You can ask best quadratic guess etc. You come to Taylor expansion. In case of random variables too, we could have asked for the best quadratic guess

$a + bX + cX^2$ for Y . We did not do so. If you were doing statistics or data analysis, you would do.

In our case there is the third question Q3, you do not have this in calculus.

mgf:

We shall return to the pgf mood.

Let now X be any random variable not necessarily integer valued. Say, takes values $x_i, i \geq 1$ with respective probabilities $p_i, i \geq 1$. We have defined $E(X) = \sum x_i p_i$. and $E(X^2) = \sum x_i^2 p_i$. These are called the first and second moments, μ_1 and μ_2 . These were, to some extent, give a feeling for the distribution – mean and spread. We have also defined n -th moment.

$$\mu_n = E(X^n) = \sum x_i^n p_i \quad \text{provided} \quad \sum |x_i^n| p_i < \infty$$

We put $\mu_0 = 1$.

The moment generating function, mgf, of a random variable X is the function defined by

$$M_X(t) = E(e^{tX}) \quad \text{defined for} \quad t \in \{s : E(e^{sX}) < \infty\}$$

Unlike the pgf which is defined for at least $-1 \leq s \leq 1$, it may so happen that M is defined only for $t = 0$, even if X is integer valued. Note that $M(0) = 1$ always. Thus if X takes values $\{x_i\}$ with corresponding probabilities $\{p_i\}$ then

$$M(t) = \sum e^{tx_i} p_i.$$

Note the sum always makes sense, at the worst it may be infinity. This is because all terms are nonnegative and hence partial sums increase.

For example, a straight forward calculation gives

$$X \sim B(n, p); \quad M(t) = (q + pe^t)^n$$

$$X \sim P(\lambda); \quad M(t) = e^{-\lambda(1-e^t)}$$

$$X \sim G(p); \quad M(t) = 1/(1 - qe^t) \quad \text{if} \quad qe^t < 1$$

When X and Y are independent then so are e^{tX} and e^{tY} . Since $e^{t(X+Y)} = e^{tX}e^{tY}$ we see

$$X, Y \text{ independent : } M_{X+Y}(t) = M_X(t)M_Y(t) \quad t \in \text{Dom}(M_X) \cap \text{Dom}(M_Y)$$

where Dom is domain of the function.

mgf does actually generate the moments. Suppose a random variable has mgf defined for t in an interval $(-a, +a)$. Then we can show

$$\mu_n = M^{(n)}(0)$$

where $M^{(n)}$ denotes the n -th derivative. One can show that under the assumed hypothesis, the function M is differentiable any number of times and the above equality holds.

If X has mgf $M_X(t)$, then $Y = aX$ has $M_Y(t) = M_X(at)$. These functions play fundamental role in certain contexts. However our purpose is not to discuss these in detail. My main purpose is to prove a beautiful result.

Chernoff bound:

Theorem (Chernoff bound): Let X be a random variable and $a > 0$. Then for any $t > 0$ such that $M_X(t)$ is finite, we have

$$P(X \geq a) \leq M_X(t) e^{-ta}$$

Proof is simple, the events $(X \geq a)$ and $(e^{tX} \geq e^{ta})$ are same because $t > 0$ and exponential function is increasing. Now Chebyshev does it. ■

But then why is it so important. You have a handle t with you and you can make best use of it. Here is an illustration.

Let us consider $X_1, X_2, \dots, X_n$ independent and each assuming values ± 1 with equal probability. Thus mean is zero. Also variance is one. Let $S_n = \sum X_i$. Thus S_n has variance n , Thus variance of S_n/n is $1/n$. Usual Chebyshev gives, for any $a > 0$

$$P(|\frac{S_n}{n}| \geq a) \leq \frac{1}{na^2} \quad (\clubsuit)$$

Let us see how our handle helps us. Let $Y = S_n/n$. Then

$$M_Y(t) = M_{S_n}(t/n) \quad M_{S_n}(t) = [M_{X_1}(t)]^n$$

the last equality is by independence.

$$M_{X_1}(t) = \frac{(e^t + e^{-t})}{2} = 1 + \frac{t^2}{2!} + \frac{t^4}{4!} + \dots + \frac{t^{2n}}{(2n)!} + \dots$$

Since

$$(2n)! \geq 2 \times 4 \times 6 \times \dots \times 2n = 2^n n!$$

we get

$$\begin{aligned} M_{X_1}(t) &\leq 1 + \frac{t^2/2}{1!} + \frac{(t^2/2)^2}{2!} + \frac{(t^2/2)^3}{3!} + \dots + \frac{(t^2/2)^n}{n!} + \dots \\ &= e^{t^2/2} \end{aligned}$$

Thus

$$M_{S_n}(t) \leq e^{nt^2/2} \quad M_{S_n/n}(t) \leq e^{t^2/2n}$$

Chernoff gives

$$P\left(\frac{S_n}{n} \geq a\right) \leq e^{-ta} e^{t^2/2n}$$

We choose $t = na$ so that $ta = na^2$ and $(t^2/2n) = na^2/2$ giving

$$P\left(\frac{S_n}{n} \geq a\right) \leq e^{-na^2/2}$$

Since $-S_n$ and S_n have same distribution we get

$$P\left(\frac{S_n}{n} \leq -a\right) \leq e^{-na^2/2}$$

Thus

$$P\left(\left|\frac{S_n}{n}\right| \geq a\right) \leq 2e^{-na^2/2} \quad (\spadesuit)$$

This is an exponential bound compared to (). This precision is very important especially when you simulate you need to know after how many steps you should stop. This has other uses too.

In the above example we chose that value of t which gave the best bound. As Neelabh was saying, since in general mgf may not be defined for all values of t , we choose that value of t where mgf is defined and gives us best inequality.

These type of inequalities are the standard tool kit in several areas not only in probability but also computer science, physics and so on.

By Union bound means: using $P(\cup A_i) \leq \sum P(A_i)$, for events $\{A_i\}$.

First moment method means: using $P(|X| \geq a) \leq E(|X|)/a$ for $a > 0$.

Second moment method means: using $P(|X| \geq a) \leq E(X^2)/a^2$ for $a > 0$.

By Chernoff bound means: using Chernoff inequality, good choice of t .

CLT:

suppose I toss a fair coin 10000 times. You expect the number of heads X to be around 5000, but you also know that it is unlikely to be exactly 5000. Want to know how close is it to 5000, say within ± 100 . Of course

$$P(4901 \leq x \leq 5100) = \sum_{k=4901}^{5100} \binom{10000}{k} (1/2)^{10000}$$

This is no doubt exact result, but what is its worth? Do we have any idea how large or small it is? You need to have a good approximation which is

understandable and useful.

Recall that a sequence of random variables $(X_n, n \geq 1)$ is independent if for each n the variables $X_1, X_2, \dots, X_n$ are independent. A sequence of random variables $(X_n, n \geq 1)$ is identically distributed if they all have the same distribution. A sequence is iid – independent identically distributed – if both hold.

We came across infinite sequence of variables in WLLN. Of course at this moment we are only studying discrete random variables. Here is Central Limit Theorem which is one of the central results, which is of great importance both theoretically and in practice; not only in probability but in computer science and physics and other areas.

Theorem:

Suppose $X_1, X_2, \dots$ is a sequence of iid random variables with mean μ and variance $\sigma^2 > 0$. Then for any $b \in R$

$$P\left(\frac{X_1 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \leq b\right) \rightarrow \int_{-\infty}^b \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

For any $a < b \in R$

$$P\left(a < \frac{X_1 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \leq b\right) \rightarrow \int_a^b \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \quad \blacksquare$$

Observe that

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = 1$$

You must appreciate that no serious assumption is made about the distribution except that the mean and variance are finite. You should understand that the quantity $Z_n = (\sum X_i - n\mu)/\sigma\sqrt{n}$ is simply the sum standardized; which means make mean zero and variance one. Since $E(X_i) = \mu$ we see $E(\sum X_i) = n\mu$ so that $E(\sum X_i - n\mu) = 0$. By independence variance adds up so that $var(\sum X_i) = n\sigma^2$ and $var(\sum X_i - n\mu) = n\sigma^2$. Thus $E(Z_n) = 0$ and $var(Z_n) = 1$

In the earlier example of Binomial probability, observe that $B(10000, p)$ variable is sum of 10000 iid variables each taking values 1, 0 with probabilities $p, (1 - p)$ (we took $p = 1/2$ there) respectively. And as usual, the above theorem, in practice means you can approximate the probability on left side with the area under the curve of the left side between a and b . In particular you can calculate

$$P(4900 < X \leq 5100) = P\left(\frac{4900 - 5000}{\sqrt{10000 \cdot \frac{1}{2} \cdot \frac{1}{2}}} < \frac{X - 5000}{\sqrt{10000 \cdot \frac{1}{2} \cdot \frac{1}{2}}} \leq \frac{5100 - 5000}{\sqrt{10000 \cdot \frac{1}{2} \cdot \frac{1}{2}}}\right)$$

$$\sim \int_{-2}^2 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

We shall return to most of these concepts — Chebychev, Chernoff, WLLN, CLT — again later when we discuss random variables with density.

Just to give you a feeling that you already know a similar convergence result, let me mention the following. Suppose X_n is a point selected at random from among $\{1/n, 2/n, \dots, n/n\}$. Fix two numbers a, b with $0 < a < b < 1$. For large n , How much is $P(a < X_n \leq b)$?

Answer: $P(a < X_n \leq b) \rightarrow b - a$. You can easily prove this.

If you denote $f(x) = 1$ for $0 \leq x \leq 1$ and $f(x) = 0$ otherwise, then you can rewrite the above as

$$P(a < X_n \leq b) \rightarrow \int_a^b f(x) dx.$$

Even though there is much that can be discussed about discrete experiments and random variables, we shall postpone to the third act of the course. Now we enter the second act, to model and understand more general experiments.

Experiments with tooooo many outcomes:

We have been discussing experiments/rvs that take countably many values, usually called discrete rvs. We shall now discuss experiments that have toooooo many outcomes and rvs that take toooooo many values.

For example, when you want to model life time of this bulb, any number $0 < x < \infty$ is a possible value. The same when you want to model the time between two disintegrations of a radio active material. That is, imagine a radio active material placed near a Geiger counter. As soon as a particle is emitted, the counter beeps. Consider the time between two beeps. It is not fixed. This can be anything in $(0, \infty)$.

Philosophy is same in analyzing these experiments. First you write down all possible outcomes: Ω , this may be $(0, \infty)$ or R or $(0, 1)$ etc. Because there are many many outcomes we can not imitate earlier simple minded modelling and try to assign probabilities to outcomes and define probability of an event as sum of probabilities of outcomes in that event. Think about it. The plan is to assign ‘at one stroke’ probabilities for events. Ideally $P(A)$ denotes the probability of ending up with an outcome in A when you perform the experiment. So what are events? Any subset of Ω as earlier? There are technical problems and I do not want to divert your attention now (shall do

so later when I think you are able to follow).

As usual a random variable is a measurement — associates with every outcome a real number. In other words it is a real valued function X defined on Ω . If I am not going to tell you about sample space and events etc, how are you going to answer questions about X ? Well we devise a mechanism which allows you to answer. That mechanism is ‘density function’.

We say $f : R \rightarrow [0, \infty)$ is a density function if $f \geq 0$ and $\int_R f(x)dx = 1$. We say that a random variable X obeys this density if for every $a < b$ the following is true: $P(a < X \leq b) = \int_a^b f(x)dx$

Thus we are assuming our functions are Riemann integrable. Do not worry, we shall not cook up esoteric density functions, most will be continuous on an interval (or union of few intervals) and zero outside. Thus a rv X obeys a density f means if you ever want to calculate probability that values of X lie in an interval, you just calculate area under f over that interval. In the discrete case we did exactly this. Once we had probability mass function, to calculate probability that X is in a set, we ‘summed’ values of mass function over this set. Pause and think.

You already came across two density functions.

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

$$f(x) = 1, \quad 0 < x < 1 \quad f(x) = 0 \quad x \notin (0, 1)$$

These are clearly non-negative and from analysis you know their integral over R is one.

We are now analyzing experiments with tooo many outcomes. As usual Sample space is still the set of all possible outcomes of the experiment (what else can it be!). For example if you pick a number at random from the interval $(0, 1)$ then Ω is the set of all numbers, x with $0 < x < 1$. If you are waiting for the Gieger counter to beep then Ω is the set of all numbers x with $0 < x < \infty$. Unfortunately we can not imitate the discrete set up and try to associate a number to each outcome and proceed. For instance for a number picked at random from the interval $(0, 1)$ each outcome should have the same probability and then it has to be zero, because if it is $1/1000$ then an event consisting of 1001 points would have probability larger than one. It necessitates following a different route. We shall not discuss the details now, shall do so later. We will have to associate probability $P(A)$ for each event A at one stroke. We see later that we also need to modify our concept of events. We assume the following for now.

Theorem:

(i) $P(\emptyset) = 0$ and $P(\Omega) = 1$.

(ii) For a sequence of disjoint events $P(\cup A_n) = \sum P(A_n)$

(iii) If $A_n \uparrow A$ then $P(A_n) \uparrow P(A)$; if $A_n \downarrow A$ then $P(A_n) \downarrow P(A)$.

Here $A_n \uparrow$ means $A_n \subset A_{n+1}$ for all n and $A_n \uparrow A$ means $A_n \uparrow$ and $\cup A_n = A$. Similarly, $A_n \downarrow$ means $A_{n+1} \subset A_n$ for all n and $A_n \downarrow A$ means $A_n \downarrow$ and $\cap A_n = A$.

Are these new rules? No, we had them earlier too, but we only noted (i) and finite additivity (and also the inclusion-exclusion formula). since we did not need others we did not spell out. Let us see why they are true in case of discrete experiments we have been considering so far.

Suppose (A_n) is a sequence of disjoint events. Since $\cup_1^k A_n \subset \cup A_n$ we see by finite additivity, $\sum_1^k P(A_n) = P(\cup_1^k A_n) \leq P(\cup A_n)$. As a consequence, $\sum_1^\infty P(A_n) \leq P(\cup A_n)$ (*).

On the other hand, since $P(\cup A_n)$ is sum of probabilities of outcomes in $\cup A_n$ and an infinite sum is limit of increasing partial sums, we conclude the following. Given any $\epsilon > 0$ there are finitely many outcomes $\omega_1, \dots, \omega_l$ such that $\sum_1^l P(\omega_i) > P(\cup A_n) - \epsilon$. But then you can find N so that these finitely many points are in one of $A_1, \dots, A_N$. Thus $\sum_1^N P(A_n) \geq \sum_1^l P(\omega_i) > P(\cup A_n) - \epsilon$. Thus for every $k > N$ we have $\sum_1^k P(A_n) > P(\cup A_n) - \epsilon$, thus $\sum_1^\infty P(A_n) \geq P(\cup A_n) - \epsilon$. This being true for every $\epsilon > 0$, $\sum_1^\infty P(A_n) \geq P(\cup A_n)$ (**). Thus (*) and (**) show (ii).

(iii) is also proved in a similar way. Since A_n is increasing $P(A_n)$ increases and so first of all it has a limit. Since $A_n \subset \cup A_k$ for each n , we see $P(A_n) \leq P(\cup A_k)$ and hence $\lim P(A_n) \leq P(\cup A_k)$ (*). On the other hand given $\epsilon > 0$ we can find many outcomes $\omega_1, \dots, \omega_l \in \cup A_k$ such that $\sum_1^l P(\omega_l) \geq P(\cup A_k) - \epsilon$. Since all these outcomes must be in some A_N , we see $P(A_N) \geq P(\cup A_k) - \epsilon$ and hence $\lim P(A_n) \geq P(\cup A_k) - \epsilon$. Thus $\lim P(A_n) \geq P(\cup A_k)$ (**). Thus as earlier (*) and (**) complete proof. To prove the decreasing case just note that if $A_n \downarrow A$ then $A_n^c \uparrow A^c$.

A random variable is, as earlier, a measurement. That is, associates with each outcome a number. Thus it is a function $X : \Omega \rightarrow R$. There is one extra condition which does not bother us now. Our data is random variable along with its density function and we answer questions about the rv using the density function it obeys.

A density function or probability density function (pdf) f is a non-negative function on R with $\int_R f(x)dx = 1$. We say a rv X obeys a pdf f and write $X \sim f$ if

$$P(a < X \leq b) = \int_a^b f(x)dx \quad a, b \in R \quad a < b$$

In a sense this is same the discrete case. There we had ‘probability mass function’ and for any subset A of sample space its probability $P(A)$ is the sum pmf at those points. Here it is the pdf and probability of an event like $(a < X < b)$ is the integral of pdf. thus calculating probabilities reduces to calculating areas.

Convention: If we prescribe f over a set A and do not say anything for points outside, then it is to be understood that $f(x) = 0$ for $x \in A^c$.

If $X \sim f$, we define its mean/expectation as $E(X) = \mu_X = \int xf(x)dx$ provided $\int |x|f(x)dx$ is finite. More generally, n -th moment of X is defined by $E(X^n) = \int x^n f(x)dx$ provided the integral $\int |x|^n f(x)dx$ is finite. Variance of X is defined as $\sigma_X^2 = E(X^2) - [E(X)]^2$.

Thus again, this is essentially same as the discrete case. There we multiply value probability and add, to get expected value. Now we multiply value with density and integrate.

Example 1: Normal densities

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad -\infty < x < \infty$$

This is called standard pdf and a rv X obeying this density is called standard normal variable denoted $X \sim N(0, 1)$. It is very easy to see for such a rv we have $E(X) = 0$ and variance one.

More generally, Fix a number $\mu \in R$ and $\sigma > 0$.

$$g(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-(x-\mu)^2/(2\sigma^2)} \quad -\infty < x < \infty$$

This is called $N(\mu, \sigma^2)$ density. If $X \sim N(\mu, \sigma^2)$ then it is easy to show $E(X) = \mu$ and variance of X is σ^2 . This appears in several contexts. We have already seen its appearance in CLT. This is a good model for errors.

Example 2: Uniform densities:

$$f(x) = 1; \quad 0 < x < 1$$

Remember according to our convention, the above definition implies $f(x) = 0$ if $x \notin (0, 1)$. This is called uniform $(0, 1)$ density if X has this pdf we say $X \sim \text{unif}(0, 1)$. For such an X , we see $E(X) = 1/2$ and variance equals $1/12$.

More generally fix numbers $a < b$

$$g(x) = \frac{1}{b-a} \quad a < x < b$$

is a pdf called uniform (a, b) density a rv with this pdf is called $\text{unif}(a, b)$ random variable. If $X \sim g$ then $E(X) = (a+b)/2$ and variance equals $(b-a)^2/12$. This corresponds to picking a number at random from the interval (a, b) . This is because if you take two intervals I, J of **same length** contained in (a, b) , then $P(X \in I) = P(X \in J)$.

Thus in particular a $\text{unif}(0, 1)$ variable is model for a point picked at random from the unit interval. You can manufacture an infinite sequence of independent Bernoulli rvs — each taking values zero/one with equal probability. Let X be such a uniform $(0, 1)$ variable. Thus X is a function defined on a sample space Ω with values in $(0, 1)$ such that for any interval $I \subset (0, 1)$; $P(X \in I)$ is length of I .

Define $Z_1(\omega)$ on the same sample space as the first binary digit of $X(\omega)$. Thus $Z_1(\omega) = 0$ or one according as $X(\omega)$ is in $(0, 1/2]$ or is in $(1/2, 1)$. (We are considering non-terminating binary expansion). It is immediate that Z_1 takes values zero/one with equal probabilities.

Define $Z_2(\omega)$ on the same sample space as the second binary digit of $X(\omega)$.

Thus $Z_2(\omega)$ is zero or one according as $X(\omega) \in (0, 1/4] \cup (1/2, 3/4]$ or not. Thus again Z_2 is Bernoulli. Moreover, it is clear that Z_1, Z_2 are independent:

$$P(Z_1 = \epsilon_1, Z_2 = \epsilon_2) = \frac{1}{4} = P(Z_1 = \epsilon_1)P(Z_2 = \epsilon_2) \quad \epsilon_i = 0/1$$

In general, for $n \geq 1$, if you define $Z_n(\omega)$ on the same sample space as the n -th binary digit of $X(\omega)$ then $Z_1, Z_2, \dots, Z_n$ are independent Bernoulli 0/1 variables with probabilities $1/2$ and $1/2$. Since this can be continued forever, we have thus constructed an infinite sequence of independent Bernoulli variables — on the same sample space where we had our $\text{Unif}(0, 1)$ variable.

exponential densities:

Consider the following function: $f(x) = e^{-x}$ for $x > 0$. Thus $f(x) = 0$ for $x \leq 0$. you can see area under the curve is one. This density function is called **exponential density**, more precisely $\exp(1)$ density function and a random variable having this density is called **exponential random variable**, more precisely, exponential rv with parameter 1; $X \sim \exp(1)$ random variable.

If X is such a random variable then $P(X \leq 0) = 0$ where as for any $0 \leq a < b$, we have

$$P(a < X < b) = e^{-a} - e^{-b}.$$

More generally fix $\lambda > 0$. Consider

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0; \quad f(x) = 0, \quad x \leq 0.$$

This is called $\exp(\lambda)$ **density** and a random variable having this density is called $\exp(\lambda)$ **random variable**. Here $\lambda > 0$ is a parameter and so this is called exponential variable with parameter λ .

This density is useful in modelling life time of electric bulbs. This is also a good model for inter-disintegration time for radio active material. This is also good model for inter arrival times of customers at a service station.

Gamma densities:

Fix $a > 0$. Let

$$\varphi(x) = \frac{1}{\Gamma(a)} e^{-x} x^{a-1} \quad x > 0; \quad \varphi(x) = 0; \text{ for } x \leq 0$$

From Analysis we know that $\int f(x)dx = 1$ and thus this is a density function. This is called **Gamma density**, more precisely gamma density with parameter a . A random variable which obeys this density is called a **gamma random variable**.

It is instructive to sketch the curves $f(x)$ for $a = 1/2$; for $a = 1$; for $a = 2$. Integration by parts immediately gives that $\Gamma(a+1) = a\Gamma(a)$. This will immediately give $\Gamma(n) = (n-1)!$; $n \geq 1$.

Beta densities:

Fix $a > 0, b > 0$. Let

$$\varphi(x) = \frac{1}{\beta(a, b)} x^{a-1}(1-x)^{b-1} \quad 0 < x < 1; \quad \varphi(x) = 0; \quad \text{if } x \notin (0, 1).$$

it is called beta density with parameters a and b . A random variable obeying this density is called **beta random variable**.

It is instructive to sketch these curves for $a = 1/2, b = 1/2$; for $a = 1, b = 1$; for $a = 2, b = 1/2$; for $a = 1/2, b = 2$; for $a = 2, b = 3$.

The gamma and beta functions arise in several contexts, not only probability but also in Physics, number theory, differential equations, special functions and so on. This is a good model for proportion. Thus if you have a coin whose chance of heads is p ($0 < p < 1$), then we may not be totally ignorant and may have some idea, like p is close to zero/close to one/can be anything etc. Then beta distribution will help is model our belief. This is essentially useful in 'Bayesian inference', which we shall not discuss.

Double exponential density:

$$f(x) = \frac{1}{2}e^{-|x|} \quad -\infty < x < \infty$$

Then $\int f = 1$. This is called double exponential or Laplace density. A rv having this density is called a double exponential rv or Laplace variable.

Let us discuss one subtle point about densities.

$$f(x) = 1; \quad x \in (0, 1) \quad \text{and} \quad f(x) = 0 \quad x \notin (0, 1)$$

$$g(x) = 1; \quad x \in (0, 1] \quad \text{and} \quad g(x) = 0 \quad x \notin (0, 1]$$

$$h(x) = 1; \quad x \in [0, 1] \quad \text{and} \quad h(x) = 0 \quad x \notin [0, 1]$$

We see that for any interval (a, b) we have

$$\int_a^b f(x)dx = \int_a^b g(x)dx = \int_a^b h(x)dx$$

Thus all these functions appear eligible to be uniform(0,1) densities. Yes right. We shall not distinguish. Same is the case with the function

$$\varphi(x) = 1; \quad x \in [0, 1/2) \cup (1/2, 1] \quad \text{and} \quad \varphi(x) = 0 \quad x \notin [0, 1/2) \cup (1/2, 1]$$

However we shall not consider such functions when good functions like the one above are available. here good means continuous on an interval and zero outside.

Same applies for standard normal density. For example

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad -\infty < x < \infty$$

$$h(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad h \neq 0; \quad h(0) = 20,000$$

Easy to see that for any interval (a, b) we have

$$\int_a^b f(x)dx = \int_a^b h(x)dx$$

However we shall never consider h as normal density simply because we have an equivalent better function f . This is a subtle point with densities.

Change of variables, dimension one:

Suppose X is a random variable with pdf f . let $\mu \in R$. Set $Y = X + \mu$. Then the density of Y is given by

$$g(y) = f(x - \mu); \quad x \in R$$

This follows from observing that

$$\begin{aligned} P(Y \leq a) &= P(X + \mu \leq a) = P(X \leq a - \mu) \\ &= \int_{-\infty}^{a-\mu} f(x)dx = \int_{-\infty}^a f(x - \mu)dx. \end{aligned}$$

where we changed the variable for the last equality. Thus by subtractions. This shows that g is density of Y . More generally, suppose that X has density f . Let $\mu \in R$ and $\sigma > 0$ be numbers. Set $Y = \sigma X + \mu$. then density of Y is given by

$$g(y) = \frac{1}{\sigma} f\left(\frac{y - \mu}{\sigma}\right).$$

This can be seen exactly as above. Many-a-times we wish to have density of a function of X , when we know density of X . For example you may know pdf of velocity and want to know pdf of kinetic energy; or you pick a number X at random from $(0, 1)$ and make a square with this side and want to know pdf of the area of the square. This problem has. a simple solution which we

already solved in analysis.

■ Suppose $\Omega = (a, b)$ is an open interval and φ is a function on this interval onto an interval $\Omega^* = (c, d)$. To fix ideas and not to get worried about irrelevant things, imagine that these are bounded intervals (though they need not be). Assume:

(i) φ is C^1 , this means, it is differentiable and its derivative φ' is a continuous function on Ω .

(ii) φ' is non-singular, that is $\varphi'(x) \neq 0$ for all $x \in \Omega$.

(iii) φ is a one-one function on Ω onto Ω^* .

Then

(A) φ has inverse $\psi : \Omega^* \rightarrow \Omega$ which is C^1 .

(B) Let X be a random variable with pdf f which is zero outside Ω . Let Y be the random variable defined by $Y(\omega) = \varphi(X(\omega))$ for any sample point ω . Then Y has density given by $g(y) = f(\psi(y))|\psi'(y)|$ for $y \in \Omega^*$ and $g(y) = 0$ for $y \notin \Omega^*$. ■

This is precisely substitution rule. First of all $f = 0$ on Ω^c implies that $P(X \in \Omega^c) = 0$ and thus Y is well defined. Remember φ is defined only on Ω . Think about this carefully. Observe for any interval $I \subset \Omega^*$ we have by the substitution rule

$$P(Y \in I) = P(X \in \psi^{-1}(I)) = \int_{\psi^{-1}(I)} f = \int_I f \circ \psi |\psi'| = \int_I g$$

and hence g is pdf of Y . ■

First, I said you can imagine bounded intervals, simply because integration over unbounded intervals is defined via bounded intervals increasing to that unbounded interval. We do apply this result for unbounded intervals when needed.

Second, I used substitution rule which we knew only for continuous functions etc (remember we are using only Riemann integration). We do deal with densities which are continuous or at least 'piecewise continuous'.

Third point is the following. having said that φ' is continuous and never zero why did we say that the function φ is one-one. is this not a consequence? yes. By continuity, you see that φ' is either through out strictly positive or through out strictly negative. Hence it has to be one-one. I am looking into the future and did not want to take advantage of the fact that we are on $\mathbb{R}$. One trouble with calculus is that you have a luxurious life in one dimension. If you do not accept those luxuries, you see that multi-variable results are exactly the same as one variable case.

Joint distributions:

Exactly as in the discrete case if you have two random variables and if you look at their densities separately, you will not have any idea of relations that may be existing among the variables. One should look at their joint distribution.

A density function or probability density function (pdf) on R^2 f is a non-negative function on R^2 with $\int_{R^2} f(x, y) dx dy = 1$. We say a pair of rvs (X, Y) obeys a pdf f and write $(X, Y) \sim f$ if

$$P(a < X \leq b, c < Y \leq d) = \int_{y=c}^d \int_{x=a}^b f(x, y) dx dy \quad a < b, c < d$$

Here

$$\int_{R^2} f(x, y) dx dy = \int_y \left[\int_x f(x, y) dx \right] dy$$

integration: Two dimensions

To discuss joint densities, we need integration of functions of two variables. But, for all the functions we come across in this course, you need not wait till analysis develops these tools. We can use our knowledge of integration of one variable functions and no more is needed. If $f(x, y)$ is a real valued function on R^2 we define

$$\int_{R^2} f(x, y) dx dy = \int \left[\int f(x, y) dy \right] dx = \int \left[\int f(x, y) dx \right] dy$$

For fixed x we are integrating the function of one variable $y \mapsto f(x, y)$. You get a number depending on the x you fixed, name this number $h(x)$. Now you integrate this function of one variable to get a number $\int h(x) dx$. This number is by definition $\int_{R^2} f(x, y) dx dy$.

Several questions need to be answered.

- Is the function $y \mapsto f(x, y)$ integrable for every x ? If not what should I do for those x ? Yes, this can happen, but in our course it does not happen.
- Once you integrate and get the number $h(x)$, is this function of x integrable? Yes, this may not be integrable; but in our course it does not happen and $h(x)$ is integrable..
- In the display above we said this is also same as integrating the function $x \mapsto f(x, y)$ for each fixed y and integrating the resulting function. is it always true? No not always, but in our course, for the functions that we handle, there is no problem. This interchange of order of integration can be done.
- Why those big brackets? Should we always put? No, you need not. The brackets are only to convey to you what we are doing first and what we are doing next. There is no need to put brackets. Sometimes you integrate only on some subsets instead of on all of R^2 . For example you may want to integrate f on the rectangle $S = \{(x, y) : 0 \leq x \leq 8; 20 \leq y \leq 30\}$. Thus

$$\int_{20}^{30} \int_0^8 f(x, y) dx dy = \int_{20}^{30} \left[\int_0^8 f(x, y) dx \right] dy$$

Putting brackets will help not to confuse between limits of x and limits of y . another way of writing is

$$\int_{y=20}^{30} \int_{x=0}^8 f(x, y) dx dy = \int_{x=0}^8 \int_{y=20}^{30} f(x, y) dy dx$$

- should we prove existence of integrals etc? No you need not prove existence or non-existence in this course.
- are we then being rigorous? No, not really, but certain things have to wait till you pick up ‘rules’ from analysis to be used for justification. However whatever we are doing can be justified, if you pick up those laws. This has been done. Only thing is, I am not discussing that in this course.
- Does Analysis teacher follow the above definition of integration for functions of two variables? Most probably not. However, for the functions we discuss in our course, whatever definition of integration you learn, it agrees with what we are doing. Though such a statement is not always true for functions in general. For the simple functions we come across in this course, no problem.

For example if you want to integrate a function f on the open unit disc $S = \{x^2 + y^2 < 1\}$ you recognize

$$\begin{aligned} S &= \{(x, y) : -1 < y < 1; -\sqrt{1-y^2} < x < +\sqrt{1-y^2}\} \\ &= \{(x, y) : -1 < x < 1; -\sqrt{1-x^2} < y < +\sqrt{1-x^2}\} \end{aligned}$$

Thus we calculate

$$\int_{y=-1}^1 \int_{x=-\sqrt{1-y^2}}^{\sqrt{1-y^2}} f(x, y) dx dy = \int_{x=-1}^1 \int_{y=-\sqrt{1-x^2}}^{\sqrt{1-x^2}} f(x, y) dy dx$$

Let us integrate the function $f = 1$ on the unit disc. remember if you integrate the function $g = 1$ on an interval, you get length of that interval.

$$\int_{y=-1}^1 \int_{x=-\sqrt{1-y^2}}^{\sqrt{1-y^2}} 1 dx dy = \int_{y=-1}^1 2\sqrt{1-y^2} dy = 4 \int_0^1 \sqrt{1-y^2} dy = \pi$$

area of the unit disc.

change of variable, dimension two:

Exactly the same result as in dimension one holds in two dimensions too.

Suppose $\Omega \subset R^2$ is an open region (open connected set) and φ is a function on this region onto a region Ω' . To fix ideas and not to get worried about irrelevant things, as earlier, imagine that these are bounded regions (though they need not be).

Let us clearly understand that now φ associates with every point of Ω a point of Ω' ; which means there are two real valued functions $\varphi_1(x_1, x_2)$ and $\varphi_2(x_1, x_2)$ on Ω such that

$$\varphi(x_1, x_2) = (\varphi_1(x_1, x_2), \varphi_2(x_1, x_2)).$$

In other words $\varphi_1(x_1, x_2)$ is the first coordinate of the point $\varphi(x_1, x_2)$ whereas $\varphi_2(x_1, x_2)$ is the second coordinate of $\varphi(x_1, x_2)$. That is all. Note φ_1 and φ_2 are real valued functions on Ω .

We assume three things about φ as earlier.

(i) φ is continuously differentiable. This means the following. The real valued function φ_1 has both partial derivatives

$$\frac{\partial \varphi_1}{\partial x_1}; \quad \frac{\partial \varphi_1}{\partial x_2}$$

at each point of Ω and they are continuous (real valued) functions on Ω . Similarly the real valued function φ_2 has both partial derivatives

$$\frac{\partial \varphi_2}{\partial x_1}; \quad \frac{\partial \varphi_2}{\partial x_2}$$

at each point of Ω and they are continuous (real valued) functions on Ω . We denote

$$\varphi'(x_1, x_2) = \begin{pmatrix} \frac{\partial \varphi_1}{\partial x_1}(x_1, x_2) & \frac{\partial \varphi_1}{\partial x_2}(x_1, x_2) \\ \frac{\partial \varphi_2}{\partial x_1}(x_1, x_2) & \frac{\partial \varphi_2}{\partial x_2}(x_1, x_2) \end{pmatrix}$$

Thus now φ' at each point of Ω is a 2×2 matrix.

(ii) φ' is non-singular, that is $\varphi'(x_1, x_2)$ is a nonsingular matrix for all points $(x_1, x_2) \in \Omega$. remember this is same as saying that its determinant is non zero.

(iii) φ is a one-one function on Ω onto Ω' .

Then conclusion:

(A) φ has an inverse $\psi : \Omega' \rightarrow \Omega$ which is again continuously differentiable. In other words if

$$\psi(y_1, y_2) = (\psi_1(y_1, y_2), \psi_2(y_1, y_2))$$

then both partial derivatives of each of these ψ_1 and ψ_2 are continuous functions.

We denote

$$\psi'(y_1, y_2) = \begin{pmatrix} \frac{\partial \psi_1}{\partial y_1}(y_1, y_2) & \frac{\partial \psi_1}{\partial y_2}(y_1, y_2) \\ \frac{\partial \psi_2}{\partial y_1}(y_1, y_2) & \frac{\partial \psi_2}{\partial y_2}(y_1, y_2) \end{pmatrix}$$

We denote $|\psi'|$ to be the modulus of the determinant of the 2×2 matrix ψ' .

(B). Let now f be a nice real valued function on Ω , for example bounded continuous function. Define the composed function $g(y) = f(\psi(y))$ on Ω' . Then the following holds:

$$\int_{\Omega} f(x) dx = \int_{\Omega'} g(y) |\psi'(y)| dy. \quad (\spadesuit)$$

That is

$$\int_{\Omega} f(x_1, x_2) dx_1 dx_2 = \int_{\Omega'} g(y_1, y_2) |\psi'(y_1, y_2)| dy_1 dy_2. \quad (\spadesuit)$$

This is called change of variable formula or substitution rule or Jacobian formula. ψ' is called the Jacobian matrix.

You may find the change of variable formula to be a little boring, because I made the statement longer than what it is. This is just to make clear to you what exactly the hypothesis is and the meaning of the terms involved. Remember, we shall not prove everything we use. But we should be clear about what we are using. You can not say ‘by such and such a theorem’ without even knowing the statement of the theorem. You are excused if you do not know the proof. But you must know the meaning of what you are saying.

OK, what is the use of the change of variable? We shall see two.

Normal integral again:

Let

$$\int_{-\infty}^{\infty} e^{-x^2/2} dx = a$$

Consider R^2 and the function $f(x, y) = e^{-(x^2+y^2)/2}$. Then

$$\int \int f dy dx = \int \left[\int e^{-x^2/2} e^{-y^2/2} dy \right] dx = \int a e^{-x^2/2} dx = a^2 \quad (\spadesuit)$$

If you denote

$$\Omega = R^2 \setminus \{(x, y) : y = 0, x \geq 0\} \quad \Omega^* = (0, \infty) \times (0, 2\pi)$$

$$\begin{aligned}\varphi(x, y) &= (\sqrt{x^2 + y^2}, \tan^{-1}(y/x)) = (r, \theta) \quad \text{say} \\ \psi(r, \theta) &= (r \cos \theta, r \sin \theta)\end{aligned}$$

Then easy to see that the conditions of the theorem are satisfied. Ω, Ω^* are open connected sets. φ is one-one and onto; in fact ψ is its inverse. Note that $\tan^{-1}(y/x)$ is the angle the line joining $(0, 0) \rightarrow (x, y)$ makes with positive x axis. It is between 0 to $\pi/2$ in the first quadrant; between $\pi/2$ to π in the second quadrant etc. Since we removed the positive x axis, this is a continuous function. It is also C^1 , better way to see this is to see ψ is so and hence so is its inverse φ . Further

$$\psi'(r, \theta) = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$$

So change of variable tells us

$$\int_{\Omega} f = \int_0^{2\pi} \int_0^{\infty} v e^{-r^2/2} r dr d\theta = \int_0^{2\pi} 1 d\theta = 2\pi \quad (\clubsuit)$$

Observe that, integrating w.r.t. y on $(-\infty, \infty)$ is same as integrating on $(-\infty, 0) \cup (0, \infty)$ and so

$$\int_{\Omega} f = \int \int f dy dx$$

Now $(\spadesuit)$ and $(\clubsuit)$ give $a^2 = 2\pi$ or $a = \sqrt{2\pi}$ a result we saw in analysis already by just using high school methods but longer and uses Walli's product.

Gamma/beta:

We shall now prove

$$\beta(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}; \quad \text{i.e.} \quad \Gamma(a)\Gamma(b) = \Gamma(a+b)\beta(a, b) \quad (\clubsuit)$$

Take

$$\begin{aligned}\Omega &= (0, \infty) \times (0, \infty); & \Omega' &= (0, \infty) \times (0, 1) \\ \varphi(x_1, x_2) &= (x_1 + x_2, \frac{x_1}{x_1 + x_2}) = (y_1, y_2); & \psi(y_1, y_2) &= (y_1 y_2, y_1(1 - y_2))\end{aligned}$$

$$\psi'(y_1, y_2) = \begin{pmatrix} y_2 & y_1 \\ (1 - y_2) & -y_1 \end{pmatrix} \quad |\psi'| = y_1$$

$$f(x_1, x_2) = e^{-x_1} x_1^{a-1} e^{-x_2} x_2^{b-1}; \quad g(y_1, y_2) = e^{-y_1} y_1^{a-1} y_2^{a-1} y_1^{b-1} (1 - y_2)^{b-1}$$

$$\iint f = \Gamma(a)\Gamma(b); \quad \iint g|\psi'| = \Gamma(a+b)\beta(a,b)$$

Change of variable formula shows ($\clubsuit$).

Transformation of densities:

Here is analogue of the result in one dimension concerning change of densities for functions of random variables.

Suppose (X_1, X_2) have joint density given by $f(x_1, x_2)$ which is zero for points not in an open region Ω . Suppose $\varphi : \Omega \rightarrow \Omega'$ as in the change variable theorem satisfying those three conditions. Then $g(y_1, y_2)|\psi'(y_1, y_2)|$ for points in Ω' and zero for points not in Ω' is joint density of $(Y_1, Y_2) = \varphi(X_1, X_2)$.

The justification is simple. First observe that since $(X_1, X_2) \in \Omega$ we conclude that $(Y_1, Y_2) = \varphi(X_1, X_2)$ takes values in Ω^* . Need to show $P\{(Y_1, Y_2) \in S^*\} = \int_{S^*} g(y_1, y_2)|\psi'(y_1, y_2)|dy_1dy_2$ for any rectangle $S^* \subset \Omega^*$. Let us denote $S = \psi(S^*)$ so that $\varphi(S) = S^*$. First observe that $(X_1, X_2) \in S$ iff $(Y_1, Y_2) \in S^*$. The change of variable formula above tells $\int_S f dx_1 dx_2 = \int_{S^*} g|\psi'| dy_1 dy_2$. Here we used the fact mentioned after the definition of pdf: Even though definition only says that $\int_{\square} f = P\{(X_1, X_2) \in \square\}$ for rectangles $\square$, one can prove that this holds for more general sets, in particular for all regions.

marginal densities:

If (X, Y) have joint density $f(x, y)$ we define the marginal densities by

$$f_1(x) = \int f(x, y)dy \text{ for } x \in R; \quad f_2(y) = \int f(x, y)dx \text{ for } y \in R$$

Clearly f_1 is pdf of X and f_2 is pdf of Y . Indeed

$$\begin{aligned} P(a < X \leq b) &= P(a < X \leq b, -\infty < Y < \infty) = \int_{x=a}^b \int_{y \in R} f(x, y)dydx \\ &= \int_a^b f_1(x)dx. \end{aligned}$$

if you feel uncomfortable using unbounded interval for Y , write the same thing for $(a < X \leq b, -n < Y \leq +n)$ and then take limit of this equation as $n \uparrow \infty$. Note that by definition of integral

$$\int_{-n}^n f(x, y)dy \rightarrow \int_{-\infty}^{\infty} f(x, y)dy$$

and by properties of probability we have

$$P(a < X < b, -n < Y \leq n) \uparrow P(a < X < b, -\infty < Y < \infty)$$

Thus marginal densities are nothing but densities of these variables X and Y respectively.

Independence:

Imitating the definition in the discrete case, we define independence of two rvs having densities. Recall that two discrete rvs are independent if their joint pmf is product of individual pmfs, that is, $P(X = i, Y = j) = P(X = i)P(Y = j)$.

Definition: X_1, X_2 defined on a space are independent if their joint density equals product of marginals, that is, $f(x_1, x_2) = f_1(x_1)f_2(x_2)$. equivalently, the function $g(x_1)h(x_2)$ is a joint density of (X_1, X_2) where g and h are densities of X and Y respectively.

This is same as saying for any intervals (a, b) and (c, d)

$$P\{X_1 \in (a, b); X_2 \in (c, d)\} = P\{X_1 \in (a, b)\} \times P\{X_2 \in (c, d)\}$$

This is also same as saying for any two numbers u, v

$$P(X \leq u, Y \leq v) = P(X \leq u) \times P(Y \leq v)$$

The calculation $\Gamma(a)\Gamma(b) = \Gamma(a+b)\beta(a, b)$ done above can be easily converted to show, painlessly, the following:

If $X \sim \Gamma(a)$ and $Y \sim \Gamma(b)$ are independent, then $X + Y \sim \Gamma(a + b)$ and $Y/(X + Y) \sim \beta(a, b)$ and these are again independent.

change of densities: orthogonal transform

suppose that X_1, X_2 are independent standard normal. Let

$$Y_1 = \frac{X_1 + X_2}{\sqrt{2}}; \quad Y_2 = \frac{X_1 - X_2}{\sqrt{2}}.$$

By independence we see that joint density of (X_1, X_2) is

$$f(x_1, x_2) = \frac{1}{2\pi} e^{-(x_1^2 + x_2^2)/2}$$

Here $\Omega = R^2 = \Omega'$.

$$\varphi_1(x_1, x_2) = \frac{x_1 + x_2}{\sqrt{2}}; \quad \varphi_2(x_1, x_2) = \frac{x_1 - x_2}{\sqrt{2}}$$

$$\begin{aligned}\psi_1(y_1, y_2) &= \frac{y_1 + y_2}{\sqrt{2}}; & \psi_2(y_1, y_2) &= \frac{y_1 - y_2}{\sqrt{2}} \\ \psi'(y_1, y_2) &= \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}; & |\psi'| &= 1\end{aligned}$$

Finally

$$h(y_1, y_2) = \frac{1}{2\pi} e^{-(y_1^2 + y_2^2)/2}$$

Thus Y_1, Y_2 are independent standard normal again.

In fact we can take any orthogonal transformation: $Y = AX$ where $AA^t = A^tA = I$. For example

$$Y_1 = X_1 \cos 17^\circ + X_2 \sin 17^\circ; \quad Y_2 = -X_1 \sin 17^\circ + X_2 \cos 17^\circ$$

are also standard normal and are independent

Just as in one dimension we have the following special case of transformations.

Change of densities: translation/scaling

Suppose $X = (X_1, X_2)$ has joint density $f(x) = f(x_1, x_2)$. Let $\mu = (\mu_1, \mu_2) \in R^2$. Define $Y = (Y_1, Y_2) = (X_1 + \mu_1, X_2 + \mu_2)$. Then density of Y is given by $g(y_1, y_2) = f(x_1 - \mu_1, x_2 - \mu_2)$.

More generally, suppose A is a non-singular 2×2 matrix and $Y = AX + \mu$ then Y has density $g(y) = f(A^{-1}(y - \mu))|\det A^{-1}|$. You only need to see that $\varphi(x) = Ax + \mu$ is a bijection of R^2 ; it is C^1 , with inverse $\psi(y) = A^{-1}(y - \mu)$ and $\psi' = |\det A^{-1}|$.

In particular, Let us take any nonsingular matrix A . Let X_1, X_2 independent standard normals and $Y = AX + \mu$. Then

$$f(x_1, x_2) = \frac{1}{2\pi} e^{-x^t x / 2}$$

so that density of Y is given by

$$\Phi(y_1, y_2) = \frac{1}{2\pi} e^{-(y-\mu)^t (A^{-1})^t A^{-1} (y-\mu) / 2} |\det A^{-1}|$$

Note: $(A^{-1})^t = (A^t)^{-1}$ and $B^{-1}C^{-1} = (CB)^{-1}$ and $|\det A^{-1}| = 1/|\det A|$. Thus if you denote $\Sigma = (AA^t)$ we see it is symmetric positive definite and $|\det A| = |\det \Sigma|^{1/2}$ we see that the joint density of (Y_1, Y_2) is

$$\Phi(y_1, y_2) = \frac{1}{2\pi |\Sigma|^{1/2}} e^{-(y-\mu)^t \Sigma^{-1} (y-\mu) / 2}; \quad y \in R^2$$

Definition: Let Σ be a symmetric positive definite 2×2 matrix and $\mu \in R^2$. Then

$$\Phi(y_1, y_2) = \frac{1}{2\pi|\Sigma|^{1/2}} e^{-(y-\mu)^t \Sigma^{-1} (y-\mu)/2}, \quad y \in R^2$$

is a pdf called bivariate normal density $N_2(\mu, \Sigma)$ and if $Y = (Y_1, Y_2)$ has this density we say Y is bivariate normal with mean vector μ and covariance matrix Σ . We express: $Y \sim N_2(\mu, \Sigma)$

There is only one subtle point, we started with non-singular A and arrived at this Σ above. But is every non-singular symmetric positive definite matrix Σ like this? Yes In fact by spectral theorem if the given symmetric positive definite matrix is $\Sigma = UDU^{-1}$ then $A = U\sqrt{D}U^{-1}$ would do.

Expectation:

Let (X, Y) have joint density $f(x, y)$. Let $\varphi(x, y)$ be a real valued function R^2 . Then we can think of the random variable

$$Z = \varphi(X, Y)$$

Again, this means that for a sample point ω , the number $Z(\omega)$ equals $\varphi(X(\omega), Y(\omega))$. Assuming $\int |\varphi(x, y)| f(x, y) dx dy$ is finite we define

$$E[\varphi(X, Y)] = \int \varphi(x, y) f(x, y) dx dy$$

For example if you take $\varphi(x, y) = x$ you see $E(X) = \int x f(x, y) dy dx = \int x f_1(x) dx$ as it should be.

There is a subtle issue that should be answered.

Are we eligible to define expected value this way? No. After all Z is a random variable in its own right. It may probably have a pdf, then if the pdf is $h(z)$ according to our definition of expectation already defined earlier, it should be $\int z h(z) dz$. Unless we are convinced that the above is same as this, we have no business to define as above. The fact is, we can show it is same as the earlier one.

The situation is complicated. Here is the reason. Z may have density, for example when $\varphi(x, y) = x$ then $Z = X$ and has density $f_1(z)$. It is also possible Z may be discrete. For example (X, Y) has density

$$f(x, y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2}$$

$\varphi(x, y) = +1$ if (x, y) is in the first or third quadrant and equals -1 otherwise. Then it is easy to see that $Z = \varphi(X, Y)$ takes ± 1 with probability $1/2$ and

1/2. There is a more serious issue. Z may be neither discrete, nor having a pdf. For example, take (X, Y) as above. suppose $\varphi(x, y) = x$ if $x > 0$ and $\varphi(x, y) = -1$ if $x \leq 0$. Then $P(Z = -1) = P(X \leq 0) = 1/2$ and so Z does not have a pdf. Recall, a rv Z has pdf then for each number a , $P(Z = a) = 0$. Also Z is not discrete because it takes all values in $(0, \infty)$, or to put it differently, you can not show countably many values (z_i) and argue $\sum P(Z = z_i) = 1$. Such random variables are, in a sense, mixed. There are still others which have no pdf, they are not discrete, not mixed either!

If we really want to define expectation in all cases, it needs more machinery and we shall not spend time. that is for an advanced course. You have to accept on faith that there is no clash if we define expected value of $\varphi(X, Y)$ as above. Thus for example if $\int |x|^m |y|^n f(x, y) dx dy$ is finite, then

$$E(X^m Y^n) = \int x^m y^n f(x, y) dx dy$$

Whenever we talk about these quantities, we assume that the required expectations exist.

Covariance is defined as earlier: $Cov(X, Y) = E(XY) - E(X)E(Y)$. Correlation is also as earlier $\rho_{X,Y} = Cov(X, Y)/(\sigma_X \sigma_Y)$.

Of course, as earlier, if $\sigma_X = 0$ or $\sigma_Y = 0$ then ρ is not defined. But interesting point is that this can not happen when the rvs have pdf! Indeed if μ is the mean of X , clearly $\sigma_X^2 = \int (x - \mu)^2 f_1(x) dx$ and the integrand is zero only when $f = 0$ or when $x = \mu$. (So what?)

Conditionals densities/Conditional expectations:

In a sense we are **redoing entire development** of the discrete case, the definitions and the results, for the density case. You should feel that we are doing exactly the same things the same way, but of course you should understand what is meant by 'same' – replace sums by integrals etc.

distribution: In the discrete case p_{ij} is pmf. For any set S of points (ij) we have $P((X, Y) \in S) = \sum_{(ij) \in S} p_{ij}$

In the pdf case $f(x, y)$ is pdf. For any region $\Omega \subset R^2$; we have $P((X, Y) \in \Omega) = \int_{(x,y) \in \Omega} f(x, y) dx dy$.

marginals: In the discrete case, marginal pmf of X is $p_{i\bullet} = \sum_j p_{ij}$. In the pdf case marginal of X is $f_1(x) = \int f(x, y) dy$. Similarly we have marginal

of Y , $p_{\bullet,j}$ and $f_2(y)$.

expectations: In discrete case $E(\varphi(X, Y)) = \sum \varphi(i, j)p_{ij}$. In the pdf case $E(\varphi(X, Y)) = \int \varphi(x, y)f(x, y)dxdy$. (Of course when required sum/integral converges absolutely).

In the pmf case we proved the formula because we could see sample spaces. In the pdf case we have decided not to look at the sample space.

conditionals: Conditional pmf of Y is defined for each i by $p(Y = j | X = i) = p_{ij}/p_{i\bullet}$. Conditional expectation of Y given $x = i$ is $E(Y | X = i) = \sum j p_{ij}/p_{i\bullet}$. In the pdf case the conditional pdf of Y is defined for each x such that $f_1(x) > 0$ by $f_{Y|X}(y) = f(x, y)/f_1(x)$. Conditional pdf of Y given $X = x$ is not defined if $f_1(x) = 0$. Conditional expectation of Y given $X = x$ is $E(Y | X = x) = \int y f(x, y)dy / f_1(x)$.

Similarly conditional pmf or pdf of X is defined and also conditional expectation of X .

$E(Y | X)$ is defined as the following rv Z : when $X = i$, this Z takes the value $E(Y | X = i)$. thus it is a rv. Similarly, in the pdf case $E(Y | X)$ is defined as the following rv Z : when $X = x$, this Z takes the value $E(Y | X = x)$. thus it is a rv.

The formula $E[E(Y | X)] = E(Y)$ is true. We proved in the discrete case. In the pdf case the same proof works.

$$\begin{aligned} E[E(Y | X)] &= \int E(Y | X = x)f_1(x)dx = \int \left[\int y \frac{f(x, y)}{f_1(x)} dy \right] f_1(x)dx \\ &= \int \int y f(x, y) dxdy = \int y f_2(y) dy = E(Y) \end{aligned}$$

Example: (X, Y) has density $f(x, y) = x + y$ for $0 < x < 1, 0 < y < 1$. As always, it is understood that for points outside this set pdf is zero. this is pdf because it is non-negative and

$$\int_0^1 \int_0^1 (x + y) dxdy = \int_0^1 \left(\frac{1}{2} + y \right) dy = \frac{1}{2} + \frac{1}{2} = 1$$

Marginal of X is

$$f_1(x) = \int_0^1 (x + y) dy = x + \frac{1}{2} \quad 0 < x < 1.$$

Conditional pdf of Y given $X = x$ is defined only for $0 < x < 1$ and then for such an x

$$f_{Y|X}(y) = \frac{x+y}{(1/2)+x} \quad 0 < y < 1$$

Conditional expectation of Y given $X = x$ (here $0 < x < 1$) is given by

$$\begin{aligned} E(Y|X=x) &= \int_0^1 y \frac{x+y}{(1/2)+x} dy = \frac{(1/2)x + (1/3)}{(1/2)+x} \\ E[E(Y|X=x)] &= \int_0^1 \frac{(1/2)x + (1/3)}{(1/2)+x} f_1(x) dx \\ &= \int_0^1 [(1/2)x + (1/3)] dx = (1/2)(1/2) + (1/3) = \frac{7}{12} \quad (\bullet) \end{aligned}$$

Let us calculate $E(Y)$

$$\begin{aligned} E(Y) &= \int_0^1 y f_2(y) dy = \int_0^1 y[y + \frac{1}{2}] dy \\ &= \int_0^1 [y^2 + \frac{1}{2}y] dy = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}. \quad (\bullet\bullet) \end{aligned}$$

When you workout such examples, maths is routine, except you should pay attention to the ranges and pay attention to what you are trying to do; is it X or Y etc. It is important for you to understand the discrete case thoroughly and then you can easily master the pdf case.

integration: k -dimensions

Suppose $f(x) = f(x_1, x_2, \dots, x_k)$ is a nice real valued function of k variables. We define its integral as follows. First calculate for each fixed $x_1, \dots, x_{k-1}$:

$$\int_{-\infty}^{\infty} f(x_1, \dots, x_{k-1}, x_k) dx_k = f_{k-1}(x_1, \dots, x_{k-1}); \quad (\text{say})$$

Then for each fixed $(x_1, \dots, x_{k-2})$ calculate

$$\int_{-\infty}^{\infty} f_{k-1}(x_1, \dots, x_{k-2}, x_{k-1}) dx_{k-1} = f_{k-2}(x_1, \dots, x_{k-2}); \quad (\text{say})$$

Continue, continue. to get

$$\int_{-\infty}^{\infty} f_1(x_1) dx_1 = f_0 \quad (\text{say})$$

Observe that each integral above is integral of function of one variable only which we understand. Further f_0 is a number. This number is called integral of f denoted

$$\int f \quad \text{or} \quad \int f(x) dx \quad \text{or} \quad \int_{R^k} f(x) dx \quad \text{or}$$

$$\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \dots) dx_1 dx_2 \cdots dx_k$$

There might be confusion with $\int f(x) dx$, because you may think it is one dimensional integral. So $\int f(\mathbf{x}) d\mathbf{x}$ is used. But we shall not. You ought to know which function you are talking about. If you have to integrate f over a region Ω , then you consider $g = f$ on Ω and $g = 0$ outside Ω and integrate g as above. Thus

$$\int_{\Omega} f = \int_{R^k} g$$

where $g(x) = f(x)$ for $x \in \Omega$ and $g(x) = 0$ for $x \notin \Omega$. As in two dimensions, for functions we ever come across in our course, the order does not matter, you can integrate variables one by one — your convenience!

Example: f is indicator of a cube. That is fix numbers $(a_i \leq b_i : 1 \leq i \leq k)$ Let S be the cube $\prod (a_i, b_i)$ — set of all points $x = (x_1, \dots, x_k) \in R^k$ such

that $a_i < x_i < b_i$ for each i . f is the function which is one on S and zero outside. Execute the above integration. You see volume of the box.

$$\int_S f = \prod (b_i - a_i)$$

Example: Suppose

$$f(x) = \frac{1}{(2\pi)^{k/2}} e^{-x^t x / 2} = \frac{1}{(2\pi)^{k/2}} e^{-\sum x_i^2 / 2}$$

Then

$$\int_{R^k} f = 1$$

you have only to integrate variables one-by-one using normal integral in mind.

Example: $k = 2$ Fix numbers $a > 0, b > 0, c > 0$.

$$f(x, y) = x^{a-1} y^{b-1} (1 - x - y)^{c-1} \quad 0 < x, y, x + y < 1$$

We are following the convention, that $f = 0$ for points (x, y) not satisfying above condition. If you fix any $x \notin (0, 1)$ and integrate you get zero. Let us fix $0 < x < 1$. Then $f(x, y) = 0$ unless $0 < y < 1 - x$ and then

$$\int_0^{1-x} x^{a-1} y^{b-1} (1 - x - y)^{c-1} dy = x^{a-1} \int_0^{1-x} y^{b-1} (1 - x - y)^{c-1} dy$$

Put $y = (1 - x)z$ and change y to z (remember x is fixed, it is not a variable). You see

$$\int_0^{1-x} x^{a-1} y^{b-1} (1 - x - y)^{c-1} dy = \beta(b, c) x^{a-1} (1 - x)^{b+c-1}$$

so that

$$\begin{aligned} \int f &= \beta(b, c) \int_0^1 x^{a-1} (1 - x)^{b+c-1} dx = \beta(b, c) \beta(a, b + c) \\ &= \frac{\Gamma(a) \Gamma(b) \Gamma(c)}{\Gamma(a + b + c)} \end{aligned}$$

Example: Fix numbers $a_1, a_2, \dots, a_{k+1} > 0$.

$$f(x_1, \dots, x_k) = x_1^{a_1-1} x_2^{a_2-1} \dots x_k^{a_k-1} (1 - x_1 - x_2 - \dots - x_k)^{a_{k+1}-1}$$

$$0 < x_1, x_2, \dots, x_k, \quad \sum_1^k x_i < 1$$

Induction shows

$$\int f = \frac{\Gamma(a_1) \Gamma(a_2) \cdots \Gamma(a_{k+1})}{\Gamma(\sum a_i)}$$

change of variable, dimension k :

Exactly the same result as in one and two dimensions holds. Even though there is nothing new, I shall spell out completely. Recall: instead of writing $dx_1 dx_2 \cdots dx_k$ we shall simply write dx and you have to understand from the context how many variables are there.

Suppose $\Omega \subset R^k$ is an open region and φ is a function on this region onto a region $\Omega^* \subset R^k$ such that (i) φ is C^1 , (ii) φ' nonsingular, and (iii) φ is a bijection. Then its inverse $\psi : \Omega^* \rightarrow \Omega$ is also C^1 and for any nice real valued function f on Ω

$$\int_{\Omega} f(x) dx = \int_{\Omega^*} f(\psi(y)) |\det \psi'(y)| dy \quad \blacksquare$$

To expand the Hypothesis: Now φ associates with every point of Ω a point of Ω^* ; which means there are k real valued functions $\varphi_1(x), \varphi_2(x) \cdots \varphi_k(x)$ on Ω such that for $x \in \Omega$ we have $\varphi(x) = (\varphi_1(x), \dots, \varphi_k(x))$. In other words $\varphi_i(x)$ is the i -th coordinate of the point $\varphi(x)$. Note φ_i are real valued functions on Ω .

(i) φ is C^1 means the following. For each i , The real valued function φ_i has all k partial derivatives $\frac{\partial \varphi_i}{\partial x_j}$ for $1 \leq j \leq k$, at each point of Ω and they are continuous functions on Ω . We denote $\varphi'(x)$ the $k \times k$ matrix of functions: (i, j) -th element is $\frac{\partial \varphi_i}{\partial x_j}$. Thus i -th row consists of the partial derivatives of i -th function φ_i . Thus now φ' at each point of Ω is a $k \times k$ matrix.

(ii) φ' is non-singular, that is $\varphi'(x)$ is a nonsingular matrix for all points $x \in \Omega$ — equivalently, its determinant is non zero.

(iii) φ is a one-one function on Ω onto Ω^* .

To expand the conclusion: φ has an inverse $\psi : \Omega^* \rightarrow \Omega$, Nothing new because φ is a bijection. Important conclusion is that ψ is again continuously differentiable. In other words if $\psi(y) = (\psi_1(y), \dots, \psi_k(y))$ then all the k partial derivatives of all the k functions, ψ_i are continuous functions. We denote $\psi'(y)$ the $k \times k$ matrix whose (i, j) -th element is $\frac{\partial \psi_i}{\partial x_j}$. Thus i -th row consists of the partial derivatives of i -th function ψ_i . We denote $|\psi'|$ to be the modulus of the determinant of the $k \times k$ matrix ψ' .

Let now f be a nice real valued function on Ω , for example bounded continuous function. Then the following holds:

$$\int_{\Omega} f(x)dx = \int_{\Omega^*} f(\psi(y)) |\psi'(y)|dy. \quad (\spadesuit)$$

Here are special cases.

translation in dimension k : Let f be a nice function on R^k . Let $\mu \in R^k$. Define $g(x) = f(x - \mu)$. Then $\int_{R^k} f = \int_{R^k} f \circ \psi$. Use the change of variable with $\Omega = \Omega^* = R^k$ and $\varphi(x) = x + \mu$ so that $\psi(y) = y - \mu$ and $\psi' = I$, the identity matrix. Thus

$$\int f(x)dx = \int f(x - \mu)dx$$

scaling: Let f be a nice function on R^k . Let A be a non-singular $k \times k$ matrix. Define $g(x) = f(Ax)$. Then $\int f = |A| \int g$. Use the change of variable with $\Omega = \Omega^* = R^k$ and $\varphi(x) = A^{-1}x$ so that $\psi(y) = Ay$ and $\psi' = A$. Thus

$$\int f(x)dx = \int f(Ax)|A|dx$$

Usually you say, put $Ax = y$ on right side so that $|A|dx = dy$ to get left side.

k dimensional pdf:

A pdf (probability density function) in k variables is a non-negative function on R^k such that $\int_{R^k} f(x)dx = 1$. We say k variables $(X_1, \dots, X_k)$ defined on a probability space has pdf f if for any a_i, b_i with $-\infty < a_i < b_i < \infty$

$$P\{a_i < X_i \leq b_i \quad \forall i\} = \int_{\prod(a_i, b_i]} f(x)dx.$$

Here $\prod(a_i, b_i]$ is the cube consisting of all points $x = (x_1, \dots, x_k)$ such that $a_i < x_i \leq b_i$ for each i . Even though it makes no difference whether integrate over $(a_i, b_i]$ or (a_i, b_i) , we shall continue writing $(a_i, b_i]$ for the sides of the box; just to maintain our earlier notation in one and two dimensional case.

Actually we should say f is the density function of the vector $X = (X_1, \dots, X_k)$ because the order is important. The pdf of (X, Y, Z) need not be same as pdf of (Z, X, Y) . Thus instead of saying density of the variables $(X_1, \dots, X_k)$ better to say density of the vector $(X_1, \dots, X_k)$. Also we are writing row vectors, it is customary to think of R^k as column vectors. It occupies too much space if we start writing column vectors.

pdf of functions of rv:

Suppose $(X_1, \dots, X_k)$ has density $f(x)$ which is zero outside a region (Open set) Ω . Suppose $\varphi = (\varphi_1, \dots, \varphi_k) : \Omega \rightarrow \Omega^*$, a region — C^1 , nonsingular, bijection. Let $Y_i = \varphi \circ X_i$. Then density of $(Y_1, \dots, Y_k)$ is given by $g(y) = f(\psi(y))|\psi'(y)|$ for $y \in \Omega^*$ (g is zero outside Ω^*). And $|\psi'(y)|$ is the modulus of the determinant of $\psi'(y)$.

This is immediate from the change of variable formula. You have only to repeat the argument of the dimension two case using the above change of variable formula.

positive definite matrices:

We consider only symmetric matrices in what follows. A symmetric $k \times k$ matrix Σ is called positive definite if all its eigen values are strictly positive. You can easily show that this is same as saying the following: For each $v \in R^k$; $\langle v, \Sigma v \rangle \geq 0$ with equality iff $v = 0$. In other words Denoting $\Sigma = ((\sigma_{ij}))$; the quadratic form $\sum_{i,j} \sigma_{ij} v_i v_j \geq 0$; and $= 0$ iff $v = 0$. Given any such matrix, there is diagonalization: there is an orthogonal matrix P (change of basis) such that

$$\Sigma = PDP^t = PDP^{-1}$$

where D is the diagonal matrix with entries, the eigen values. Consider the matrix

$$A = P\sqrt{D}P^t = P\sqrt{D}P^{-1}$$

where $\sqrt{D}$ is the diagonal matrix with diagonal entries being Positive square roots of diagonal entries of D . square roots of eigen values makes sense because these are positive. Also this is nonsingular and symmetric. This is denoted $\sqrt{\Sigma}$, notation justified because

$$A^2 = AA^t = \Sigma; \quad A \text{ symmetric, positive definite.}$$

Multivariate normal:

Let now Σ be any $k \times k$ positive definite symmetric matrix. Let $\mu \in R^k$. Let

$$f(x) = \frac{1}{(2\pi)^{k/2} |\Sigma|^{1/2}} e^{-\frac{1}{2}(x-\mu)^t \Sigma^{-1}(x-\mu)}.$$

We claim $\int_{R^k} f(x) = 1$ This is very simple. By translation it suffices to prove

$$\int \frac{1}{(2\pi)^{k/2} |\Sigma|^{1/2}} e^{-\frac{1}{2}x^t \Sigma^{-1}x} = 1 \quad \text{or} \quad \int |\Sigma|^{-1/2} e^{-\frac{1}{2}x^t \Sigma^{-1}x} = (2\pi)^{k/2}$$

Apply change of variable: $\Omega = \Omega^* = R^k$ and $\varphi(x) = Ax$ where $A = \sqrt{\Sigma}$ so that $\psi(y) = A^{-1}y$ and $\psi' = A^{-1}$. Note $\det A = (\det \Sigma)^{1/2}$. Apply to the function

$$f^*(x) = e^{-\frac{1}{2}x_1^2} e^{-\frac{1}{2}x_2^2} \dots e^{-\frac{1}{2}x_k^2}$$

where we already know

$$\int f^* = (2\pi)^{k/2}$$

Did you realize you have evaluated a complicated integral of million variables if k is a million. and Σ is any million by million positive definite matrix!

This function f is called multivariate normal density; $N_k(\mu, \Sigma)$ -density. Also called k dimensional normal with mean vector μ and covariance matrix Σ . This is justified soon.

Normal densities are also called Gaussian densities.

Dirichlet density:

Fix numbers $a_i > 0$ for $1 \leq i \leq k+1$. Define

$$f(x_1, \dots, x_k) = C x_1^{a_1-1} \dots x_k^{a_k-1} (1 - \sum_1^k x_i)^{a_{k+1}-1}$$

$$C = \Gamma(\sum a_i) / \prod \Gamma(a_i)$$

defined on

$$\Delta_k = \{(x_1, \dots, x_k) : 0 < x_1, \dots, x_k, \sum_1^k x_i < 1\}$$

We are following the convention, that $f = 0$ for points not satisfying above condition. We know that this is a pdf. If $X = (X_1, \dots, X_k) \sim f$, then we say $X \sim D(a_1, \dots, a_k; a_{k+1})$ Dirichlet with parameters (a_i) .

You can see the analogy with multivariate distribution in the discrete case. One can calculate all marginals here very easily. This pdf is useful in several contexts, especially Bayesian inference.

Marginals, conditionals, Independence:

Suppose f is pdf of $(X_1, \dots, X_k)$. We define $f_i(x_i)$ on R by integrating all the variables other than x_i from f . Thus

$$f_i(x_i) = \int_{R^{k-1}} f(x_1, \dots, x_k) dx_1 \dots dx_{i-1} dx_{i+1} \dots dx_k$$

As in the bivariate (2 dimensional) case you can see f_i is the marginal density of X_i :

$$P(\alpha < X_i \leq \beta) = \int_{\alpha}^{\beta} f_i(x_i) dx_i$$

You can talk about many other marginals. You can define marginal of any subcollection of $(X_1, \dots, x_k)$ by integrating the other variables from f . For example marginal of (X_2, X_4, X_5) is

$$f_{2,4,5}(x_2, x_4, x_5) = \int_{R^{k-3}} f(x_1 \dots, x_k) dx_1 dx_3 dx_6 dx_7 dx_8 \dots dx_k$$

Clearly if S is the box $(\alpha_2, \beta_2] \times (\alpha_4, \beta_4] \times (\alpha_5, \beta_5]$ then

$$P(\alpha_i < X_i \leq \beta_i; i = 2, 4, 5) = \int_S f_{2,4,5}(x_2, x_4, x_5) dx_2 dx_4 dx_5$$

(Make sure you understand the ranges of the integrals dx_2 and dx_4 and dx_5 . You can also now talk about conditionals:

$$f_i|_j(x_i | x_j) = f_{i,j}(x_i, x_j) / f_j(x_j)$$

defined when $f_j(x_j) > 0$; otherwise undefined. You can now define many other conditionals. For example conditional density of (X_2, X_4) given (x_7, x_8, x_9)

$$f_{(2,4)}|_{(7,8,9)}(x_2, x_4 | x_7, x_8, x_9) = f_{2,4,7,8,9}(x_2, x_4, x_7, x_8, x_9) / f_{7,8,9}(x_7, x_8, x_9)$$

defined only when $f_{7,8,9}(x_7, x_8, x_9) > 0$.

Do not worry, we shall not do more with this; even the general definitions of conditionals you need not remember. We shall have a couple of exercises just to give you a feel. These are very important and useful concepts, especially when we discuss multivariate normal density. But since we have no time to do enough exercises, and since we do not understand without doing some exercises, I decided to omit from our syllabus. (Then why did I say all this? You must know these exist and are no more complicated).

We say $X_1, \dots, X_k$ are independent if their joint density is product of marginals.

$$f(x_1, \dots, x_k) = \prod f_i(x_i)$$

More precise way of saying is that: product of marginals is a candidate for joint pdf. (Remember, even if one dimensional case we had a discussion that if you change pdf at one point, it still acts like the same pdf — which means

its integral over intervals gives probability for the variable to belong to that interval).

Moments, inequalities:

If $X = (X_1, \dots, X_k) \sim f$ and $\varphi : R^k \rightarrow R$ a nice function, we define

$$E(\varphi(X)) = \int \varphi(x_1, \dots, x_k) f(x_1, \dots, x_k) dx_1 \cdots dx_k$$

As in the discrete case, $\text{Cov}(X_i, X_j) = \sigma_{ij} = E(X_i, X_j) - E(X_i)E(X_j)$. Variance of X_i was already discussed. correlation $\rho(X_i, X_j) = \sigma_{ij}/(\sigma_i\sigma_j)$. The result of discrete case hold good here too.

If X, Y are independent then $\text{Cov}(X, Y) = 0$

$$\sigma^2(\sum X_i) = \sum \sigma^2(X_i) + 2 \sum_{i < j} \sigma(X_i, X_j).$$

In particular if (X_i) are independent then $\sigma^2(\sum X_i) = \sum \sigma^2(X_i)$.

(Chebychev) If $X \sim f$ and $a > 0$ then $P(|X| > a) \leq E(|X|)/a$.

(Chebychev) If X has mean μ and variance σ^2 and $a > 0$ then $P(|X - \mu| > a) \leq \sigma^2/a^2$.

You can now justify that if $X \sim N_k(\mu, \Sigma)$ then indeed $E(X_i) = \mu_i$ and $\text{Cov}(X_i, X_j)$ is the (ij) -th element of Σ .

Let $X_1, \dots, X_n$ be iid $\exp(\lambda)$. Compute the density of $\sum_1^n X_i$.

Let us do for $n = 2$. $f(x_1, x_2) = \lambda^2 e^{-\lambda(x_1+x_2)}$ for $x_1, x_2 > 0$. We change variables.

Put $\Omega = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$ and $\Omega^* = \{(y_1, y_2) : 0 < y_1 < y_2 < \infty\}$.

$\varphi(x_1, x_2) = (x_1, x_1 + x_2) = (y_1, y_2)$ and $|\psi'| = 1$.

giving joint density of $(Y_1, Y_2) = (X_1, X_1 + X_2)$ as

$$f(y_1, y_2) = \lambda^2 e^{-y_2}; \quad 0 < y_1 < y_2 < \infty.$$

So marginal of y_2 is

$$\begin{aligned} f_2(y_2) &= \int_0^{y_2} \lambda^2 e^{-y_2} dy_1 = y_2 \lambda^2 e^{-y_2} \\ &= \lambda y_2 \lambda e^{-\lambda y_2} \quad 0 < y_2 < \infty. \end{aligned}$$

Let $X_1, \dots, X_n$ be iid $\exp(\lambda)$. Compute the density of $S_n = \sum_{i=1}^n X_i$.

We did density of $X_1 + X_2$ is $f_2(z) = (\lambda z)\lambda e^{-\lambda z}$ for $0 < z < \infty$. We claim that density of S_n is given by

$$f_n(z) = \frac{(\lambda z)^{n-1}}{(n-1)!} \lambda e^{-\lambda z} \quad 0 < z < \infty \quad (\spadesuit)$$

Thus this is true for $n = 1, 2$. Assuming it to be true for n we prove for $(n+1)$. Denote $Y = S_n$ and $W = X_{n+1}$ so that $Y + W$ is what we are interested.

$$f(y) = \frac{(\lambda y)^{n-1}}{(n-1)!} \lambda e^{-\lambda y}; \quad g(w) = \lambda e^{-\lambda w} \quad 0 < y, w < \infty$$

Given (X_i) are independent we accept the fact that $X_1 + \dots + X_n$ and X_{n+1} are independent. Thus Y, W are independent so that their joint density is

$$\begin{aligned} & \frac{(\lambda y)^{n-1}}{(n-1)!} \lambda e^{-\lambda y} \lambda e^{-\lambda w} \quad 0 < y, w < \infty \\ &= \frac{(\lambda y)^{n-1}}{(n-1)!} \lambda^2 e^{-\lambda(y+w)} \quad 0 < y, w < \infty \end{aligned}$$

Change variable $Z = Y + W (= S_{n+1})$ and $W = W$. Thus $\varphi(y, w) = (y+w, w)$ defined on first quadrant $\Omega = \{(y, w) : y > 0, w > 0\}$ to $\Omega^* = \{(z, w) : 0 < w < z < \infty\}$ and $\psi(z, w) = (z - w, w)$. You see $|\psi'| = 1$. Joint density of (z, w) is given by

$$h(z, w) = \frac{(\lambda(z-w))^{n-1}}{(n-1)!} \lambda^2 e^{-\lambda z} = \lambda^n \frac{(z-w)^{n-1}}{(n-1)!} \lambda e^{-\lambda z}; \quad 0 < w < z < \infty.$$

so that marginal density of $Z = S_{n+1}$ is given by

$$f_{n+1}(z) = \int_0^z \lambda^n \frac{(z-w)^{n-1}}{(n-1)!} \lambda e^{-\lambda z} dw = \frac{(\lambda z)^n}{n!} \lambda e^{-\lambda z}; \quad 0 < z < \infty$$

This proves $(\spadesuit)$.

inequalities:

Chebychev's inequality is true and same proof works.

Theorem: If $X \sim f$ and $a > 0$ then $P(|X| \geq a) \leq E(|X|)/a$.

If $X \sim f$, $E(X) = \mu$, $\text{var}(X) = \sigma^2$, $a > 0$ then $P(|X - \mu| \geq a) \leq \sigma^2/a^2$.

The second sentence follows from first, apply to $P(|X - \mu|^2 > a^2)$.
First sentence follows from:

$$\begin{aligned} E(|X|) &= \int |x|f(x)dx \geq \int_{|x| \geq a} |x|f(x)dx \\ &\geq a \int f(x)dx \geq aP(|X| \geq a) \end{aligned}$$

Several things of discrete case have analogues. Recall If $X \sim f$ then $\mu_n = E(X^n) = \int x^n f(x)dx$ is called the n -th moment of X . This is defined only when $\int |x|^n f(x)dx < \infty$. whenever we talk about any moment, we assume its existence.

We shall now show that For $N_k(\mu, \Sigma)$ the vector μ is indeed the mean vector and covariance matrix is Σ . This is easy. If you recall the definition, we took symmetric positive definite matrix A such that $A^2 = \Sigma$. we took i.i.d. standard normals $X_1, \dots, X_k$. Set $X = (X_1, \dots, X_k)$ and put $Y = AX + \mu$. That is $Y_i = \sum_j a_{ij}X_j + \mu_i$. From this it is immediate $E(Y_i) = \mu_i$. Also

$$\begin{aligned} \text{Cov}(Y_i, Y_j) &= E[(Y_i - \mu_i)(Y_j - \mu_j)] = E\left(\sum_k a_{ik}X_k\right)\left(\sum_l a_{jl}X_l\right) \\ &= E\left(\sum_k a_{ik}X_k a_{jk}X_k\right) = \sum_k a_{ik}a_{jk} = \Sigma_{ij} \end{aligned}$$

where we used, for the second equality above, symmetry of the matrix A and that X_i are independent with mean zero. The last equality is from $A^2 = \Sigma$. This proof works even for $i = j$.

■ there are several very useful and interesting things with Multivariate normal and with i.i.d. exponentials and with Dirichlet that can be developed. Especially multivariate normal is ubiquitous — marginals are normal; all conditionals are normal; all conditional expectations are ‘linear’ and so on. However, I feel that you are not yet ready for those. (yes, we are leaving out some important things!) We shall return to some aspects like CLT and Laws of Large Numbers later. ■

Distribution functions:

We shall now discuss briefly that discrete/continuous are not like east and west. They are special cases of one concept. In discrete case we assigned to a rv its mass function; in continuous case we assigned pdf. We define, instead of pmf and pdf, the concept of distribution function or cumulative

distribution function. Suppose we have a probability space and a random variable X on the space.

The cdf or simply distribution function of X is the function $F(a) = P(X \leq a)$ for $a \in R$. Thus $F(a)$ is probability of the event $\{\omega : X(\omega) \leq a\}$.

This function satisfies the following three conditions:

(i) F is monotone increasing: $a \leq b$ implies $F(a) \leq F(b)$. (ii) F is right continuous. (iii) $\lim_{a \rightarrow -\infty} F(a) = 0$ and $\lim_{a \rightarrow +\infty} F(a) = 1$.

(i) is because when $a \leq b$ the event $\{\omega : X(\omega) \leq b\}$ contains the event $\{\omega : X(\omega) \leq a\}$ and hence $F(a) \leq F(b)$. Remember even if $a < b$ we may have $F(a) = F(b)$. (ii) follows from the fact that when $a_n \downarrow a$ the sets $(-\infty, a_n] \downarrow (-\infty, a]$ and hence the events $\{\omega : X(\omega) \leq a_n\} \downarrow \{\omega : X(\omega) \leq a\}$. Remember, from our rules for probability, when $A_n \downarrow A$ then $P(A_n) \downarrow P(A)$. For later use note also that $A_n \uparrow A$ then $P(A_n) \uparrow P(A)$. Finally, when $a_n \downarrow -\infty$ then the events $\{\omega : X(\omega) \leq a_n\} \downarrow \emptyset$ so that $F(a_n) \downarrow 0$. Also when $a_n \uparrow +\infty$ then the events $\{\omega : X(\omega) \leq a_n\} \uparrow \Omega$, the sample space, so that $F(a_n) \uparrow 1$. Here is a basic theorem, we shall not prove.

Theorem: If X is a rv then its distribution function $F_X = F$ satisfies the three conditions: F is (i) monotone increasing; (ii) right continuous; and (iii) has limits 0/1 as $a \rightarrow \mp\infty$.

Conversely given any function $F : R \rightarrow [0, 1]$ satisfying the three conditions above, there is a rv X such that $F = F_X$.

This is not a tautology. The converse part involves, first of all cooking up a probability space and then a random variable on that space satisfying $F = F_X$. If F_X is given we can calculate probabilities of events defined by the rv X .

Given F_X can we calculate $P(X < a)$? Yes $P(X < a) = F(a-)$ the left limit of F at a . this is because if $a_n \uparrow a$ with each $a_n < a$, then $(-\infty, a_n] \uparrow (-\infty, a)$. As a result $F(a_n) \uparrow P(X < a)$. Thus by definition of left limits, we conclude $F(a-) = P(X < a)$.

Can we answer $P(X = 5)$ by knowing F_X ? Yes it is $[F(5) - F(5-)]$. This is simply because $P(X = 5) = P(X \leq 5) - P(X < 5)$.

Can we recognize if X is discrete by knowing F_X ? Yes. X is discrete iff you can find distinct points x_n (finite or infinite sequence) such that $\sum [F(x_n) - F(x_n-)] = 1$. This is immediate from above.

Can we recognize if X has a density by knowing F_X . Yes. Define f by

$f(x) = F'(x)$ when the derivative exists, otherwise $f(x) = 0$. X has density if $F(a) = \int_{-\infty}^a f(x)dx$ for all a . Then f is the pdf. We shall not prove this. When F is not differentiable at x we have arbitrarily taken $f(x) = 0$, you can take $f(x) = 5$ for all such x . But keep in mind it is not enough to differentiate and calculate f as said; you must check if $F(a) = \int_{-\infty}^a f(x)dx$.

If F_X is continuous can you say X has density? No. The Devil's stair case F , discussed in analysis is a continuous F satisfies the above three conditions and it makes sense to consider a rv X for which $F_X = F$. It is easy to see that F is indeed differentiable at all points outside the Cantor set with derivative zero. It is true, but not easy to prove, that at points in the Cantor set F is not differentiable. This random variable is not discrete and has no pdf.

You can have F_X which may take some values with positive probability (so has no pdf) and also may have a density in an interval. You have some examples in HA.

Example: X takes values zero and one with probabilities $1/2$ and $1/2$. Then $F_X(a) = 0$ for $a < 0$; $F_X(a) = 1/2$ for $0 \leq a < 1$ and $F_X(a) = 1$ for $a \geq 1$.

Example: If X is Poisson with parameter λ then $F(a) = 0$ for $a < 0$ and $F(a) = e^{-\lambda}$ for $0 \leq a < 1$ and $F(a) = e^{-\lambda}[1 + \lambda]$ for $1 \leq a < 2$ and $F(a) = e^{-\lambda}[1 + \lambda + \frac{\lambda^2}{2!}]$ for $2 \leq a < 3$ and so on.

Example: If $X \sim \text{unif}(0, 1)$ then $F(a) = 0$ for $a \leq 0$ and $F(a) = a$ for $0 \leq a \leq 1$ and $F(a) = 1$ for $a \geq 1$.

Example: If $X \sim \exp(\lambda)$ then $F(a) = 0$ for $a \leq 0$ and $F(a) = 1 - e^{-\lambda a}$ for $a \geq 0$.

Example: If $X \sim N(0, 1)$ there is no explicit formula for $F(a)$.

Thus cdf is the one object using which you can answer questions about your rv – it may be discrete, may have pdf, may be neither.

Markov Chains:

We shall now discuss discrete rvs. But deal with questions depending on infinitely many rvs. We already saw one such problem, namely, branching process. each generation size is a discrete rv; but question answered was about the infinite sequence, a question that depends on all (X_n) . We saw its importance. We have also discussed Chandrasekhar model. Again the number of balls on day n is a discrete rv; but the question involved all the

random variables.

Here is another example; a very instructive dynamics of historical importance. Start with a box having two compartments H, C . I have 2000 balls numbered 1 to 2000. I toss a fair coin. Heads up I put 1900 balls in H and others in C . If Tails, I put 1901 balls in H and others in C . Here is how we continue the game.

We pick a number from $S = \{1, 2, \dots, 2000\}$ at random, see where ball of that number is and move it to the other compartment. Repeat. Question: What happens in the long run?

Here X_n is the number of balls in the compartment H after n exchanges. Once you know how many balls are in H , you know $2000 - X_n$ are in the compartment C .

This model was proposed by the physicists, Ehrenfests (Paul Ehrenfest and Tatiana Ehrenfest) to clarify and explain subtle points in the phenomenon of heat exchange. The compartment $H = M$ has hot milk and $C = W$ has cold water. The balls are the molecules. Exchanging ball from M to W signifies a fast moving milk molecule colliding with a slow moving water molecule and thus increasing its momentum. Similarly exchanging ball from W to M signifies a water molecule doing the same. Keep in mind the exchange of heat is not a one way process; a water molecule which gained momentum earlier may now bump into a slow moving milk molecule and impart momentum. The number of balls in a compartment signifies its temperature.

Just as there was a steady state in the Chandrasekhar model, here too you can show (in some sense) there is a steady state: In the limit it appears as if the balls are distributed at random into the two compartments. Thus limiting distribution of number of balls in H is binomial(2000, 1/2).

$$P(X_n = k) \rightarrow \binom{2000}{k} 2^{-2000} \quad 0 \leq k \leq 2000$$

Hence on the average there are 1000 balls in each compartment. Remembering number of balls signifies the temperature of the compartment, we see both water and milk reach a common temperature.

Here is another Example, Card Shuffling:

Consider usual deck of 52 cards arranged in a stack, say top to bottom cards numbered 1 to 52. Pick two numbers (with replacement) at random, $1 \leq i, j \leq 52$. Interchange the cards at positions i, j . You have a new arrangement of the stack. Of course, if $i = j$ nothing changes. So be it. Repeat for ever.

Question: What happens in the long run?

It will be a well shuffled deck! Let us denote by X_n the stack arrangement on day n . Thus values of X_n are not real numbers, but Permutations of the deck, equivalently, elements of S_{52} . We can show

$$P(X_n = \pi) \rightarrow \frac{1}{52!} \quad \forall \pi \in S_{52}$$

Thus the deck will be in random order.

You can select i, j without replacement. Result is same.

If you want to change only one card, you can do too as follows: Given a stack, select $1 \leq i \leq 52$ at random. Put top card (position 1) at position i ; do NOT change others. the final conclusion is same!

This technique is very useful because many times you want to pick an element at random from a given finite set.

We start with special example before introducing definitions.

Random Walk:

Consider the set Z of all integers and a walk in this set as follows. Here is the rule: we start at zero. Every day we toss a fair coin and move one step forward (if at 35, move to 36) if Heads up; one step backward (if at 35, move to 34) if Tails up. Of course, there is a non-zero chance of returning to zero. For example HT or TH will bring you to zero on day two. The question is: Are we sure to return to zero? is the chance of 'ever returning' one?

This is called simple symmetric random walk in one dimension. One dimension because we are moving in Z . Random walk because the motion is not deterministic and is governed by tossing coin, random mechanism. Symmetric because there is equal chance of moving in either direction. Simple because only one unit is moved at a time.

We can consider the set Z^2 , pairs of integers. We start at the origin $(0, 0)$, which we still denote by 0. We select one of the axes at random and move in that direction forward or backward at random. Thus, from (a, b) we move to one of the four points selected at random: $(a - 1, b)$, $(a + 1, b)$, $(a, b - 1)$, $(a, b + 1)$. Again it is clear that there is a chance of returning to origin. Question: Are you sure to return? Is the chance of returning one?.

This is called simple symmetric random walk in two dimensions.

We can consider the set Z^3 , triples of integers. We start at the origin $(0, 0, 0)$, which we still denote by 0. We select one of the axes at random and move in that direction forward or backward at random. Thus, from (a, b, c) we move to one of the six points selected at random: $(a - 1, b, c)$, $(a + 1, b, c)$, $(a, b - 1, c)$, $(a, b + 1, c)$, $(a, b, c - 1)$, $(a, b, c + 1)$. Again it is clear that there is a chance of returning to origin. Question: Are you sure to return? Is the chance of returning one?.

This is called simple symmetric random walk in three dimensions. In our course we refer to these as simply random walks. The above questions are difficult because the question depends on infinitely many rvs: entire future rvs.

Let us denote our position, state, on day n by X_n which obviously is a rv. Of course $X_0 = 0$. The event of interest is

$$A = \{X_n = 0 \text{ for some } n \geq 1\}.$$

Let $p^{(n)} = P(X_n = 0)$ for $n \geq 1$. We put $p^{(0)} = 1$ simply because $P(X_0 = 0) = 1$. Clearly our event of interest A , is union of all these events over $n \geq 1$, thus the question is: Is $P(X_n = 0 \text{ for some } n \geq 1) = 1$? Unfortunately these events are not disjoint. Let us disjointify. We define $f^{(0)} = 0$ and for $n \geq 1$,

$$f^{(n)} = P(X_m \neq 0, \forall 1 \leq m < n; X_n = 0).$$

Thus $f^{(n)}$ is the chances of returning to zero for the first time on day n . Clearly, on day zero game started and no return has yet taken place and hence $f^{(0)} = 0$. Our event of interest is union of all these disjoint events. Thus the question is: Is $\sum f^{(n)} = 1$?

Here is a theorem which we shall prove soon:

$$\text{Theorem: } \sum f^{(n)} = 1 \quad \text{iff} \quad \sum p^{(n)} = \infty.$$

The point to note is that the quantities $p^{(n)}$ are easy to calculate where as the quantities $f^{(n)}$ are not easy to calculate. The question is about $\sum f^{(n)}$ and this theorem converts the problem to a problem about the easily computable quantities. Accept this for now. Let us see how this helps us.

This reduces our problem, in one dimension, to decide whether the following series converges or not.

$$\sum \binom{2n}{n} \frac{1}{2^{2n}}$$

But how do we understand these numbers? This is where we take the help of James Stirling. Stirling's formula says $n! \sim \sqrt{2\pi}e^{-n}n^{n+\frac{1}{2}}$.

Recall the notation. Suppose that (a_n) and (b_n) are sequences of strictly positive numbers. We use

$$a_n \sim b_n \quad \Longleftrightarrow \quad \frac{a_n}{b_n} \rightarrow 1.$$

From the definition you can show the following.

$$a_n \sim b_n; c_n \sim d_n \quad \Rightarrow \quad a_n c_n \sim b_n d_n; \quad \frac{a_n}{c_n} \sim \frac{b_n}{d_n}$$

As a result

$$\frac{2n!}{n!n!} \frac{1}{2^{2n}} \sim \frac{1}{\sqrt{\pi n}}$$

It is also easy to see that

$$a_n \sim b_n \implies \left(\sum a_n < \infty \leftrightarrow \sum b_n < \infty \right)$$

This can be seen by just noting that after some stage

$$\frac{1}{2} < \frac{a_n}{b_n} < 2; \quad \frac{1}{2}b_n < a_n < 2b_n$$

Now recall $\sum \frac{1}{\sqrt{n}} = \infty$. As a result we conclude that $\sum p^{(n)} = \infty$ This solves our problem. Thus

We are sure to return to zero.

Two dimensions:

As earlier, we can argue to see

$$p^{(2n+1)} = 0$$

$$p^{(2n)} = \sum_{k=0}^n \frac{(2n)!}{k!k!(n-k)!(n-k)!} \frac{1}{4^{2n}}.$$

This is because to be at origin on day $2n$, we should have made a certain number (may be zero, may be n , but not more) of right moves and same number of left moves and in the remaining $(2n - 2k)$ days we should make $(n - k)$ up moves and same number of down moves. Thus

$$\begin{aligned} p^{(2n)} &= \binom{2n}{n} \sum_{k=0}^n \frac{n!n!}{k!k!(n-k)!(n-k)!} \frac{1}{4^{2n}} \\ &= \binom{2n}{n} \binom{2n}{n} \frac{1}{4^{2n}} \sim \frac{1}{\pi n}. \end{aligned}$$

Thus again the sum $\sum p^{(n)} = \infty$. Thus

We are sure to return to the origin.

Three dimensions:

To be at origin on day $2n$ you need to make certain number k of moves in X^+ direction and same number in X^- direction; a certain number l in Y^+

direction and same number in Y^- direction; make sure $k + l \leq n$ and in the remaining $2n - 2k - 2l$ days half Z^+ moves and half Z^- moves. Thus

$$\begin{aligned} p^{(2n)} &= \sum_{0 \leq k, l, k+l \leq n} \frac{(2n)!}{k! k! l! l! (n-k-l)! (n-k-l)!} \frac{1}{6^{2n}} \\ &= \binom{2n}{n} \frac{1}{2^{2n}} \frac{1}{3^n} \sum_{0 \leq k, l, k+l \leq n} \left[\frac{n!}{k! l! (n-k-l)!} \right]^2 \frac{1}{3^n} \end{aligned}$$

If you have positive numbers a_i and $\max a_i \leq M$ then

$$\sum a_i^2 \leq M \sum a_i$$

Thus if

$$\max \left\{ \frac{n!}{k! l! (n-k-l)!} : 0 \leq k, l, k+l \leq n \right\} = M_n$$

then

$$\sum_{0 \leq k, l, k+l \leq n} \left[\frac{n!}{k! l! (n-k-l)!} \right]^2 \frac{1}{3^n} \leq M_n \sum_{0 \leq k, l, k+l \leq n} \left[\frac{n!}{k! l! (n-k-l)!} \right] \frac{1}{3^n}$$

But the last sum is just trinomial expansion

$$\left(\frac{1}{3} + \frac{1}{3} + \frac{1}{3} \right)^n = 1$$

Thus

$$p^{(2n)} \leq \binom{2n}{n} \frac{1}{2^{2n}} \frac{1}{3^n} M_n$$

We know

$$\binom{2n}{n} \frac{1}{2^{2n}} \sim \frac{1}{\sqrt{\pi n}}$$

Remembering that the multinomial terms are the largest at

$$k = l = (n - k - l) = n/3$$

(we are careless here) we see

$$\begin{aligned} \frac{1}{3^n} M_n &\leq \frac{1}{3^n} \frac{n!}{(\frac{n}{3})! (\frac{n}{3})! (\frac{n}{3})!} \\ &\sim \frac{1}{3^n} [\sqrt{2\pi} e^{-n} n^{n+(1/2)}] [\sqrt{2\pi} e^{-n/3} (n/3)^{(n/3)+(1/2)}]^{-3} \end{aligned}$$

$$= \frac{1}{2\pi} 3^{3/2} \frac{1}{n}$$

where we used Stirling. Thus ultimately

$$p^{(2n)} \leq \alpha_n; \quad \alpha_n \sim Cn^{-3/2}$$

for some constant C . Since $\sum n^{-3/2}$ and hence $\sum \alpha_n$ and hence $\sum p^{(2n)}$ converges we conclude the following.

chance of NOT returning to the origin is strictly positive.

We were careless at one point: $n/3$ may not be integer and hence saying that $(n/3, n/3, n/3)$ term is largest does not make sense. How do you rectify it? Well, the argument above definitely shows that sum of all $p^{(2n)}$ where n is multiple of 3, is finite.

Consider the sum of $p^{(2n)}$ over n of the form $1 \bmod(3)$. That is, over all n of the form $3m+1$. Now the max term is attained at $(m, m, m+1)$ Do similar Stirling analysis and show sum is finite.

Then consider n of the form $2 \bmod(3)$. The trinomial term for this case attains maximum at $(m, m+1, m+1)$ and again show this sum is finite.

Thus you deduce that $\sum p^{(2n)}$ is finite. AAha, done!

We shall now prove the theorem which involves interesting ideas; valid in more generality.

The following considerations are true in greater generality. But let us stick to RW. In what follows, we use 0 for the zero of Z , Z^2 , Z^3 .

$$p^{(0)} = 1; \quad p^{(n)} = P(X_n = 0); \text{ for } n \geq 1$$

$$f^{(0)} = 0; \quad f^{(n)} = P(X_n = 0, X_m \neq 0 \text{ for all } 0 < m < n); \quad n \geq 1$$

Recall we started from zero, so $X_0 = 0$ and hence $p^{(0)} = 1$. $p^{(n)}$ is the probability of being at zero on day n . On day zero, no return has yet taken place and hence $f^{(0)} = 0$. $f^{(n)}$ is the chances of returning to zero on day n for the first time.

Renewal Equation:

$$p^{(n)} = \sum_{k=0}^n f^{(k)} p^{(n-k)}; \quad n \geq 1.$$

The first term on right side ($k = 0$) is zero simply because $f^{(0)} = 0$. Thus right side has only p_k for $k < n$ and thus this equation describes p_n in terms of the previous ones. for aesthetic reasons we keep $k = 0$ term, it helps in relating to convolution. In case n is odd both sides are zero and you can as well assume that n is even. However renewal equation holds more generally, so to maintain the spirit of the equation, we shall not assume n is even.

The event $A = (X_n = 0)$ can be expressed as disjoint union of events depending on first return to zero.

$$A = \bigcup_{k=1}^n A_k; \quad A_k = (X_i \neq 0, 1 \leq i < k; \quad X_k = 0; X_n = 0).$$

If you take any sample point in any of the A_k it is clearly in A , because the condition $X_n = 0$ is present in all A_k . Conversely if a sample point is in A , that is $X_n(\omega) = 0$ then there must be a first $k \leq n$ such that $X_k(\omega) = 0$ and hence the sample point is in the corresponding A_k . (Even though we were not talking about sample points in infinite tosses, observe the above equation has nothing to do with infinite number of tosses, it is about tossing n times). By rules about conditional probability,

$$\begin{aligned} P(A_k) &= P(X_i \neq 0, 1 \leq i < k; X_k = 0) P(X_n = 0 | X_i \neq 0, 1 \leq i < k; X_k = 0). \\ &= f^{(k)} P(X_n = 0 | X_i \neq 0, 1 \leq i < k; X_k = 0). \end{aligned} \quad (\clubsuit)$$

Note that under the given condition, irrespective of whatever be X_i for $i < k$; if we know $(X_k = 0)$ holds then chances of $(X_n = 0)$ is just the following: chances of reaching zero on day n starting at zero on day k which is same as chances of reaching zero on day $(n - k)$; having started at zero initially. Thus

$$P(A_k) = f^{(k)} p^{(n-k)}.$$

If you do not like the conditional probability argument ($\clubsuit$), you can directly calculate the number of outcomes in the event A_k : Consider any sequence of H/T of length k such that (i) there are equal number of H and T and (ii) before k the number of H and T are not equal. (there are $f^{(k)} 2^k$ such) AND follow it up with any sequence of H/T of length $(n - k)$ having equal number of H and T (there are $p^{(n-k)} 2^{n-k}$ such). Thus, noting $f^{(0)} = 0$;

$$p^{(n)} = \sum_{k=1}^n f^{(k)} p^{(n-k)} = \sum_{k=0}^n f^{(k)} p^{(n-k)}$$

generating functions:

Let us consider the generating functions:

$$P(s) = p^{(0)} + p^{(1)}s + p^{(2)}s^2 + \cdots + p^{(k)}s^k + \cdots$$

$$F(s) = f^{(0)} + f^{(1)}s + f^{(2)}s^2 + \cdots + f^{(k)}s^k + \cdots$$

which are defined at least for $0 \leq s < 1$. Of course $f^{(n)}$ for $n \geq 1$, being probabilities of disjoint events, F is defined for $s = 1$ as well.

Observe that, irrespective of whether $\sum p^{(n)}$ is finite or not we have,

$$\lim_{s \uparrow 1} P(s) = \sum_n p^{(n)}; \quad \lim_{s \uparrow 1} F(s) = \sum_n f^{(n)}.$$

Since limit can be interchanged with finite sums and not always with infinite sums let us argue as follows to justify the above equality. First note that $\lim P(s)$ exists because P is increasing on $[0, 1)$ Fix any $k \geq 1$. Since everything is non-negative we have $P(s) \geq \sum_{n=0}^k p^{(n)} s^n$. Since we have finite sum on right side, we conclude $\lim P(s) \geq \sum_{n=0}^k p^{(n)}$. This being true for every k , we conclude $\lim P(s) \geq \sum p^{(n)}$. Of course, for every $s < 1$, we have $P(s) \leq \sum p^{(n)}$ so that $\lim P(s) \leq \sum p^{(n)}$. Both these inequalities prove the stated equality.

Now fix an s ($0 \leq s < 1$) and consider the two series defining $F(s)$ and $P(s)$.

Note these being of positive terms they are absolutely convergent and thus we see by Cauchy's theorem and renewal equation,

$$F(s)P(s) = P(s) - 1.$$

We have here used that renewal equation is valid for $n \geq 1$, not for $n = 0$. ($f^{(0)}p^{(0)} = 0; p^{(0)} = 1$.) Thus

$$P(s) = \frac{1}{1 - F(s)}; \quad 0 \leq s < 1. \quad (\spadesuit)$$

Note that $F(s) < 1$ for $0 \leq s < 1$, so the above makes sense. Moreover, if $\sum f^{(n)} = 1$ then $\lim_{s \uparrow 1} F(s) = 1$ and $(\spadesuit)$ shows that $\lim_{s \uparrow 1} P(s) = \infty$ or $\sum p^{(n)} = \infty$. On the other hand if $\sum f^{(n)} = c < 1$ then $\lim_{s \uparrow 1} F(s) = c$ and $(\spadesuit)$ shows that $\lim_{s \uparrow 1} P(s) = \frac{1}{1-c} < \infty$ or $\sum p^{(n)} < \infty$.

This completes proof of the theorem.

All the previous examples, RW, Chandrasekhar model, Ehrenfest model, Card shuffling have the following commonality.

There is a finite or countable set, to be called, State Space S . There is a probability μ on $S - \{\mu(i), i \in S\}$. This tells you how to start on day zero. Start at i with probability $\mu(i)$. This is called initial distribution. To make life simple, if you wish, you can start with some fixed state $s^* \in S$. This means $\mu(s^*) = 1$ (and $\mu(s) = 0$ for $s \neq s^*$).

For each state i , there is a probability $\{p_{ij} : j \in S\}$; that is, $p_{ij} \geq 0$ for all $j \in S$ and $\sum_{j \in S} p_{ij} = 1$. This tells that on any day if you are at i , then (do not ask how did you come to i) move to j with probability p_{ij} . These numbers p_{ij} are arranged as a $S \times S$ matrix: $P = ((p_{ij}))_{i,j \in S}$. This P is called transition matrix. We sometimes write $p(i, j)$ instead of p_{ij} . This is stochastic matrix.

A stochastic matrix is a matrix where each row is a probability vector.

We are using $P(A)$ for probability of event A ; now we are using P for transition matrix but there would be no confusion if you pay attention to what you are reading.

Definition: A sequence of random variables $\{X_n, n \geq 0\}$ is called markov chain with initial distribution μ and transition matrix P if

$$P(X_0 = i) = \mu(i) \quad i \in S$$

and

$$P(X_{n+1} = j | X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i) = p_{ij}, \quad n \geq 0; i, j \in S$$

Thus the dynamics is that on any day if you are at i you move to j with probability p_{ij} ; irrespective of how you came to i .

◆ For one dimensional random walk,

$$S = \mathbb{Z}; \quad \text{and } \mu(0) = 1.$$

$$p_{i,i\pm 1} = 1/2 \quad p_{ij} = 0 \text{ for } j \neq i \pm 1.$$

◆ For two dimensional random walk

$$S = \mathbb{Z}^2. \quad \mu\{(0,0)\} = 1$$

$$p_{(i,j),(k,l)} = 1/4 \text{ if } |i-k| + |j-l| = 1 \text{ and zero otherwise.}$$

◆ For three dimensional random walk

$$S = \mathbb{Z}^3. \quad \mu\{(0,0,0)\} = 1$$

$$p_{(i_1,j_1,k_1),(i_2,j_2,k_2)} = 1/6 \text{ if } |i_1-i_2| + |j_1-j_2| + |k_1-k_2| = 1 \text{ and zero otherwise.}$$

◆ For Chandrasekhar model

$$S = \{0, 1, 2, \dots\} \quad \mu(0) = 1$$

$$p_{ij} = \sum_{l=0}^{i \wedge j} \binom{i}{l} q^l p^{i-l} e^{-\lambda} \lambda^{j-l} \frac{1}{(j-l)!}$$

◆ For Ehrenfests model

$$S = \{0, 1, 2, \dots, 2000\} \quad \mu(1900) = \frac{1}{2} = \mu(1901)$$

$$p_{i,i+1} = \frac{2000-i}{2000} \quad p_{i,i-1} = \frac{i}{2000}$$

◆ For card shuffling

$Z = S_{52}$ the set of permutations of $\{1, 2, \dots, 52\}$ OR permutations of the 52 cards. understanding is that π means the vertical stack: top card $\pi(1)$ and bottom card $\pi(52)$.

$\mu(\pi_0) = 1$ for some permutation.

For the i, j interchange shuffle when i, j are selected with replacement, the transition matrix is:

$$p_{\pi,\pi} = 1/52$$

$$p_{\pi,\eta} = 2/(52 \times 51) \text{ if there is } i_0 \neq j_0 \text{ such that } \eta(i_0) = \pi(j_0); \quad \eta(j_0) = \pi(i_0)$$

$$\text{and } \eta(i) = \pi(i) \text{ for } i \neq i_0, j_0.$$

For the top to random shuffle

$$p_{\pi,\eta} = 1/52 \text{ if there is an } i_0 (1 \leq i_0 \leq 52) \text{ such that}$$

$$\eta(i_0) = \pi(1) \text{ and } \eta(i) = \pi(i+1) \text{ for } i < i_0 \text{ and } \eta(i) = \pi(i) \text{ for } i > i_0.$$

Note that if $i_0 = 1$ then there is no i with $i < i_0$ and $\eta = \pi$.

We do not worry as to where the random variables are defined. Our interest is to calculate probabilities and say something interesting. We see that the analysis will be of immense use; both in theory and practice. Here is how we calculate simple probabilities.

(i) $P(X_0 = i_0) = \mu(i_0)$. this follows from definition.

(ii) $P(X_0 = i_0, X_1 = i_1) = \mu(i_0)p(i_0, i_1)$ because left side equals $P(X_0 = i_0)P(X_1 = i_1 | X_0 = i_0)$

(iii) $P(X_0 = i_0, X_1 = i_1, X_2 = i_2) = \mu(i_0)p(i_0, i_1)p(i_1, i_2)$ because left side equals $P(X_0 = i_0)P(X_1 = i_1 | X_0 = i_0)P(X_2 = i_2 | X_0 = i_0, X_1 = i_1)$

(iv) In general for any n

$$P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i_n) = \mu(i_0) \prod_{m=1}^n p(i_{m-1}, i_m)$$

(v) For any $n \geq 0$ we have $P(X_{n+1} = j | X_n = i) = p(i, j)$. (♣)

You can dispose this by saying that it is a consequence of the definition of the dynamics. Even though we said that ‘no matter what the past was, if you are at i move to j with probability p_{ij} ; remember the dynamics only tells you

$$P(X_{n+1} = j | X_0 = i_0, \dots, X_{n-1} = i_{n-1}, X_n = i) = p(i, j).$$

Thus you need to justify what you wrote in symbols plus your definition of conditional probabilities agrees with what you said in words. Once you understand what is to be done, doing itself is just discipline. Left side of (♣) equals by definition

$$\frac{P(X_{n+1} = j, X_n = i)}{P(X_n = i)} = \frac{\sum P(X_{n+1} = j, X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0)}{\sum P(X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0)}$$

where the sum in both numerator and denominator is over $(i_0, \dots, i_{n-1})$.

$$\begin{aligned} &= \frac{\sum \mu(i_0)p(i_0, i_1) \cdots p(i_{n-1}, i) \quad p(i, j)}{\sum \mu(i_0)p(i_0, i_1) \cdots p(i_{n-1}, i)} \\ &= \frac{\sum \mu(i_0)p(i_0, i_1) \cdots p(i_{n-1}, i)}{\sum \mu(i_0)p(i_0, i_1) \cdots p(i_{n-1}, i)} p(i, j) = p(i, j) \end{aligned}$$

For this reason the matrix $P = ((p_{ij}))$ is also called the one step transition matrix. Let us denote the powers of this matrix by $P^2, P^3, \dots$. Also let us denote the entries of P^m by $p^{(m)}(i, j)$ or $p_{ij}^{(m)}$. We see now that P^2 is the two step transition matrix and in general P^m is the m -step transition matrix.

(vi) For any $n \geq 1$ we have $P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}) = p(i, j)$.
 More generally if $m_1 < m_2 < \dots < m_l < n < n+1$ then
 $P(X_{n+1} = j | X_{m_1} = i_1, X_{m_2} = i_2, \dots, X_{m_l} = i_{m_l}, X_n = i) = p(i, j)$.
 The proof is exactly as above. Thus any exact information about the past is irrelevant if you have yesterday's information. You can in fact strengthen this to say among all past information, the most recent information is enough. Just keep in mind we need exact information about the state, not like: the state on day 2 is either this or that. Suppose $m_1 < m_2 < \dots < m_l < m < n+1$
 $P(X_{n+1} = j | X_{m_1} = i_1, X_{m_2} = i_2, \dots, X_{m_l} = i_{m_l}, X_m = i)$
 $= P(X_{n+1} = j | X_m = i).$

(vii) For any $n \geq 0$ we have $P(X_{n+2} = j | X_n = i) = p^{(2)}(i, j)$.
 This is immediate from previous result

$$\begin{aligned} P(X_{n+2} = j | X_n = i) &= \sum_k P(X_{n+2} = j, X_{n+1} = k | X_n = i) \\ &= P(X_{n+1} = k | X_n = i) P(X_{n+2} = j | X_n = i, X_{n+1} = k) = \sum_k p_{ik} p_{kj} \end{aligned}$$

as stated.

Thus, the matrix P^2 is called two step transition matrix. Observe that $P(X_{n+2} = j | X_n = i)$ is nothing but the (ij) -th entry of the matrix P^2 .

(viii) For any $m \geq 1$ and any n ; $P(X_{n+m} = j | X_n = i) = p_{ij}^{(m)}$, the ij -th element of the matrix P^m .
 in other words P^m is the m -step transition matrix; its i -th row tells you where you are after m days if you are at i today. The proof is exactly as above. We define P^0 to be the identity matrix.

(ix) Chapman-Kolmogorov equations:

$$p_{ij}^{(m+n)} = \sum_k p_{ik}^{(m)} p_{kj}^{(n)} \quad m, n \geq 0; \quad i, j \in S$$

This follows from definition and above results,

There are several questions that can be asked and answered about such systems. One important thing to know (as we did in RW and Chandrasekhar's model) is 'what happens in the long run/ Is there any limiting distribution for our state on day n '?

Example: Consider a two state chain with states AB and transition matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Clearly, if we started at A we will be moving alternatively.

That is if $P(X_0 = A) = 1$ then for even n we have $P(X_n = A) = 1$ whereas for odd n this is zero. In other words $(X_n = A)$ does not have a limiting value. It is the periodicity in our movement that is obstructing the existence of limiting probabilities.

Example: Consider four state chain with states $ABCD$ and transition matrix $\begin{pmatrix} 1/3 & 0 & 2/3 & 0 \\ 0 & 1/7 & 0 & 6/7 \\ 2/3 & 0 & 1/3 & 0 \\ 0 & 6/7 & 0 & 1/7 \end{pmatrix}$. In this chain we can show, no matter

where you start, there is a limiting distribution, that is $P(X_n = s)$ has a limiting value, no matter what $s \in \{A, B, C, D\}$ is. But the limit depends on where you started. For example if you started at A or C you remain at those whereas if you started at B or D you remain in those states. In a sense to understand this chain you can simplify your task and understand the two simpler chains:

(i) States A, C and transition matrix $\begin{pmatrix} 1/3 & 2/3 \\ 2/3 & 1/3 \end{pmatrix}$.

(ii) States B, D and transition matrix $\begin{pmatrix} 1/7 & 6/7 \\ 6/7 & 1/7 \end{pmatrix}$.

Thus you can reduce your job to simpler problems. Thus existence of limiting probabilities irrespective how we started is obstructed by two phenomena: the chain is reducible or there is a periodicity. If this is not so then there are limiting probabilities and these reveal much more about the chain than just being 'limit of something'. this is what we explore now and prove a basic theorem. We restrict our attention to finite state chains only in what follows; though somethings we do are true for chains with infinite state space.

■ S stands for state space and P for transition matrix.

■ A chain is irreducible if from any state you can reach any other state someday. In symbols if given $i, j \in S$ there is n such that $p_{ij}^{(n)} = P(X_n = j | X_0 = i) > 0$.

■ for an irreducible chain its period is $d = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\}$. where $i \in S$. A chain is aperiodic if $d = 1$.

Is the gcd well defined and independent of the state i ? Yes. here are some simple facts.

(1°) The only subgroups of Z are $G = \{0\}$ and $G = gZ$ for some $g \geq 1$.

(2°) Let $S \subset \{1, 2, \dots\}$, finite or infinite, but non-empty. Then $\gcd S$ make sense (exists).

Indeed, if S is finite you knew in high school. In any case, let m be an element of S . If it divides all elements of S , then done, remember $m \in S$. Otherwise try $m - 1$. Continue this way, there are only finitely many integers below m and you will surely stop and that will be $\gcd S$. If you reached one already then one is $\gcd S$.

(3°) if $S \subset \{1, 2, \dots\}$ is non-empty and $\gcd S = a$ then there are elements $s_1, \dots, s_k \in S$ and integers $x_1, \dots, x_k$ such that $a = \sum x_i s_i$. Indeed the collection of all such elements $\sum x_i s_i$ is a group and both S and this have same $\gcd$, but if $G = gZ$ ($g \geq 1$) then clearly g is $\gcd$ of G .

(4°) Suppose $S \subset \{1, 2, \dots\}$ non-empty and has $\gcd 1$. Assume that S has the property: $m, n \in S$ implies $m + n \in S$. Then there exists n_0 such that $n > n_0$ implies $n \in S$.

get the $\gcd 1 = \sum x_i s_i$ let us put x as sum of those $x_i s_i$ where $x_i > 0$ and $-y$ as sum of those $x_i s_i$ where $x_i < 0$. Thus $x - y = 1$. Because of the assumed condition on S , we see $x, y \in S$. Now let $n > y^2$ then we can write $n = dy + r$ where $0 \leq r < y$ and $d \geq y$. This last one is because $n > y^2$. Thus

$$n = dy + r1 = dy + r(x - y) = (d - r)y + rx$$

Thus n is a non-negative combination of x, y and is hence in S .

(5°) For an irreducible chain period does not depend on the state, that is, $d(i) = d(j)$ for all i, j .

Indeed fix $i \neq j$. By irreducibility fix a, b such that $p_{ij}^{(a)} > 0$ and $p_{ji}^{(b)} > 0$. Observe that by C-K equations, $p_{jj}^{(b+a)} > 0$ and $p_{ii}^{(a+b)} > 0$. Thus $d(i)$ and $d(j)$ both divide $a + b$. Again by C-K, we see that whenever $p_{ii}^{(n)} > 0$, then $p_{jj}^{(b+n+a)} > 0$ so that $d(j)$ divides $b + n + a$ and hence divides n . Thus $d(j)$ divides all n with $p_{ii}^{(n)} > 0$. In other words $d(j) \leq d(i)$. Similarly $d(i) \leq d(j)$. completes proof of statement.

(6°) In an irreducible aperiodic chain, given (ij) there is n_0 such that $n > n_0$ implies $p_{ij}^{(n)} > 0$.

Using irreducibility, fix a such that $p_{ij}^{(a)} > 0$. Fix m such that $n \geq m$ implies $p_{jj}^{(n)} > 0$, by earlier result. Now C-K implies $p_{ij}^{(a+n)} > 0$ for all $n \geq m$. Thus $n_0 = m + a$ would do.

(7°) A finite state chain is irreducible and aperiodic iff some power of P has all entries strictly positive

Indeed if it is irreducible and aperiodic then for fixed (ij) there is n_0 so that $p_{ij}^{(n)} > 0$ for all $n > n_0$. State space being finite there are only finitely many such pairs and so done. Conversely, if some power of P has strictly positive entries then we immediately see irreducibility. Also if you take any i , then the set $\{n \geq 1 : p_{ii}^{(n)} > 0\}$ includes all integers after some stage and hence must have gcd one.

Here is the main theorem we are heading to.

Theorem : Consider a finite state irreducible aperiodic chain with transition matrix P . Then $P^n \rightarrow \Pi$ for some stochastic matrix Π , with all rows same. Thus if π denotes its first row, then all rows equal π . Further

- (i) π is a probability vector and each entry strictly positive.
- (ii) $\pi P = \pi$. That is π is left eigen vector of P for eigen value one. The eigen subspace for eigen value 1 is one-dimensional.
- (iii) $P(X_n = j \mid X_0 = i) \rightarrow \pi(j)$ whatever be i . Thus no matter how you start, in the long run chances of being in j are $\pi(j)$. Thus π is steady state distribution/limiting distribution/ final distribution of the chain.
- (iv) If $X_0 \sim \pi$ then $X_n \sim \pi$ for all n . More generally for all n, m , we have $(X_0, \dots, X_n) \sim (X_m, X_{m+1}, \dots, X_{n+m})$ if $X_0 \sim \pi$. In other words π is stationary initial distribution of the chain. If your initial distribution is π then the chain appears 'stationary' — not moving in probabilistic sense — no matter when you start observing the chain looks the same.
- (v) For almost all sample points,

$$\frac{\#\{0 \leq m \leq n-1 : X_m = j\}}{n} \rightarrow \pi(j)$$

Thus $\pi(j)$ is the proportion of time spent in the state j . Since motion is random, time spent in a given state is also random, so the proportion of time spent is also random; however (like the law of large numbers you saw) with probability one the limiting proportion is deterministic and equals $\pi(j)$.

- (vi) If $P(X_0 = i) = 1$, then there is surely some day m such that $X_m = i$. In other words you are sure to return to i . Moreover the expected return time is $1/\pi(i)$. That is $\sum_{n=1}^{\infty} n f_n = 1/\pi(i)$. where f_n for $n \geq 1$ is the chance that first return to i takes place on day n . ■

Now you see the importance of the vector π . Also, from practical point, if you simulate and run the chain for a long time, say N , then what you see as X_N will be a sample from the distribution π . That is $P(X_N = i)$ is 'nearly' $\pi(i)$. In particular, if π is uniform probability on S , that is, $\pi(i) = 1/|S|$ for each i , then what you see as X_N is a point picked at random from S .

In view of the facts mentioned before statement of the theorem, we can restate the theorem as simply a theorem on matrices: If P is a stochastic matrix such that all entries of some power are strictly positive then P^n converges to a matrix with identical rows.

Proof is a hands-on calculation. We need to show entries of each column converge to one number. That is, for each fixed j , $\lim p_{ij}^{(n)}$ exists and does not depend on i . Instead of handling so many sequences $\{p_{ij}^{(n)}; n \geq 1\}$ for each $i \in S$, we simplify our job by considering max and min of the column. More precisely, let us fix one j till further orders and put

$$m_n = \min_i p_{ij}^{(n)}, \quad M_n = \max_i p_{ij}^{(n)}.$$

These quantities depend on j , but j is fixed now. Clearly $m_n \leq M_n$ for each n .

Claim: $m_1 \leq m_2 \leq \dots \leq m_n \leq \dots \leq M_n \leq \dots \leq M_2 \leq M_1$ ($\spadesuit$)

This is seen as follows. For any i ,

$$p_{ij}^{(n+1)} = \sum_k p_{ik} p_{kj}^{(n)} \leq M_n \sum_k p_{ik} = M_n$$

This being true for each i , $\max_i p_{ij}^{(n+1)} \leq M_n$. In other words $M_{n+1} \leq M_n$ for each n . Similarly

$$p_{ij}^{(n+1)} = \sum_k p_{ik} p_{kj}^{(n)} \geq m_n \sum_k p_{ik} = m_n$$

This being true for each i , $\min_i p_{ij}^{(n+1)} \geq m_n$. In other words $m_{n+1} \geq m_n$ for each n . Thus claim is proved.

In view of ($\spadesuit$), if we show $M_n - m_n \rightarrow 0$, then we can conclude that the two sequences $\{m_n\}$ and $\{M_n\}$ converge to some number $\pi(j)$. This implies that for each i , the sequence $\{p_{ij}^{(n)}, n \geq 1\}$ converges to $\pi(j)$. Again, in view of ($\spadesuit$) it is enough to show a subsequence of $\{M_n - m_n\}$ converges to zero. Let us fix any one n_0 with all entries of P^{n_0} strictly positive. The subsequence we consider is $\{M_{kn_0} - m_{kn_0}; k \geq 1\}$.

We have fixed n_0 so that each entry of P^{n_0} is strictly positive. Say $p_{ij}^{(n_0)} \geq \epsilon > 0$ for each i, j . Plan to show $\{M_{kn_0} - m_{kn_0}; k \geq 1\}$ converges to zero. More precisely, we show $M_{kn_0} - m_{kn_0} \leq (1 - \epsilon)^k$.

Clearly $M_n \leq 1$ and $m_{n_0} \geq \epsilon$. Thus

$$M_{n_0} - m_{n_0} \leq (1 - \epsilon). \quad (\bullet)$$

$$\begin{aligned} p_{ij}^{(2n_0)} &= \sum_k p_{ik}^{(n_0)} p_{kj}^{(n_0)} \\ &= \sum_k (p_{ik}^{(n_0)} - \epsilon p_{jk}^{(n_0)}) p_{kj}^{(n_0)} + \sum_k \epsilon p_{jk}^{(n_0)} p_{kj}^{(n_0)} \quad (\text{add and subtract}) \\ &\geq m_{n_0} \sum_k (p_{ik}^{(n_0)} - \epsilon p_{jk}^{(n_0)}) + \epsilon p_{jj}^{(2n_0)} \quad \text{by definition of } m_{n_0}(j) \\ &= (1 - \epsilon)m_{n_0} + \epsilon p_{jj}^{(2n_0)} \quad \text{row sums are one.} \end{aligned}$$

This being true for every i , taking minimum over i , we see

$$m_{2n_0} \geq (1 - \epsilon)m_{n_0} + \epsilon p_{jj}^{(2n_0)} \quad (\diamond)$$

Similarly,

$$p_{ij}^{(2n_0)} = \sum_k (p_{ik}^{(n_0)} - \epsilon p_{jk}^{(n_0)}) p_{kj}^{(n_0)} + \sum_k \epsilon p_{jk}^{(n_0)} p_{kj}^{(n_0)} \leq (1 - \epsilon)M_{n_0} + \epsilon p_{jj}^{(2n_0)}.$$

This being true for every i , taking maximum over i , we see

$$M_{2n_0} \leq (1 - \epsilon)M_{n_0} + \epsilon p_{jj}^{(2n_0)} \quad (\blacktriangle)$$

The two inequalities $(\blacktriangle)$ and $(\diamond)$ prove

$$M_{2n_0} - m_{2n_0} \leq (1 - \epsilon)(M_{n_0} - m_{n_0}) \leq (1 - \epsilon)^2. \quad (\bullet)$$

It is now easy to see that for every k

$$M_{kn_0} - m_{kn_0} \leq (1 - \epsilon)^k. \quad (\bullet)$$

Because of the inequality $(\spadesuit)$ the above suffices to show that $p_{ij}^{(n)} \rightarrow \pi_j$, limit not depending on i . Thus there is a limit matrix and it has identical rows. Denoting the above limit by π_j and π the row vector $(\pi_j : j \in S)$ all rows of the matrix Π are π . $\blacksquare$

This π helps you to understand most of the things about the chain.

(i) Since we have finite state space, and each matrix P^n is stochastic matrix, the limit is also stochastic matrix. Hence row sums must be again one,

that is, $\sum \pi_j = 1$. Note that $\pi P = \pi$ because $P^n P = P^{n+1}$ and so (use finite state space) $\Pi P = \Pi$ or $\pi P = \pi$. To show each entry of π is strictly positive, observe $\pi P^{n_0} = \pi$. If you say $\pi_j = 0$ then $\pi_j = \sum \pi_i p_{ij}^{(n_0)}$ implies each $\pi_i = 0$ because each $p_{ij}^{(n_0)}$ is strictly positive. But π is a probability vector. Thus $\pi_j > 0$ for all j .

(ii) Above already shows $\pi P = \pi$. To show that the eigen subspace is one dimensional, Let η be another eigen vector, $\eta P = \eta$, not a multiple of π . Since $\pi P = \pi$ we see that for any integer k , $(\eta + k\pi)P = (\eta + k\pi)$. Since each entry of π is strictly positive, by choosing k large we can ensure that $\eta + k\pi$ has all entries positive. Observe that this is not a multiple of π , because if it were then η itself is so. Thus, to start with we can take η to be non-negative. Again, since η is non-zero vector, $c = \sum \eta_j \neq 0$ and η/c is a probability vector and eigen vector. This is also not a multiple of π , otherwise η would be so.

Thus we can safely assume that η is a probability vector. Now $\eta P = \eta$ gives $\eta P^n = \eta$ so that $\eta \Pi = \eta$ or $\sum_i \eta_i \pi_j = \eta_j$ or $\pi_j = \eta_j$.

(iii) $P(X_n = j | X_0 = i) \rightarrow \pi_j$ is just a restatement of $P^n \rightarrow \Pi$.

(iv) Suppose $X_0 \sim \pi$. Thus initially we have $P(X_0 = j) = \pi_j$. From what was discussed earlier $P(X_m = j) = \sum_i \pi_i p_{ij}^{(m)}$ which is the j -th entry of $\pi P^m = \pi$ and hence $P(X_m = j) = \pi_j$ or $X_m \sim \pi$. Not only for one time point, but any consecutive time points of the same length look same. Fix any n, m . Keep in mind $X_0 \sim \pi$ and we also now know $X_m \sim \pi$.

$$\begin{aligned} P(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) &= \\ P(X_0 = i_0)P(X_1 = i_1 | X_0 = i_0)P(X_2 = i_2 | X_0 = i_0, X_1 = i_1) \cdots \\ &= \pi_{i_0} p_{i_0 i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n} \\ P(X_m = i_0, X_{m+1} = i_1, \dots, X_{m+n} = i_n) &= \\ P(X_m = i_0)P(X_{m+1} = i_1 | X_m = i_0)P(X_{m+2} = i_2 | X_m = i_0, X_{m+1} = i_1) \cdots \\ &= \pi_{i_0} p_{i_0 i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n} \end{aligned}$$

where for the last equality we used the relations we proved earlier, loosely speaking, no matter what were the states in the past only the most recent matters.

Thus the process, no matter on which day you start looking at it, day zero or day m , it looks the same; provided its initial distribution is π .

(v) and (vi) involve calculation, not complicated, but need patience and concentration. We shall not do completely. First we explain (vi).

Let us start from j . We show that you surely return to j . That is

$$P(X_n \neq j, \forall n \geq 1 \mid X_0 = j) = 0. \quad (\bullet)$$

This is as follows. Get as earlier n_0 and ϵ such that all entries of P^{n_0} are larger than ϵ . Define events:

$$A_1 = \{X_1 \neq j, X_2 \neq j, \dots, X_{n_0} \neq j\}$$

$$A_2 = \{X_1 \neq j, X_2 \neq j, \dots, X_{n_0} \neq j, \dots, X_{2n_0} \neq 0\}$$

$$A_3 = \{X_1 \neq j, X_2 \neq j, \dots, X_{n_0} \neq j, \dots, X_{2n_0} \neq 0, \dots, X_{3n_0} \neq 0\}$$

etc. We show

$$P(A_k \mid X_0 = j) \leq (1 - \epsilon)^k \quad (\star)$$

If A is the event $\{X_n \neq j, \forall n \geq 1\}$, then clearly $A_n \downarrow A$ and so the above implies $P(A \mid X_0 = j) = 0$ as claimed.

$$\begin{aligned} P(A_1 \mid X_0 = j) &\leq P(X_{n_0} \neq j \mid X_0 = j) = 1 - p_{jj}^{(n_0)} \leq 1 - \epsilon \\ P(A_2 \mid X_0 = j) &\leq P(X_{n_0} \neq j, X_{n_0+1} \neq j, \dots, X_{2n_0} \neq j \mid X_0 = j) \\ &= \sum_{k \neq j} P(X_{n_0} = k, X_{n_0+1} \neq j, \dots, X_{2n_0} \neq j \mid X_0 = j) \\ &= \sum_{k \neq j} P(X_{n_0} = k \mid X_0 = j) P(X_{n_0+1} \neq j, \dots, X_{2n_0} \neq j \mid X_0 = j, X_{n_0} = k) \\ &= \sum_{k \neq j} P(X_{n_0} = k \mid X_0 = j) P(X_{n_0+1} \neq j, \dots, X_{2n_0} \neq j \mid X_{n_0} = k) \\ &\leq \sum_{k \neq j} P(X_{n_0} = k \mid X_0 = j) P(X_{2n_0} \neq j \mid X_{n_0} = k) \\ &\leq \sum_{k \neq j} P(X_{n_0} = k \mid X_0 = j) (1 - p_{kj}^{(n_0)}) \\ &\leq (1 - \epsilon) \sum_{k \neq j} P(X_{n_0} = k \mid X_0 = j) \leq (1 - \epsilon)^2. \end{aligned}$$

The same proof can be repeated for every k leading to $(\star)$

When the chain starts from j , that is, $P(X_0 = j) = 1$ then it makes sense to define the first return time T as:

$T = n$ iff $\{X_n = j; \quad \forall(1 \leq m < n) X_m \neq j\}$.
 and put $P(T = n) = f^{(n)}$ as was done in the Random walk discussion;
 with $f^{(0)} = 0$. Now from what was proved above, we conclude $\sum f^{(n)} = 1$.
 Just keep in mind that $f^{(n)}$ depends on j we fixed. Thus T is a legitimate
 random variable taking value n with probability $f^{(n)}$ and $E(T) = \sum n f^{(n)}$.
 We shall not keep writing $X_0 = j$, but keep this in mind when you talk about
 probabilities. question: is $E(T)$ finite? Answer: Yes, seen as follows. First
 observe from the earlier calculations, $P(T > n) \leq 1$ for $n = 1, 2, \dots, n_0$.
 $P(T \geq n) \leq (1 - \epsilon)$ for $n = n_0 + 1, n_0 + 2, \dots, 2n_0$
 $P(T \geq n) \leq (1 - \epsilon)^2$ for $n = 2n_0 + 1, 2n_0 + 2, \dots, 3n_0$ and so on.
 By one of the formulas we had earlier

$$\begin{aligned}
 E(T) &= \sum_1^{\infty} P(T > n) = \sum_1^{n_0} + \sum_{n_0+1}^{2n_0} + \sum_{2n_0+1}^{3n_0} + \dots \\
 &\leq n_0 + (1 - \epsilon)n_0 + (1 - \epsilon)^2 n_0 + \dots < \infty
 \end{aligned}$$

Thus mean return time to j is finite, denote by m_j (usually this is denoted
 by m_{jj} expected time to return to j starting from j). The claim is that
 $\pi(j) = 1/m_j$.

Here is the idea which we shall not execute. Start the chain from j , that
 is $X_0 = j$. Let T be the random time when we return to j . Consider the
 chain until this time, not including this time. That is we consider the cycle,
 $X_0, X_1, \dots, X_{T-1}$. (This is called cycle because $X_T = j$ and so from time
 T onwards you are starting from j once again and running the show). For
 each i , let $\eta(i)$ be the expected number of visits to i during this cycle. Note
 $\eta(j) = 1$, during this period you visited j only at time zero and never again
 — we have not included X_T in the present calculation. Moreover during
 times $0, 1, 2, \dots, T - 1$ you are at some state or other and so the total time
 spent in all states during this period adds to T , in other words expected
 time spent in all states during this period adds to $E(T) = m_j$. Thus η is
 non-negative vector with $\eta(j) = 1$ and $\sum_i \eta(i) = m_j$. The crucial thing is
 $\eta P = \eta$. By uniqueness of eigen vector (for eigen value one) we see η is a
 multiple of π . Or $\eta(i) = m_j \pi(i)$. In particular, $1 = \eta(j) = m_j \pi(j)$. That is
 $m_j = 1/\pi(j)$ as asserted.

Proof of (v) is a little more complicated. We show that, not only we are
 sure to return to j , but we return infinitely many times. Thus if $T_1, T_2, \dots$
 are successive return times then the cycles $Y_1 = (X_n : 0 \leq n < T_1), Y_2 =$
 $(X_n : T_1 \leq n < T_2), Y_3 = (X_n : T_2 \leq n < T_3), \dots$ are i.i.d. There is a

conceptual barrier here, because Y_n are not random variables. They are not even multidimensional variables. Their lengths are ‘random’. Some times Y_1 may have length 4, sometimes 8, sometimes 1000! Once you mature to cross this barrier, then proof is indeed trivial. One applies the Strong law of large numbers. We shall not do. However we shall do one example of strong law of large numbers for you to see. ■

Thus for finite state chain which is irreducible and aperiodic we know the details about the chain, what is its ultimate fate, what proportion of time does it spend in each state, how long does it take to return to a state etc. What if it is not aperiodic. We already saw that the distribution of X_n need not converge. However, not all is lost. Everything else can be said exactly as before. Here is the second fundamental theorem in Markov chains.

Theorem: For a finite state irreducible chain with transition matrix P ,

$$\frac{I + P + \cdots + P^{n-1}}{n} \rightarrow Q$$

for some stochastic matrix Q . Further $QP = PQ = QQ = Q$.

Further

- (i) Q has identical rows. If π is the first row, then it is a probability vector and $\pi P = \pi$.
- (ii) all entries of π are strictly positive. Eigen subspace corresponding to eigen value one is one dimensional, in particular, π is the only probability vector with $\pi P = \pi$.
- (iii) If $X_0 \sim \pi$ then $X_n \sim \pi$ for all n . More generally for all n, m , we have $(X_0, \dots, X_n) \sim (X_m, X_{m+1}, \dots, X_{n+m})$ if $X_0 \sim \pi$. Thus π is stationary or invariant distribution for the chain. We may not say steady state because the chain may not have a steady state, for example when period is not one and initial distribution is deterministic.
- (iv) For almost all sample points,

$$\frac{\#\{0 \leq m \leq n-1 : X_m = j\}}{n} \rightarrow \pi(j)$$

- (v) If $P(X_0 = i) = 1$, then there is surely there is an m such that $X_m = i$ and the expected return time equals $1/\pi(i)$. ■

Here is the proof. Let us denote

$$Q_n = \frac{I + P + \cdots + P^{n-1}}{n}$$

Since there are finitely many sequences, $\{q_n(i, j) : n \geq 1\}$, one for each pair (ij) in the finite state space we can take common convergent subsequence. Thus select $n_1 < n_2 < n_3 < \cdots$ such that

$$Q_{n_1}, Q_{n_2}, Q_{n_3}, \cdots$$

converges to say Q . Note that

$$PQ_n = \frac{P + P^2 + P^3 + \dots + P^n}{n} = Q_n - \frac{I}{n} + \frac{P^n}{n}$$

writing the above equation for our subsequence

$$PQ_{n_r} = Q_{n_r} - \frac{I}{n_r} + \frac{P^{n_r}}{n_r}$$

and taking limits we get $PQ = Q$. We used the fact that entries of P^n are all bounded between zero and one.

Similarly $QP = Q$. This in turn tells $QP^n = Q$ and taking averages we get $QQ_{n_r} = Q$, now taking limits we see $QQ = Q$. Thus

$$PQ = QP = QQ = Q \quad (\spadesuit)$$

It remains to show that the entire sequence (Q_n) converges. Suppose that for some (i^*j^*) , the entries $q_n(i^*, j^*)$ do not converge to $q_{i^*j^*}$ where $Q = (q_{ij})$ is the above sub-sequential limit matrix. So there is an $\epsilon > 0$ such that infinitely many terms are outside the interval $(q_{i^*j^*} - \epsilon, q_{i^*j^*} + \epsilon)$. Again boundedness of the sequence tells us that we can get a convergent subsequence of those infinitely many that lie outside this interval. of course they can not converge to $q_{i^*j^*}$. The upshot is that we can get a subsequence $m_1 < m_2 < m_3 < \dots$, such that $\{q_{i^*j^*}^{(m_r)}; r \geq 1\}$ converges to a number different from $q_{i^*j^*}$. By taking a further subsequence of this, we can safely assume that the matrices $Q_{m_1}, Q_{m_2}, \dots, Q_{m_r}, \dots$ converge, say to Q^* .

Of course Q^* has also similar properties as Q , namely,

$$PQ^* = Q^*P = Q^* \quad (\clubsuit)$$

Using $(\spadesuit)$, we see $P^m Q = Q$ and taking averages we see $Q_{m_r} Q = Q$ and taking limits we see

$$Q^* Q = Q \quad (\bullet).$$

Using $(\clubsuit)$, we see $Q^* P^n = Q^*$ and taking averages we see $Q^* Q_{n_r} = Q^*$ and taking limits we see

$$Q^* Q = Q^* \quad (\bullet\bullet).$$

Now $(\bullet)$ and $(\bullet\bullet)$ tell that $Q = Q^*$ which is a contradiction because they differ in the (i^*j^*) -th term.

This shows that the entire sequence Q_n converges. Since we have a finite matrix, row sum of limit equals limit row sum and hence equals one. Non-negativity is clear. Thus Q is stochastic matrix.

Other statements are proved in the same way as earlier. Because each Q_n is stochastic, so is Q . Let π be first row of Q . Clearly it is a probability vector. $QP = Q$ tells $\pi P = \pi$. each entry of π is positive can be seen as follows. Suppose $\pi(j) = 0$. Consider any i . By irreducibility, get m so that $P_{ij}^{(m)} > 0$. Use $\pi = \pi P^m$ to see, $\pi(j) = \sum_k \pi_k P_{kj}^{(m)}$. If $\pi(i) > 0$ then $\pi(i) P_{ij}^{(m)} > 0$ so that $\pi(j) > 0$. Thus each $\pi(i) = 0$ which is a contradiction because π is a probability vector.

Finally to prove π is unique proceed as earlier, if $\eta P = \eta$ then $\eta P^n = \eta$ and $\eta Q = \eta$ etc. We shall not discuss the last two statements of the theorem. ■

Of course in the irreducible case you can get this theorem from previous one. This is simply Cesaro's theorem or numbers: if $a_n \rightarrow a$ then $(a_1 + a_2 + \cdots + a_n)/n \rightarrow a$.

Ehrenfest revisited:

All the above discussion tells you that

(i) the Ehrenfest chain is heading to a steady state: as if each ball is placed at random in the two compartments.

(ii) Each state is visited infinitely often: That is you see k balls in H infinitely many times in (almost) any run of the chain for $0 \leq k \leq 2000$

if you are not careful, you think that these two conclusions are contradictory: a steady state in (i) and a chaotic behaviour in (ii). The point is that (i) is macroscopic behaviour, distribution of the state and not any specific run of the chain; whereas (ii) is a microscopic description in each run of the game. There is no contradiction. This was one of the sources of confusion in the initial days of the Boltzmann Theory. There are others objections too like: have you seen atoms/molecules? In mechanics if a force makes a particle move from A to B , then you can make it move from B to A by applying same force in reverse direction. So if you wish to explain heat through mechanics of motion then can you reverse from steady state of common temperature to original mixture of hot milk and cold water?

Graph Walk:

Consider a connected (undirected) graph G on a finite set of vertices V . Eventhough what we are going to describe works perfectly well even with loops and multiple edges, let us assume that there are no loops and multiple edges. Say that two vertices are neighbours if they are joined. Here is the walk on V : From any vertex move to one of its neighbours chosen at random. Thus if d_v is the degree of v , then $p_{vw} = 1/d_v$ if w is a neighbour of v and $p_{vw} = 0$ otherwise. This is irreducible. Its invariant distribution is given by

$$\pi_v = d_v/d; \quad d = \sum d_v$$

That this is invariant can be easily verified by using a simple useful fact.:
 For a chain with transition matrix P , if you can find a vector π with positive entries with $\pi_i p_{ij} = \pi_j p_{ji}$ for all i, j , then $\pi P = \pi$. This is because

$$\sum_i \pi_i p_{ij} = \sum_i \pi_j p_{ji} = \pi_j$$

Thus if you normalize π then it is an invariant probability vector. In the present case, $\pi_v p_{vw} = \pi_w p_{wv} = 1/d$ or zero according as v, w are neighbours or not.

Fun with Random Walk:

Now let us return to simple symmetric random walk (X_n) starting at zero in dimension one ($X_0 = 0$.) We start with a simple but powerful result. (we shall omit showing the condition $X_0 = 0$)

Theorem: $P(X_1 > 0, X_2 > 0, \dots, X_{2n} > 0) = \frac{1}{2} P(X_{2n} = 0)$. (♠)
 This immediately gives,

Theorem: $P(X_1 \neq 0, X_2 \neq 0, \dots, X_{2n} \neq 0) = P(X_{2n} = 0)$.

To see this you only have to note that the event on left is union of two disjoint sub events: all strictly positive or all strictly negative and both have same number of outcomes (reflection in x -axis) and now use previous Theorem.

This in turn leads to

Theorem: $f^{(2n)} = p^{(2n-2)} - p^{(2n)} = \frac{1}{2n-1} p^{(2n)}$ and $\sum f^{(2n)} = 1$.

Observe that this gives another proof of recurrence because of the last equation. To prove the first equation,

$$f^{(2n)} = P(X_1 \neq 0, X_2 \neq 0, \dots, X_{2n-1} \neq 0, X_{2n} = 0)$$

$$= P(X_1 \neq 0, \dots, X_{2n-2} \neq 0) - P(X_1 \neq 0, \dots, X_{2n-1} \neq 0, X_{2n} \neq 0)$$

note $X_k \neq 0$ for odd k . Now use earlier Theorem

$$= p^{(2n-2)} - p^{(2n)} \text{ as stated.}$$

Last sentence follows because $\sum f^{(2n)}$ is a telescopic sum

$$[p^{(0)} - p^{(2)}] + [p^{(2)} - p^{(4)}] + [p^{(4)} - p^{(6)}] + \dots = p^{(0)} = 1$$

. (Think: the following is also a telescopic sum but the equation is false

$$(1 - 2) + (2 - 4) + (4 - 6) + (6 - 8) + \dots = 1$$

why is it false here and why is it true for us?)

To prove the middle equation of the theorem

$$p^{(2n-2)} = 2^{-(2n-2)} \frac{(2n-2)!}{(n-1)!(n-1)!} = \frac{2n}{2n-1} 2^{-2n} \frac{(2n)!}{n!n!}$$

so that

$$f^{(2n)} = \frac{2n}{2n-1} p^{(2n)} - p^{(2n)} = \frac{1}{2n-1} p^{(2n)}$$

. as required.

Theorem: The Expected return time is infinity.

This is easy because the return time distribution is given by $f^{(2n+1)} = 0$ and $f^{(2n)} = \frac{1}{2n-1} p^{(2n)}$ and $f^{(0)} = 0$. Thus expected return time equals

$$\sum 2n f^{(2n)} = \sum \frac{2n}{2n-1} p^{(2n)} = \infty$$

the last equation is from the fact that

$$\frac{2n}{2n-1} p^{(2n)} \sim p^{(2n)}; \quad \sum p^{(2n)} = \infty$$

. This completes the proof.

We can also show the following: Let u, v be any two states. Starting at u , you will surely return to v ; however the expected return time is infinite.

Now let us prove theorem (♠):

Left side equals

$$\begin{aligned} & 2^{-2n} \times \text{number of paths from } (0, 0) \text{ of length } 2n \text{ strictly above } x\text{-axis} \\ &= 2^{-2n} \times \sum_{r \geq 1} [\text{number of paths from } (0, 0) \rightarrow (2n, 2r) \text{ strictly above } x\text{-axis}] \\ &= 2^{-2n} \times \sum_{r \geq 1} [\text{number of paths from } (1, 1) \rightarrow (2n, 2r) \text{ strictly above } x\text{-axis}] \\ &= 2^{-2n} \times \sum_{r \geq 1} [\# \text{ paths } (1, 1) \rightarrow (2n, 2r) \text{ minus } \# (1, -1) \rightarrow (2n, 2r)] \end{aligned}$$

used reflection principle.

$$= 2^{-2n} \times \sum_{r \geq 1} [\# (0, 0) \rightarrow (2n-1, 2r-1) \text{ minus } \# (0, 0) \rightarrow (2n-1, 2r+1)]$$

sum is telescopic, last entry is zero, only first entry remains

$$\begin{aligned}
&= 2^{-2n} \times [\text{number of paths from } (0, 0) \rightarrow (2n-1, 1)] \\
&= 2^{-2n} \times \binom{2n-1}{n} \quad \text{use } \binom{2n-1}{n} = \binom{2n}{n}/2 \\
&= P(X_{2n} = 0)/2 \text{ as stated.}
\end{aligned}$$

There are several interesting results. We shall discuss one more, namely the proportion of time spent on positive side in $2n$ tosses.

In tosses of a fair coin, you expect that half of the time you will be leading and half of the time you will be losing. But interestingly, even though it is true for ‘infinitely many’ tosses; it is not true for any finite number of tosses. In fact in finite number of tosses this has the least probability. We shall now see this. But first a companion to Theorem (♠).

Theorem: $P(X_1 \geq 0, X_2 \geq 0, \dots, X_{2n} \geq 0) = P(X_{2n} = 0)$.

This is observed as follows. By earlier Theorem,

$$P(X_{2n} = 0) = 2P(X_1 > 0, \dots, X_{2n} > 0)$$

so that

$$\begin{aligned}
2^{2n} P(X_{2n} = 0) &= 2 \text{ number of paths } (0, 0) \rightarrow (2n, -) \text{ strictly above } x\text{-axis} \\
&= 2 \text{ number of paths } (1, 1) \rightarrow (2n, -) \text{ strictly above } x\text{-axis} \\
&= 2 \text{ number of paths } (0, 0) \rightarrow (2n-1, -) \text{ on or above } x\text{-axis} \\
&= 2 \cdot 2^{2n-1} P(X_1 \geq 0, \dots, X_{2n-1} \geq 0) \\
&= 2 \cdot 2^{2n-1} P(X_1 \geq 0, \dots, X_{2n-1} \geq 0, X_{2n} \geq 0) \\
&\text{Observe if } X_{2n-1} \geq 0, \text{ then } X_{2n-1} > 0 \text{ and hence } X_{2n} \geq 0. \\
&= 2^{2n} P(X_1 \geq 0, \dots, X_{2n-1} \geq 0, X_{2n} \geq 0) \\
&\text{completing the proof. } \blacksquare.
\end{aligned}$$

Let us consider $2n$ tosses of a fair coin. We say that during $[m, m+1]$ we are on positive side if $X_m \geq 0$ and $X_{m+1} \geq 0$. Note that then the entire path during this period is on positive side. Clearly, the time spent on positive side is even: $0, 2, 4, \dots, 2n$. Let $t_{2k, 2n}$ denote the probability that we spend $2k$ units on positive side, $0 \leq k \leq n$.

Theorem: Discrete arc sine law $t_{2k, 2n} = p^{(2k)} p^{(2n-2k)}$.

Recall $p^{(2m)} = P(X_{2m} = 0) = \binom{2m}{m}/2^{2m}$. Why is the above theorem interesting and why is it called arc sine law? A simple calculation with binomial coefficients shows you that these numbers $t_{2k, 2n}$ decrease as k increases from zero to $n/2$ and then start increasing back in a symmetrical fashion.

If n is odd, then this means you have same answer for $k = (n - 1)/2$ and $k = (n + 1)/2$.

Suppose we take $0 < a < b < 1$ and ask: What are the chances that the proportion of time spent on the positive side is between a and b ? The answer is

$$\begin{aligned} \sum_{a \leq \frac{k}{n} \leq b} t_{2k, 2n} &= \sum_{a \leq \frac{k}{n} \leq b} p^{(2k)} p^{(2n-2k)} \\ &\sim \sum_{a \leq \frac{k}{n} \leq b} \frac{1}{\sqrt{\pi k}} \frac{1}{\sqrt{\pi(n-k)}} \end{aligned}$$

(this was seen using Stirling)

$$\sim \frac{1}{\pi} \sum_{a \leq \frac{k}{n} \leq b} \frac{1}{n} \frac{1}{\sqrt{\frac{k}{n}(1 - \frac{k}{n})}} \sim \frac{1}{\pi} \int_a^b \frac{1}{\sqrt{x(1-x)}} dx$$

because the sum is nearly a Riemann sum for the integral

$$= \frac{2}{\pi} [\sin^{-1} \sqrt{b} - \sin^{-1} \sqrt{a}]$$

Of course with so many approximations, you may doubt if it is correct. Yes. The following is true but we shall not prove. To rigorously show this you need an idea for the error involved in using Stirling.

Theorem: Let $0 < a < b < 1$. If α_n is the probability that, in $2n$ tosses of a fair coin, the proportion of time spent on positive side is between a and b ; then

$$\alpha_n \rightarrow \frac{1}{\pi} \int_a^b \frac{1}{\sqrt{x(1-x)}} dx = \frac{2}{\pi} [\sin^{-1} \sqrt{b} - \sin^{-1} \sqrt{a}]$$

You might wonder my loose statement between a, b . Do I include a, b or not? This makes no difference. We shall not prove this.

Let us now prove the discrete arcSine law. Whatever be n , For $k = 0$ it follows from the previous theorem, remember $p^{(0)} = 1$. For $k = n$ it follows by reflection argument (not reflection principle).

Consider the statement: for $0 \leq k \leq n$, $t_{2k, 2n} = p^{(2k)} p^{(2n-2k)}$.

We prove this statement by induction on n , (there is no induction on k). For $n = 1$, we have already proved because then the only possibilities are $k = 0$ and $k = n$, (why bother, do it directly!). So let $n \geq 2$. Suppose we proved it for $1, 2, \dots, n-1$. We prove it for n . So take any k as stated. If $k = 0$ or $k = n$ it is done. So assume that $1 \leq k \leq n-1$. If A is the set of

paths satisfying the condition: spend exactly $2k$ units on positive side, then we need to calculate $|A|$. Remember that every path in A must hit zero at sometime after starting because $1 \leq k \leq n-1$. Let A_r be those paths that hit zero for the first time at $2r$. There are two possibilities: till $2r$ the path is either on positive side or on the negative side. Let A_r^+ be those paths that stay on positive side till $2r$ and A_r^- be those that stay on negative side till $2r$. Observe that, since path spends exactly $2k$ units on positive side, we have $A_r^+ = \emptyset$ for $r > k$ and $A_r^- = \emptyset$ for $r \geq n-k$. Thus

$$A = \bigcup_{r=1}^k A_r^+ \cup \bigcup_{r=1}^{n-k} A_r^-$$

The above union is a disjoint union, so that

$$|A| = \sum_{r=1}^k |A_r^+| + \sum_{r=1}^{n-k} |A_r^-|$$

All paths in A_r^+ are obtained as follows: take any path that returns to zero for the first time at time $2r$ (and there are $2^{2r} f^{(2r)}$ such paths) which stay on positive side till then; there are $2^{2r-1} f^{(2r)}$ such paths. Combine this with any path of length $(2n-2r)$ that spends exactly $(2k-2r)$ units on the positive side – there are $2^{(2n-2r)} p^{(2k-2r)} p^{(2n-2k)}$ such paths by induction hypothesis, note $n-r < n$ to use induction. Thus

$$\begin{aligned} \sum_r |A_r^+| &= \sum_r 2^{2r-1} f^{(2r)} 2^{2n-2r} p^{(2k-2r)} p^{(2n-2k)} \\ &= \frac{1}{2} 2^{2n} p^{(2n-2k)} \sum f^{(2r)} p^{(2k-2r)} = \frac{1}{2} 2^{2n} p^{(2n-2k)} p^{(2k)} \end{aligned}$$

where we used the renewal equation. Similarly

$$\sum_r |A_r^-| = \frac{1}{2} 2^{2n} p^{(2k)} p^{(2n-2k)}$$

so that

$$t_{2k,2n} = \frac{|A|}{2^{2n}} = p^{(2n-2k)} p^{(2k)}$$

completing the proof.

There are several other interesting results on RW and Markov chains. you can consider random walks with asymmetric probabilities. You can think of moving pretty fast, rather than once an hour, but with smaller jumps. ask

if there is a limit which can be considered ‘continuous random walk’. Yes. You can consider Markov chains on infinite state space.

There is an interesting concept of self avoiding walk. Self avoiding walk is a random walk path with no repetition of vertices. An n -length self avoiding path is a nearest neighbour random walk path of length n where there are no repetitions of states – starting from 0. This mathematical concept of self avoiding walks is due to Nobel Laureate Paul Flory who was a polymer chemist. The number of such paths of length n is very important. In one dimension, there are only two such paths. In dimension d , one can show that there is a number $c(d)$ such that the number of such paths is of the order of $[c(d)]^n$.

We shall conclude our discussion of Markov chains and Random walks here.

Matters to think:

When we initially discussed random variables, they were real valued. Then we discussed joint distributions, say joint pdf of X, Y , or in the discrete case the bivariate table. I said we can think of (X, Y) as ‘one’ quantity taking values in R^2 rather than two things taking real values. In other words, you can think of (X, Y) as a random vector, two dimensional.

When we came to markov chains, I defined it as a sequence $(X_n, n \geq 0)$ of random variables; X_0 is the initial starting random variable. there is some preassigned rule how to proceed. you can calculate several probabilities and joint probabilities (see what we did). There are two new things that happened here.

I stressed only one thing, namely, we shall not look at probability space (this already happened when we discussed branching process, when we discussed random variables with pdf). Our purpose is to understand the phenomenon and calculate probabilities and answer questions, rather than seeing where these random variables are defined. (Just as our purpose is to learn probability rather than focussing on the place where we are learning it, like lecture hall etc) Thus we have discussed how to calculate $P(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n)$. we have also discussed how to answer complicated questions using these. For example in RW chances of ‘ever returning’ is a complicated question, depends on entire walk, on all the infinitely many steps. But we did it using what we know. If you look back, the same thing happened in Branching process.

The second thing that happened which I left quietly low key is this:

The random variables of the Markov chain (X_n) take values in the state space S . Thus these could be taking values in the set of all permutations of $\{1, 2, \dots, 52\}$ or the set of all graphs or the set of chairs in this room. Thus the second thing that happened is random variables are no longer real valued, not even euclidean space valued. They can take values anywhere! this ‘anywhere’ has to be qualified when you learn advanced material, but for the things we study, there is no need to complicate life; a random variable can take values anywhere, not necessarily real.

We return to continuous random variables. Many of you still seem to think that if X is a random variable with pdf f , then $P(X = a) = f(a)$. You yourself will realize how meaningless is the statement if you consider, for example, $\text{uniform}(0, 1/2)$ random variable. Its pdf is $f(x) = 2$ for $0 < x < 1/2$. More meaningless is the statement when X, Y have joint pdf $f(x, y)$ then

$$P(Y = b | X = a) = P(X = a, Y = b) / P(X = a)$$

The right side has zero both on top and bottom. Make sure you write meaningful things.

Let us recapitulate about our life before Markov chains.

We first considered experiments with finitely many or countably infinitely many outcomes, defined Sample space, outcomes, events, probability of outcomes, of events, conditional probability of events and their rules, independence of events. We then defined random variables as real valued functions on the sample space (random variable need not take real values, but we do not complicate life unless necessary). To answer questions about rvs we described the concept of distribution of rv so that we need not look at the sample space and then consider all outcomes that satisfy the question asked about the random variable and then calculate probability of that event. Several rvs that arise in practice are discussed like; Bernoulli, binomial, geometric, negative binomial, Poisson, Hypergeometric. To summarize the complicated probability mass function (pmf) into a few understandable numbers, we defined mean, variance and moments. Discussed Chebyshev's inequality, Chernoff bound, and some uses.

We realized that to understand relations between two random variables, it is not enough to look at their probability mass functions separately. Defined joint distribution. One example is the Multinomial. Concepts for events are then imitated for random variables, like conditional distribution, independence of rvs. Rules of Conditional expectation allowed us to calculate some complicated-looking expectations. Defined covariance and correlation and saw their importance in prediction. We discussed WLLN and CLT. We concluded with pgf, mgf and discussion of branching process. This is the first time we really calculated probability of an event that is described using the entire infinite sequence of rvs. This was done by using pgf and not by looking at any sample space.

We realized that many rvs in practice take too many values and discussed random variables with probability density function (pdf), where probabilities are given by areas under the pdf curve. Such a rv takes any one prespecified value with probability zero. Whereas the discrete setup earlier needed summations and no more; this setup with pdf needs integration. One essential difference is that earlier (except in branching processes) we saw the sample space, we knew what function on the space — rv — we are talking about, but now the sample space is out of the picture. We look at the pdf and answer all questions asked. However there is a sample space, there is a probability for

events satisfying: (i) zero for empty set, (ii) one for the whole sample space, and (iii) adds up for countably many disjoint events. And rvs are still functions defined on the sample space. We only look at their distributions and not at where they are defined. We then discussed standard rvs: Uniform, Beta, Exponential, Gamma, Cauchy, Normal etc. Defined usual quantities like expectation, variance, moments etc. Then we discussed joint densities, marginals, conditionals etc. Change of variables is an important tool that allowed us to calculate distributions of functions of independent variables. we Briefly discussed multivariate normal.

Are there two theories: discrete rvs and rvs with pdf? NO. We could cook up rvs which are neither discrete nor have a pdf. Then we defined the concept of distribution function for any rv (as opposed to density function or mass function). Calculated for many known rvs. We observed distribution functions are non-decreasing, with limits zero and one at the two ends of real line, and right continuous at every point. We accepted without proof: given any such function, there is a random variable with the given function as its distribution function. Thus the concept of distribution function subsumes discrete rvs, rvs with pdf, and all others too. We have seen how to answer questions about the rv using its distribution function; how to recover pmf or pdf.

Here are some results in the pdf case, analogous to discrete case. Of course these are true for rvs which are none of these two kinds too; but I need to define expectation etc. We shall not.

Chebyshev: If X is a rv with pdf, and $a > 0$ then $P(|X| \geq a) \leq E(|X|)/a$.
If $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$ then $P(|X - \mu| \geq a) \leq \sigma^2/a^2$

Indeed if f is the pdf of X , then

$$\begin{aligned} E(|X|) &= \int |x|f(x)dx = \int_{(-a,a)^c} |x|f(x) + \int_{(-a,a)} |x|f(x) \\ &\geq \int_{(-a,a)^c} |x|f(x) \geq a \int_{(-a,a)^c} f(x) = aP(|X| \geq a) \end{aligned}$$

For the second part you use $P(|X - \mu| \geq a) = P(|X - \mu|^2 \geq a^2)$ ■

Just as in the discrete case this leads to WLLN:

WLLN: If X_n is a sequence of iid rvs having pdf, with expectation μ and finite variance, then for any given $\epsilon > 0$

$$P\left(\left|\frac{\sum_1^n X_i}{n} - \mu\right| > \epsilon\right) \rightarrow 0$$

The proof remains the same: Chebyshev and the facts variance of sum is sum of variances and variance of aX is a^2 times variance of X .

Just as in the discrete case we can define mgf for a rv X with density f by

$$M_X(t) = E(e^{tX}) = \int e^{tx} f(x) dx$$

Again, as in the discrete case, this may be finite only for some values of t , always $M_X(0) = 1$.

Chernoff bound: If X is a rv with pdf, and $a > 0$ then

$$P(X \geq a) \leq M_X(t)e^{-ta} \quad t > 0$$

This is because $P(X \geq a) = P(e^{tX} \geq e^{ta})$ because $t > 0$ and now use Chebyshev.

Here is an application: Suppose (X_n) be i.i.d. rvs taking values ± 1 with equal probability. We have already seen that the above inequality gives us, denoting $S_n = \sum_1^n X_i$ and choosing a specific t ,

$$P\left(\left|\frac{S_n}{n}\right| \geq a\right) \leq 2e^{-na^2/2} \quad (\spadesuit)$$

We shall now invoke a nice theorem due to Borel and also due to Cantelli and use it to prove a profound statement.

Borel-Cantelli Lemma: If $(A_n, n \geq 1)$ are events in a probability space, and $\sum P(A_n) < \infty$ then $P(\limsup A_n) = 0$. (\clubsuit)

For any sequence of sets (A_n) we define $\limsup A_n$ to be the set of all those points which belong to infinitely many of these sets. For example if $A_n = (0, 1/2]$ when n is odd and $A_n = [1/2, 1)$ when n is even, then $\limsup A_n = (0, 1)$. If $A_n = \left(\frac{n}{n+1}, \frac{n^2}{n^2+1}\right)$ for all n , then $\limsup A_n = \emptyset$.

Thus to understand the event $\limsup A_n$, Take an outcome ω . See whether it is in A_n or not. Do it for all n . If it is in only finitely many, then we do not put it in $\limsup A_n$. if it is in infinitely many then we put it in $\limsup A_n$. Proof of the lemma above you can all discover! Whatever m you take, we have $\limsup A_n \subset \cup\{A_i : i \geq m\}$ and hence $P(\limsup A_n) \leq \sum_{m=\infty}^{\infty} P(A_i)$. Since the grand sum $\sum P(A_n)$ is finite, the tail sums converge to zero and

can be made as small as we please. On the other hand $P(\limsup A_n)$ has nothing to do with m . Hence $P(\limsup A_n) = 0$.

Take $a = 1$ in $(\spadesuit)$ and $A_n = (|\frac{S_n}{n}| \geq 1)$ and $B_1 = \limsup A_n$. then $(\spadesuit)$ tells us $\sum P(A_n) < \infty$ and so $(\clubsuit)$ tells us $P(B_1) = 0$. Observe that if an outcome ω is not in B_1 , then there is a stage (may depend on your ω) after which $(|\frac{S_n(\omega)}{n}| \leq 1)$. (★1)

Take $a = 1/2$ in $(\spadesuit)$ and $A_n = (|\frac{S_n}{n}| \geq 1/2)$ and $B_2 = \limsup A_n$. then $(\spadesuit)$ tells us $\sum P(A_n) < \infty$ and so $(\clubsuit)$ tells us $P(B_2) = 0$. Observe that if an outcome ω is not in B_2 , then there is a stage (may depend on your ω) after which $(|\frac{S_n(\omega)}{n}| \leq 1/2)$. (★2)

In general, Take and integer $p > 1$ and $a = 1/p$ in $(\spadesuit)$ and the event $A_n = (|\frac{S_n}{n}| \geq 1/p)$ and $B_p = \limsup A_n$. then $(\spadesuit)$ tells us $\sum P(A_n) < \infty$ and so $(\clubsuit)$ tells us $P(B_p) = 0$. Observe that if an outcome ω is not in B_p , then there is a stage (may depend on your ω) after which $(|\frac{S_n(\omega)}{n}| \leq 1/p)$. (★p)

Now consider $B = \cup B_p$. As we have already seen earlier, if you take countably many events each having probability zero, then their union has probability zero. Thus $P(B) = 0$ so that $P(B^c) = 1$ Now if an outcome $\omega \in B^c$ then it is not in any of the B_p ; so whatever $p \geq 1$ you take, after some stage $(|\frac{S_n(\omega)}{n}| \leq 1/p)$. This means that for such an outcome ω , we have $\frac{S_n(\omega)}{n} \rightarrow 0$. Thus we have

$$P\left(\omega : \frac{S_n(\omega)}{n} \rightarrow 0\right) = 1$$

This is known as the Strong Law of Large Numbers for our sequence of Bernoulli variables (X_n) (which have mean zero and are i.i.d). Even if we consider iid rvs (Y_n) taking values zero/one with equal probability so that the mean is $1/2$, you can say

$$P\left(\omega : \frac{\sum_1^n Y_i(\omega)}{n} \rightarrow 1/2\right) = 1$$

Indeed then $X_n = 2Y_n - 1$ takes values ± 1 with equal probability and

$$\frac{\sum_1^n Y_i(\omega)}{n} \rightarrow 1/2 \quad \text{iff} \quad \frac{\sum_1^n X_i(\omega)}{n} \rightarrow 0.$$

We have shown that the binary digits are iid zero/one with equal probability, for a number selected at random from $(0, 1)$. Thus those numbers $x \in (0, 1)$ for which the frequency of each binary digit (in its binary expansion) equals $1/2$ has probability one.

You can make similar statements about decimal expansions too and soon realise how little we know: Plenty of numbers have nice properties about digit frequency, but if we are asked to show/exhibit one such specific number, we have hard time.

Here are two fundamental theorems of (elementary) probability: Laws of Large Numbers which appears in two flavours weak/strong and Central Limit Theorem. There is a third one Law of Iterated Logarithm which we shall not discuss.

WLLN: If $(X_n, n \geq 1)$ are iid mean μ and finite variance, then

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{X_1 + \cdots + X_n}{n} - \mu\right| > \epsilon\right) = 0 \quad \forall \epsilon > 0.$$

SLLN: If $(X_n, n \geq 1)$ are iid mean μ then

$$P\left(\lim_{n \rightarrow \infty} \frac{X_1 + \cdots + X_n}{n} = \mu\right) = 1$$

CLT: If $(X_n, n \geq 1)$ are iid mean μ and finite non-zero variance σ^2 , then

$$P\left(a < \frac{X_1 + \cdots + X_n - n\mu}{\sqrt{n} \sigma} < b\right) \rightarrow \int \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad a < b \in R.$$

We proved WLLN both in the discrete and pdf case. But it is true always. We proved SLLN for Bernoulli variables taking two values with equal probabilities. But it is true always. CLT we have not proved.

WLLN tells you that given any precision ϵ , you can show a large event where the observed average is close to the common expectation μ upto the required precision. Here large means probability as close to one as required; of course the closer to one you want, the larger n I need to take. The event where the required precision is achieved changes with n and it does not assure us of any one specific outcome where the averages converge to μ . On the other hand SLLN produces a probability one event in which for every outcome, the observed averages converge to μ .

The advantage with WLLN is that you are calculating probabilities of event that depends on only finitely many rvs at a time. Your assertion is

that the sequence of numbers so obtained by calculating probabilities converges to zero. Whereas in SLLN, even to describe the event you need to look at all the random variables at one go and check if the averages are converging to zero, only then put such a sample point in the event. Having done this, you calculate probability of the event.

CLT tells that any standardized large sum behaves like a standard normal. Large sum means $X_1 + \dots + X_n$ where n is large. Standardizing means subtract its mean and divide by its standard deviation so the resulting random quantity has mean zero and variance one. Behaves means the chance this lies in an interval is nearly equal to the area under standard normal curve.

With valid reasons these are fundamental.

If (X_n) iid rvs then this is also referred to as a sample from their common distribution as explained in the discrete case earlier.

LLN at work:

(A) math agrees with our intuition

We said that we are going to build a mathematical theory of chance experiments. We defined probability, independence etc and developed mathematics of tossing a coin 10000 times independently and so on. When we felt chance of heads is p we are aware that the proportion of heads need not exactly be p , but it is close to p — in the sense that if you toss a ‘large’ number of times then the proportion is close to p . This is precisely what WLLN tells you. SLLN tells you that if you do indeed toss infinitely many times then the observed partial proportions do actually converges to p .

(B) Estimation

Many things we cannot observe but we need to know. Laws of large numbers gives you a justification for taking the observed average as a good estimate for the unknown value. Let us be clear, most of the things we do not know, What we know is only through observations.

(C) understanding numbers

Let us assume that we have a mechanism to pick a number at random from the interval $(0, 1)$ — a uniform number generator. An algorithm that picks a number in the interval $(0, 1)$ and you can show: the chances that the number given by the algorithm falls in a subinterval equals length of that subinterval.

Then pick two such numbers independently, say X, Y . I say (X, Y) is

uniform over the square $(0, 1) \times (0, 1)$ = set of points (x, y) in the plane with $0 < x < 1$ and $0 < y < 1$. If you take any sub-rectangle $(a, b) \times (c, d)$, the chances that selected point falls in this rectangle equals (by independence) product of the two quantities: chances that X falls in (a, b) and chances Y falls in (c, d) . This is $(b - a) \times (d - c)$, area of the rectangle. From this you can actually show that the chances of (X, Y) falling in a triangle equals the area of the triangle; the chances that it falls in any ‘region’ equals area of that region.

Now pick a large number of points $(x_1, y_1), \dots, (x_{10,000}, y_{10,000})$. Calculate the proportion f of these points which satisfy $x_i^2 + y_i^2 \leq 1$. Then this is a good approximation for the number $\pi/4$. Because area of the region $(x^2 + y^2 \leq 1)$ in the square is $\pi/4$. SLLN tells you the observed proportion is close to the expectation. We are now considering the random variables $Z_i = 1$ if (X_i, Y_i) falls in the quarter circle above and $Z_i = 0$ if it does not. Then $E(Z_i) = \pi/4$. This gives you an idea of the number $\pi/4$ and hence of π . Such methods are called Monte-Carlo methods or simulations.

(D) understanding integrals

Evaluate

$$\int_0^1 \frac{\sqrt[3]{x} \sin(x^{29}) e^{57x}}{1 + x^{32}} dx$$

We can evaluate only very very few integrals. But in some calculations it is important ‘to get an idea’ of the value. So what do we do? Let X be uniform $(0, 1)$ random variable. Consider the random variable

$$Z = \frac{\sqrt[3]{X} \sin(X^{29}) e^{57X}}{1 + X^{32}}$$

Our problem is simply to find $E(Z)$. Remember $Z = \varphi(X)$ where φ is the integrand in the above integral and so

$$E(Z) = E(\varphi(X)) = \int_0^1 \varphi(x) dx$$

But now SLLN tells you if you take an independent sample $X_1, \dots, X_n$ from $(0, 1)$ and set $Z_i = \varphi(X_i)$ then this will be i.i.d. random variables having the same distribution as that of Z . Thus $(Z_1 + \dots + Z_n)/n$ is a good estimate of the integral. This is another example of Monte-Carlo method.

(E). Binary digit frequency:

If X is uniformly chosen from $(0, 1)$ then its binary digits (X_i) are also random takes values 0/1 with probability 1/2 each. Clearly SLLN now tells

us

$$\frac{X_1 + \cdots + X_n}{n} \rightarrow \frac{1}{2}$$

with chance one.

Same holds even for decimal digits. Eventhough there are plenty and plenty of numbers in whose decimal expansion several digits do not appear at all, SLLN tells us that those numbers for which each of the ten digits appears with frequency $1/10$ has probability one, when you pick a number at random from unit interval. Nothing special with binary and decimal. Let $r > 1$ be an integer. You know from analysis that any number $0 < x < 1$ can be written as

$$x = \frac{x_1}{r} + \frac{x_2}{r^2} + \cdots + \frac{x_n}{r^n} + \cdots$$

where each x_i is one of the integers $\{0, 1, \dots, r-1\}$. These x_i are called digits of x expanded to the base r and we also write

$$x = 0 . x_1 x_2 x_3 \cdots x_n \cdots$$

Now x_i is called the i -th digit of x to base r . As in above expansion, no number has more than two expansions; if a number has two expansions then one must end with all zeros and the other with all $(r-1)$. We take this later nonterminating expansion. If X is picked at random from the interval $(0, 1)$ then, these digits X_i are also random variables, taking r values $\{0, 1, 2, \dots, r-1\}$ with equal probabilities and they are i.i.d. You can show, with chance one each of the r digits appears with frequency $1/r$.

We can say much more. For example go back to decimal expansion. We saw frequency of ‘single’ digits. You can ask frequency of patterns. For example take pattern (187). I say this pattern occurs at k -th place in the decimal expansion of x , if $(x_{i-2}, x_{i-1}, x_i) = (1, 8, 7)$. That is, $x_{i-2} = 1, x_{i-1} = 8, x_i = 7$. You can count frequency of this pattern till n . If you pick a number at random from $(0, 1)$ then with chance one, this frequency has a limit as $n \rightarrow \infty$ and the limit equals $1/1000$. Note that there are 1000 patterns of length three. Thus you can think of any pattern of any length in any expansion and climb a beautiful peak without any trouble.

The interesting point is the following: For the numbers we know like e or π does every decimal digit occur with frequency $1/10$. I have not met anyone who knows the answer.

CLT at work:

(A) universality:

The first thing is that CLT shows you universality of the Normal distribution. No matter what your random variables are (if they are not too large) if they are independent, then their standardized sum looks like a standard normal. This explains the central role played by the normal distribution in statistics.

(B) Calculating probabilities:

We have seen it earlier while discussing discrete random variables. We can use CLT to calculate probabilities approximately. If $X \sim B(n, p)$ then it is sum of n i.i.d. random variables each taking values one and zero with probabilities p and $(1 - p)$ respectively. Hence CLT tells you

$$P\left(a < \frac{X - np}{\sqrt{np(1-p)}} < b\right) \sim \int_a^b \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

For example I toss a fair coin 10,000 times. The expected number of Heads is 5,000; but never is it exact (in general). So wish to know the chances that number of heads is within ± 100 of expected value. That is want $P(4900 < X < 5100)$. Since we have binomial probabilities, we can write the exact answer

$$\sum_{k=4901}^{5099} \binom{10,000}{k} 2^{-10,000}$$

But what is the worth of the exact answer. Do you get any idea how large or how small it is? It is better to have a close approximate answer that we can understand rather than a complicated exact answer that we do not understand. CLT tells you

$$\begin{aligned} P(4900 < X < 5100) &= \\ P\left(\frac{4900 - 5000}{\sqrt{10,000 \times 1/4}} < \frac{X - 5000}{\sqrt{10,000 \times 1/4}} < \frac{5100 - 5000}{\sqrt{10,000 \times 1/4}}\right) \\ P\left(-2 < \frac{X - 5000}{\sqrt{10,000 \times 1/4}} < +2\right) &\sim \int_{-2}^2 \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \\ &\sim 0.95 \end{aligned}$$

The amazing thing about this is that we do not ask for the common distribution of our Z 's. They could be binomial or uniform or exponential etc.

(C) confidence intervals:

We can use CLT to provide how much confidence we put in our estimate being close to the true value. For example, Suppose Z is a random variable

with mean μ and variance σ^2 . We said (WLLN and SLLN) that we make independent observations on Z , say, $Z_1, Z_2, \dots, Z_n$ and estimate the unknown mean μ by observed average $\bar{Z} = (Z_1 + \dots + Z_n)/n$. How much confidence you will put in your prediction?

Let us rewrite what CLT says. First note that for the standard normal curve the area under $(-1, 1)$ is approximately 0.67; area under $(-2, 2)$ is approximately 0.95; area under $(-3, 3)$ is approximately 0.99. By CLT, with $a = -2$ and $b = 2$

$$P\left(-\frac{2\sigma}{\sqrt{n}} < (\bar{Z} - \mu) < +\frac{2\sigma}{\sqrt{n}}\right) \sim 0.95$$

You can write $-a < (\bar{Z} - \mu) < b$ as either $\mu - a < \bar{Z} < \mu + b$ or as $(\mu < \bar{Z} + a \ \& \ \mu > \bar{Z} - b)$, that is $(\bar{Z} - b < \mu < \bar{Z} + a)$. We choose the later way and express the earlier display as (take $a = b = 2\sigma/\sqrt{n}$) ;

$$P\left(\bar{Z} - 2\frac{\sigma}{\sqrt{n}} < \mu < \bar{Z} + 2\frac{\sigma}{\sqrt{n}}\right) \sim 0.95$$

In other words, with 95% confidence we can say that the **unknown** value μ is within $2\sigma/\sqrt{n}$ of the **known** average $\bar{Z}$. Of course if σ is unknown then you estimate it too!

(D) Testing and wrong decisions:

I have a coin. I am not estimating p . Tejas says that $p = 1/3$ and Sidharth says $p = 1/2$. It is known that p is one of these two values. I want to test who is close to the truth. I toss a large number of times, say 10,000 and observe the proportion of Heads f . If f is close to $1/3$, I say T is correct; if it is close to $1/2$, I say S is right. What is this 'close'? It is reasonable to take the midpoint $[(1/3) + (1/2)]/2$ which is $5/12$ and decide as follows:

If $f < 5/12$ then T is correct; if $f > 5/12$ the S is right. ($f = 5/12$ decide at random).

Now there are two kinds of errors that are possible. First is this. T is right, but when I did the experiment I got $f > 5/12$ and I decided S is correct. What are the chances of this error? Thus you want to find out $P(X > 5/12)$ when $X \sim B(10,000; 1/3)$.

Second kind of error is this. S is right but when I did the experiment I got $f < 5/12$ and so decided T is right. What are the chances of this error? Thus you want to find out $P(X < 5/12)$ when $X \sim B(10,000; 1/2)$.

CLT helps you with both.

Many times you suspect whether the estimating etc has any meaning when

matters depend on chance. How reliable are the so called estimates. While accepting that the estimate may not always be close to the true value, we are able to quantify the chances of errors. Think about it.

(E) modelling errors:

There is yet another way of looking at the CLT. For the moment assume that mean μ is zero and variance is one. Then what CLT says is that

$$\frac{X_1}{\sqrt{n}} + \frac{X_2}{\sqrt{n}} + \frac{X_3}{\sqrt{n}} + \cdots + \frac{X_n}{\sqrt{n}}$$

is approximately standard normal. Thus a large number of small quantities when added is nearly normal.

When you make measurements, there are lots of possible reasons for errors (measuring instrument, your observation, the inherent fluctuations in what you are measuring etc); each reason causing a small error. Thus it is reasonable to assume that the total error is a normal variable. This you will encounter when you do regression, which in its simplest form is the following. I have two variables X and Y and I believe that Y is a function of X , say $\varphi(X)$. How do I model it? We say $Y = \varphi(X) + \epsilon$ where ϵ is a normal variable with mean zero.

Characteristic functions:

There is a close relative of mgf, called Characteristic function. If X is a random variable then its characteristic function φ_X is a complex valued function defined on R by

$$\varphi_X(t) = E(e^{itX}) \quad t \in R$$

Thus if X has a pdf f then

$$\varphi_X(t) = \int e^{itx} f(x) dx = \int \cos(tx) f(x) dx + i \int \sin(tx) f(x) dx$$

and if X is discrete with values (x_n) and corresponding probabilities (p_n)

$$\varphi_X(t) = \sum e^{itx_n} p_n = \sum \cos(tx_n) p_n + i \sum \sin(tx_n) p_n$$

Example: $X \sim \text{unif}(0, 1)$

$$\varphi_X(t) = \int_0^1 e^{itx} dx = \frac{e^{it} - 1}{it}$$

Example: $X \sim \text{unif}(-1, 1)$ Use $\sin tx$ is an odd function on $(-1, 1)$

$$\varphi_X(t) = \int_{-1}^1 \cos xt \frac{1}{2} dx = \frac{\sin x}{x}$$

Example: $X \sim B(n, p)$

$$\varphi_X(t) = \sum \binom{n}{k} e^{itk} p^k q^{n-k} = (q + pe^{it})^n$$

Example $X \sim \text{Poisson}(\lambda)$

$$\varphi_X(t) = \sum e^{itk} e^{-\lambda} \frac{\lambda^k}{k!} = e^{-\lambda(1-e^{it})}$$

Just like mgf we have $\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$ if X, Y are independent. The advantage with cfs is that they always exist unlike mgf. Because $|e^{itX}| \leq 1$, cf $\varphi_X(t)$ is defined for all $t \in R$. Also this is a faithful representative of the distribution. In other words if $\varphi_X(t) = \varphi_Y(t)$ for all t then X and Y have the same distribution. There are formulae explaining how to recover the pdf or pmf of X using its cf. One great advantage is that this puts all the tools of complex analysis at our disposal.

sampling distributions:

As explained earlier iid random variables are called sample from the common distribution. Distributions of functions of iid rvs are called sampling distributions. For example the sums of independent rvs we were discussing are examples, but you can talk about others too. For example $\sum_1^n (X_i - \bar{X})^2$ where $\bar{X} = \sum_1^n X_i/n$. These are useful in statistical inference and better studied then, though we shall do a little bit while discussing normal variables.

poisson process:

Suppose $(X_i, i \geq 1)$ are iid $\exp(\lambda)$ random variables. For example you can think of them as life times of bulbs. You made first replacement at time X_1 , next replacement at time $X_1 + X_2$, next replacement at time $X_1 + X_2 + X_3$ and so on. Or you can think of radio active disintegration. X_1 is the time of first beep of Gieger counter; $X_1 + X_2$ is the time of second beep; $X_1 + X_2 + X_3$ is the time of third beep. We calculated their density functions earlier. If $S_n = \sum_1^n X_i$ then its pdf is given by

$$f_n(x) = \frac{(\lambda x)^{n-1}}{(n-1)!} \lambda e^{-\lambda x} \quad x > 0$$

Their distribution functions $F_n(t)$ are easy to calculate using integration by parts For $t \leq 0$ they are zero. For $t > 0$ they are given as follows

$$F_1(t) = 1 - e^{-\lambda t}, \quad F_2(t) = 1 - e^{-\lambda t} - \lambda t e^{-\lambda t}$$

and in general

$$F_n(t) = \int_0^t f_n(t)dt = 1 - e^{-\lambda t} - \frac{(\lambda t)}{1!}e^{-\lambda t} - \dots - \frac{(\lambda t)^{n-1}}{(n-1)!}e^{-\lambda t} - \lambda e^{-\lambda t}$$

Now you think at time t , how many replacements have I made so far? or how many beeps did the counter make so far? Let N_t be this number. Clearly $N_0 = 0$ and N_t takes values $0, 1, 2, 3 \dots$

$$P(N_t = 0) = P(X_1 > t) = 1 - F_1(t) = e^{-\lambda t}$$

$$\begin{aligned} P(N_t = 1) &= P(X_1 \leq t, X_1 + X_2 > t) = P(X_1 \leq t) - P(X_1 + X_2 \leq t) \\ &= F_1(t) - F_2(t) = \lambda t e^{-\lambda t} \end{aligned}$$

In general

$$\begin{aligned} P(N_t = n) &= P\left(\sum_1^n X_i \leq t; \sum_1^{n+1} X_i \geq t\right) \\ &= P\left(\sum_1^n X_i \leq t\right) - P\left(\sum_1^{n+1} X_i \leq t\right) = \frac{(\lambda t)^n}{n!} e^{-\lambda t} \end{aligned}$$

In other words N_t is a Poisson rv with parameter λt . The collection of random variables $(N_t, t \geq 0)$ is called Poisson process and has many useful applications.

Normal rvs:

Suppose X, Y are independent $X \sim N(0, \sigma^2)$ and $Y \sim N(0, \tau^2)$. What is the distribution of $X + Y$? If only someone tells you that $X + Y$ is again normal, you can quickly derive (because means and variances add up) $X + Y \sim N(0, \sigma^2 + \tau^2)$. You can deduce this going to multivariate normal. But let us directly solve the problem from change of variable formula. Joint density of X, Y is

$$f(x, y) = \frac{1}{\sqrt{2\pi} \sigma} \frac{1}{\sqrt{2\pi} \tau} e^{-\frac{1}{2}[\frac{x^2}{\sigma^2} + \frac{y^2}{\tau^2}]}$$

Put $Z = X + Y$ and $W = X$. Thus $X = W$ and $Y = Z - W$. Jacobian is one. Joint density of Z, W is

$$g(z, w) = \frac{1}{\sqrt{2\pi} \sigma} \frac{1}{\sqrt{2\pi} \tau} e^{-\frac{1}{2}H}$$

where

$$H = \frac{w^2}{\sigma^2} + \frac{(z - w)^2}{\tau^2} = w^2\left[\frac{1}{\sigma^2} + \frac{1}{\tau^2}\right] - 2\frac{wz}{\tau^2} + \frac{z^2}{\tau^2}$$

$$= c(w - \frac{z}{\tau^2 c})^2 - \frac{z^2}{\tau^4 c} + \frac{z^2}{\tau^2}$$

where

$$c = \frac{1}{\sigma^2} + \frac{1}{\tau^2} = \frac{\sigma^2 + \tau^2}{\sigma^2 \tau^2}$$

note

$$\frac{z^2}{\tau^2} - \frac{z^2}{\tau^4 c} = z^2 \frac{\tau^2(\frac{1}{\sigma^2} + \frac{1}{\tau^2}) - 1}{\tau^4(\frac{1}{\sigma^2} + \frac{1}{\tau^2})} = z^2 = z^2 \frac{1}{\sigma^2 + \tau^2}$$

so that

$$H = c(w - \frac{z}{\tau^2 c})^2 + z^2 \frac{1}{\sigma^2 + \tau^2}$$

Thus

$$g(z, w) = \frac{1}{\sqrt{2\pi}} \frac{1}{\sigma} \frac{1}{\sqrt{2\pi}} \frac{1}{\tau} e^{-\frac{1}{2}c(w - \frac{z}{\tau^2 c})^2} e^{-\frac{1}{2}z^2 \frac{1}{\sigma^2 + \tau^2}}$$

Integrating w we get

$$h(z) = \frac{1}{\sqrt{2\pi(\sigma^2 + \tau^2)}} e^{-\frac{1}{2(\sigma^2 + \tau^2)}z^2}$$

showing $Z \sim N(0, \sigma^2 + \tau^2)$. You can now easily deduce if $X \sim N(\mu, \sigma^2)$ and $Y \sim N(\nu, \tau^2)$ independent then $X + Y \sim N(\mu + \nu, \sigma^2 + \tau^2)$

Suppose $X_1, X_2, \dots, X_n$ are iid $N(0, 1)$ so that their joint density is

$$f(x) = \frac{1}{(2\pi)^{n/2}} e^{-x^t x/2} \quad x \in R^n$$

If A is an orthogonal matrix then change of variable formula tells us that the joint density of $Y = AX$ is given by

$$g(y) = \frac{1}{(2\pi)^{n/2}} e^{-y^t y/2} \quad y \in R^n$$

Thus $Y_1, Y_2, \dots, Y_n$ are again iid $N(0, 1)$. let us now consider a special case. Take an orthogonal matrix A where first row is

$$\left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}, \dots, \frac{1}{\sqrt{n}} \right)$$

Since this is a vector of norm one, you can make such an A . Let us see $Y = AX$. Clearly

$$Y_1 = \frac{X_1 + \dots + X_n}{\sqrt{n}} = \sqrt{n} \bar{X} \quad (\clubsuit)$$

and noting that A being orthogonal, we have $\sum X_i^2 = \sum Y_i^2$;

$$\sum (X_i - \bar{X})^2 = \sum X_i^2 - 2 \sum X_i \bar{X} + n \bar{X}^2 = \sum Y_i^2 - n \bar{X}^2$$

now use ($\clubsuit$)

$$\sum (X_i - \bar{X})^2 = \sum Y_i^2 - Y_1^2 = \sum_2^n Y_i^2 \quad (\spadesuit)$$

Since (Y_i) are independent and $\bar{X}$ is a function of Y_1 and $\sum (X_i - \bar{X})^2$ is a function of $Y_2, \dots, Y_n$ we conclude these are independent. Thus we have $\bar{X}$ and $\frac{1}{n-1} \sum (X_i - \bar{X})^2$ are independent. If we use the interpretation that $X_1, \dots, X_n$ is a sample from standard normal; the quantity $\bar{X}$ is called sample mean and the quantity $s^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ is called sample variance. Yes, some use $\frac{1}{n} \sum (X_i - \bar{X})^2$ as sample variance. take anything, you like, no problem. The reason I took $(n-1)$ is that then we have $E(\bar{X}) = 0$ the ‘population’ mean and $E(s^2) = 1$, the population variance. Thus what we have is:

For a standard normal sample, the sample mean and sample variance are independent.